



AI for Software Engineers Course
Core AI + Machine Learning Foundations (Engineer-Oriented)
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Breakthrough

➤ **AlexNET - 2012**

- *Alex Krizhevsky, Ilya Sutskever and Geoffrey Hinton*
- *8 Layer CNN Architecture with 60M parameters,*
- *Used GPU to calculate the results faster.*
- *More than 10.8% above the runner-up in ImageNet Large Scale Visual Recognition Challenge (ILSVRC)*



Core AI + Machine Learning Foundations (Engineer-Oriented)

- 1. AI landscape for engineers*
- 2. How ML systems actually work*
- 3. Core ML types (practical framing)*
- 4. Neural networks intuition*
- 5. Hands-on (important for engineers)*



1 - AI Landscape for Engineers

1. *AI vs ML vs Deep Learning vs Generative AI*
2. *Where classical software ends and ML systems begin*
3. *Why ML is “data + optimization + models”*
4. *Roles*



1.1 - AI vs ML vs Deep Learning vs Generative AI

ARTIFICIAL INTELLIGENCE

MACHINE LEARNING

NEURAL NETWORKS

Fully Connected

RNN

LSTM

CNN

GAN

DBN

VAE

KAN

RBM

GNN

PINN

Diffusion

TRANSFORMERS

BERT
ENC

BART
ENC-DEC

GPT
DEC

LLM

AGENTIC

REGRESSION

SVM

DECISION TREE

KNN

PCA

RANDOM FOREST

CLUSTERING

Reinforcement Learn

.....

Prolog

Comp. Vision

Automation

Robotics

NLP

...



1.1 - AI vs ML vs Deep Learning vs Generative AI

AI

Input

MODEL

Code : Rule Based

No Training

Prolog/Lisp/CV/MFCC

Coded Human Experience

Directly Explainable

Output

ML

Input

MODEL

Code : Parameter Based

Train Model Parameters

Python / SkLearn / ...

Human Experience + Data

Easy to Explain

Output

DL

Input

MODEL

Code : No Code

Train Model Parameters

Python / Torch / TensorFlow

Full Data

Hard to Explain

Output

GEN

Input

Random
Numbers

MODEL

Code : No Code

Train Model Parameters

Python / Torch / TensorFlow

Full Data

Very Hard to Explain

Output



1.2 - Where classical software ends and ML systems begin

Counter Example : Fibonacci Numbers : $a_n = a_{n-1} + a_{n-2}$ $a_{100000} = ?$

Math : Binet's Formula (Generating Functions) (De Moivre 1730)

$$F(n) = \frac{\phi^n - (1-\phi)^n}{\sqrt{5}}$$

$$\phi = \frac{1+\sqrt{5}}{2}$$

Algorithm :

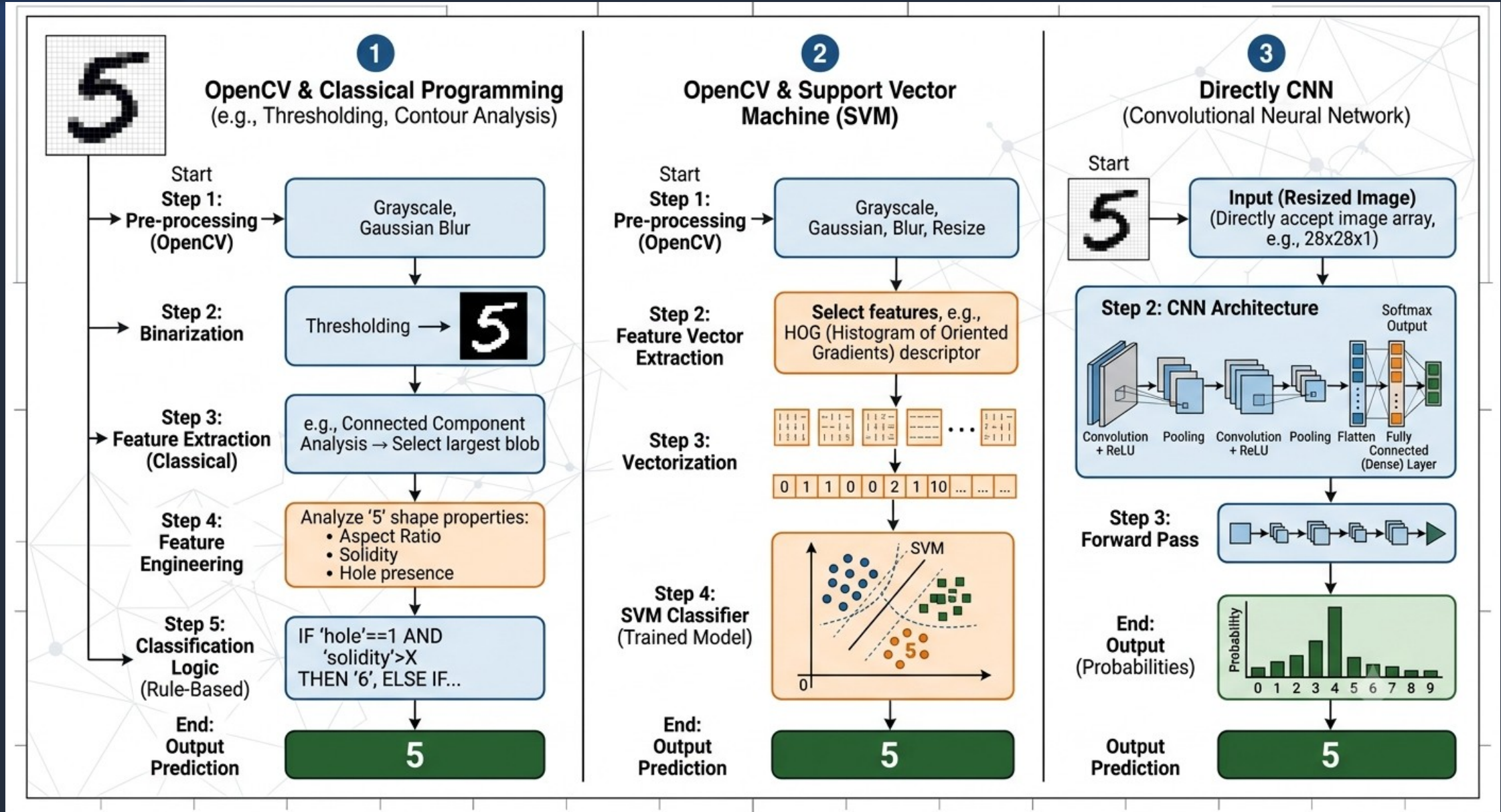
```
F_n=0 ; F_n_1=2; F_n_2=1
n=5
for i in range(n):
    F_n = F_n_1 + F_n_2
    F_n_2=F_n_1
    F_n_1=F_n
print(F_n)
```

O(n)

Machine Learning : Generate 1000 data points , train model using these data, try to guess a_{100000} (test the model).



1.2 - Where classical software ends and ML systems begin



1.2 - Where classical software ends and ML systems begin

Solving problems with Classical Software :

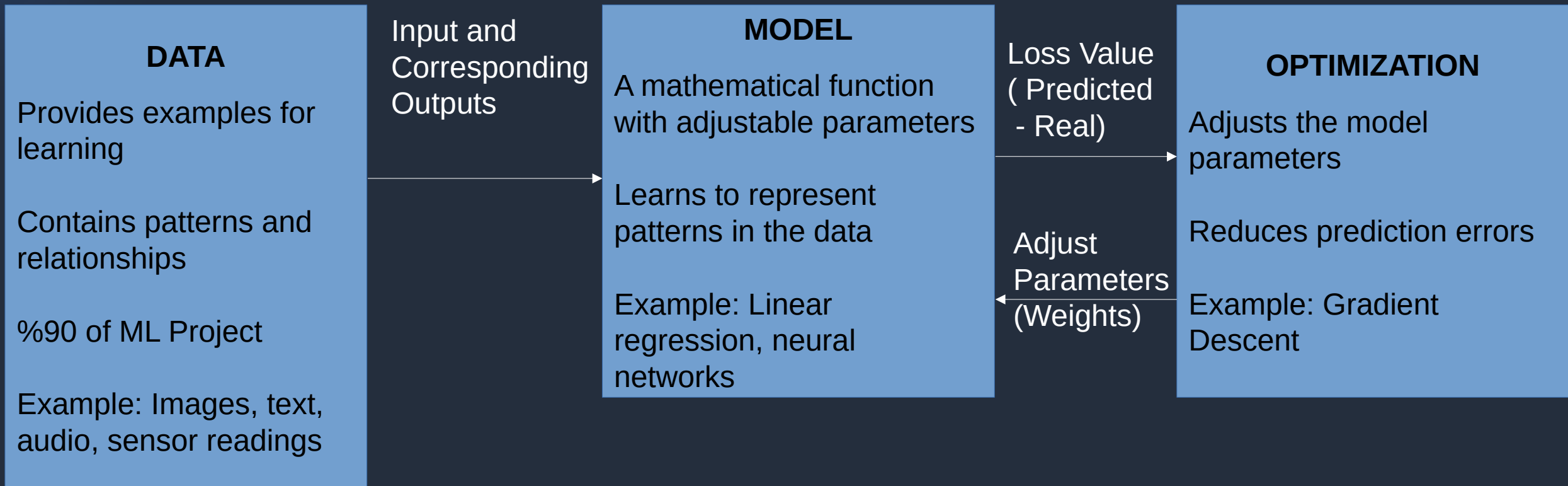
- Analyze the problem mathematically
- Read previous works and try to generalize the previous solutions to cover your problem.
- **Test your solution on various input/output data of the problem.**
- **Analytical proof exists BUT can not be easily generalized !**

Solving problems with ML/DL :

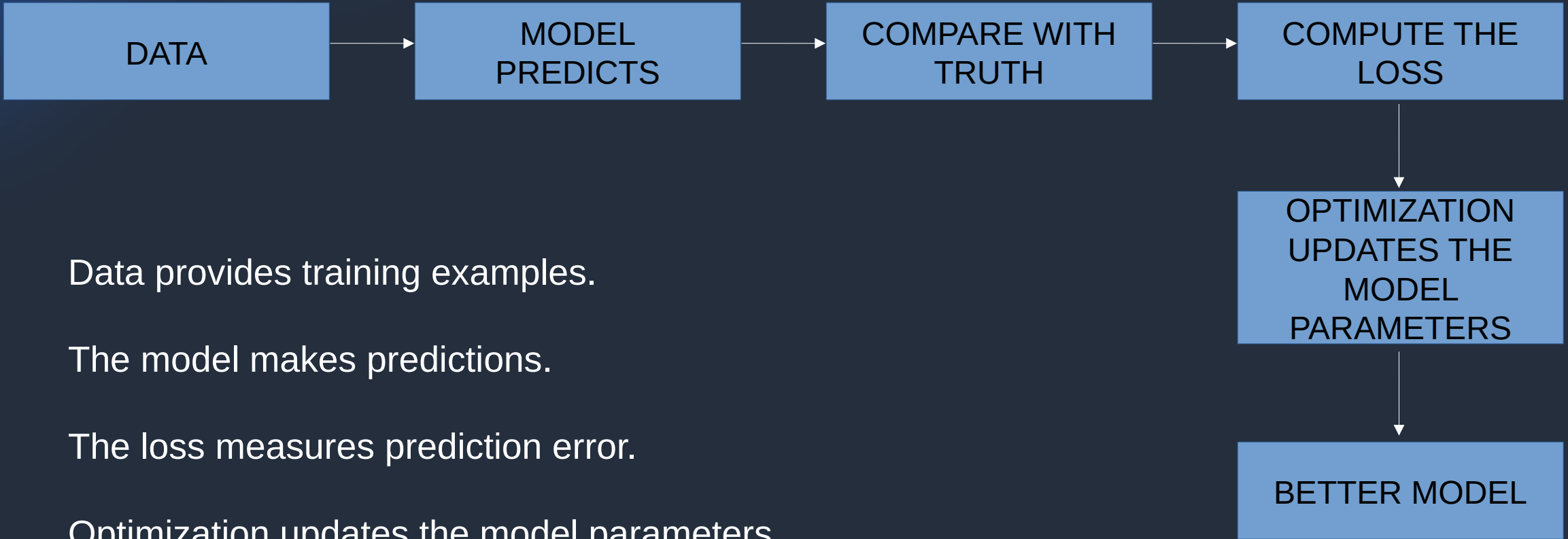
- Collect real data for the problem.
- Choose the ML/DL Architecture to use. (SVM, CNN, Transformer,)
- Train, Validate and Test
- **Good generalization BUT no analytical proof and DEPENDS ON THE DATA QUALITY !**



1.3 – Why ML is “data + optimization + models”



1.3 – Why ML is “data + optimization + models”



Data provides training examples.

The model makes predictions.

The loss measures prediction error.

Optimization updates the model parameters.

The process repeats until the error is minimized



1.3 – Why ML is “data + optimization + models”

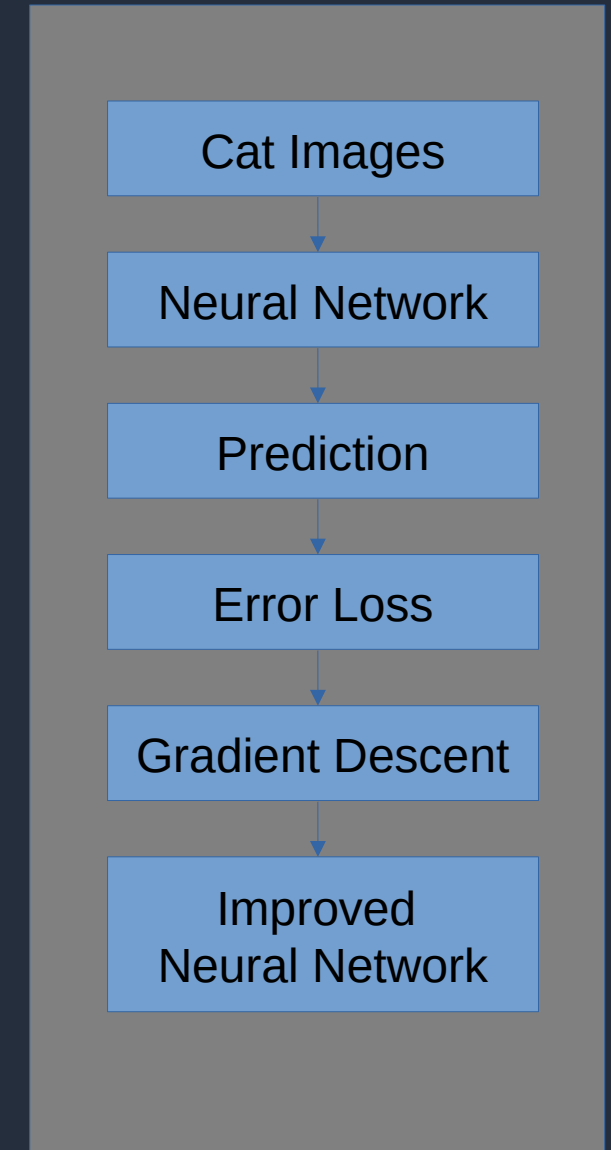
Without	What Happens
Data	No Information to learn from
Model	No function to represent the solution
Optimization	Model parameters never improve

Data tells the algorithm what to learn.

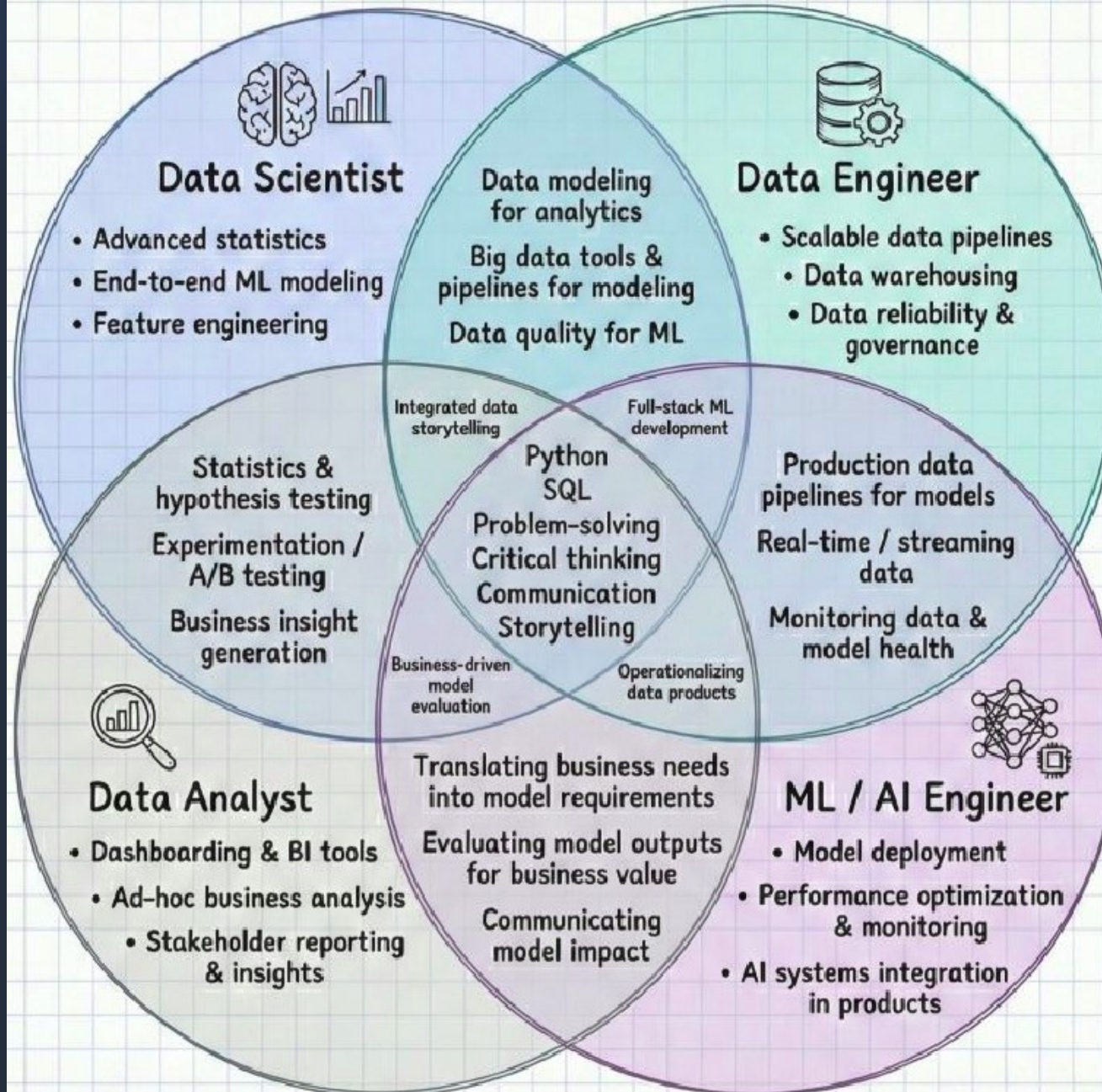
Model defines how knowledge is represented.

Optimization determines how the model improves.

EXAMPLE



1.4 – Roles



2 - How ML Systems Actually Work

1. *Dataset → training → model → inference pipeline*
2. *Train/test split, overfitting (intuition, not math-heavy)*
3. *Loss function intuition*
4. *Feature vs representation learning*
5. *ML Model Evaluation : False Positive, Precision, Recall, Accuracy, Confusion Matrix*



2.1 - Dataset → training → model → inference pipeline

PROBLEM DEFINITION

DATA COLLECTION

DATA CLEANING

FEATURE EXTRACTION

TRAINING

VALIDATION

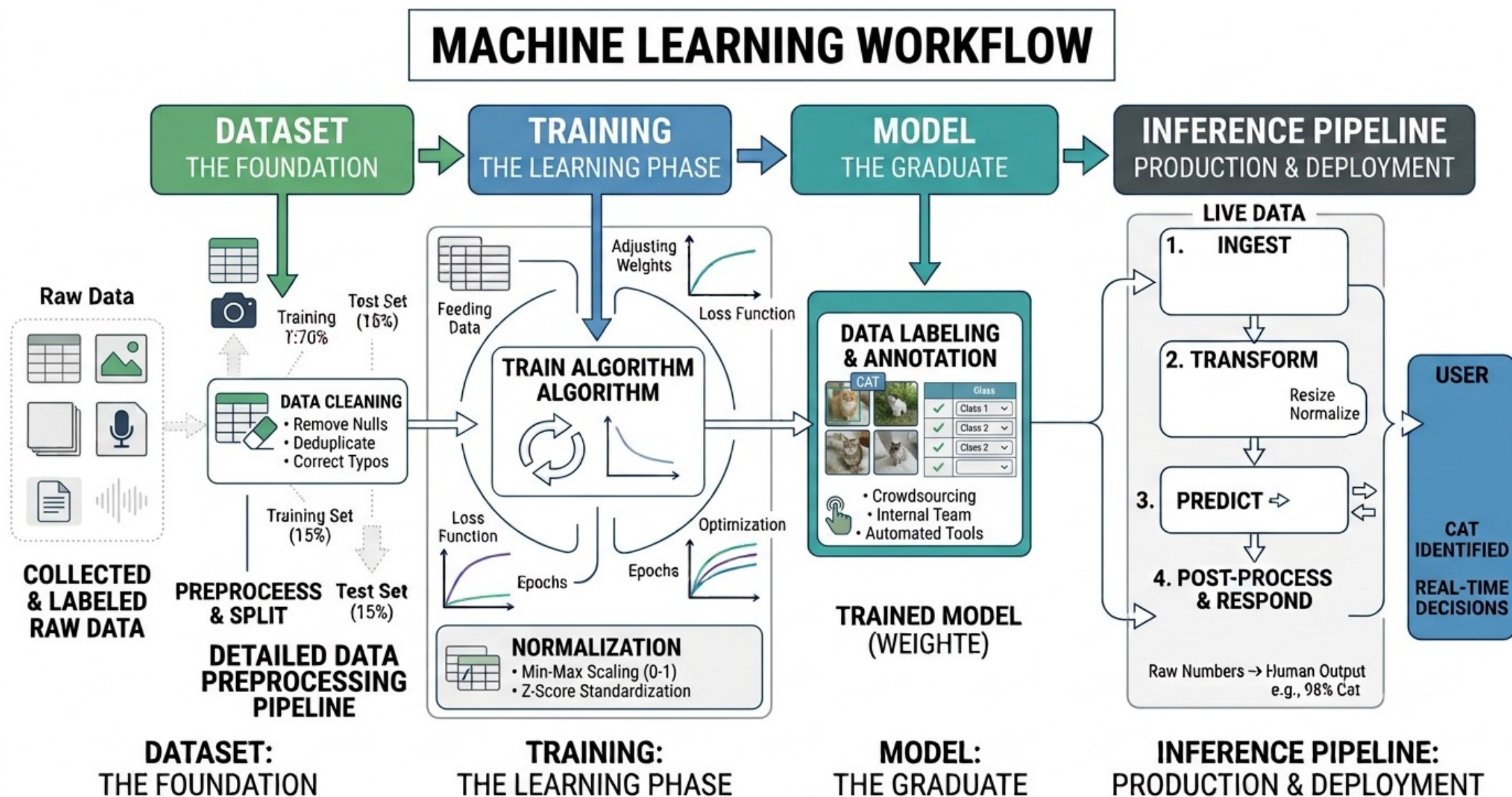
TESTING

DATA VISUALIZATION

REPORT GENERATION



2.1 - Dataset → training → model → inference pipeline



2.1 - Dataset → training → model → inference pipeline

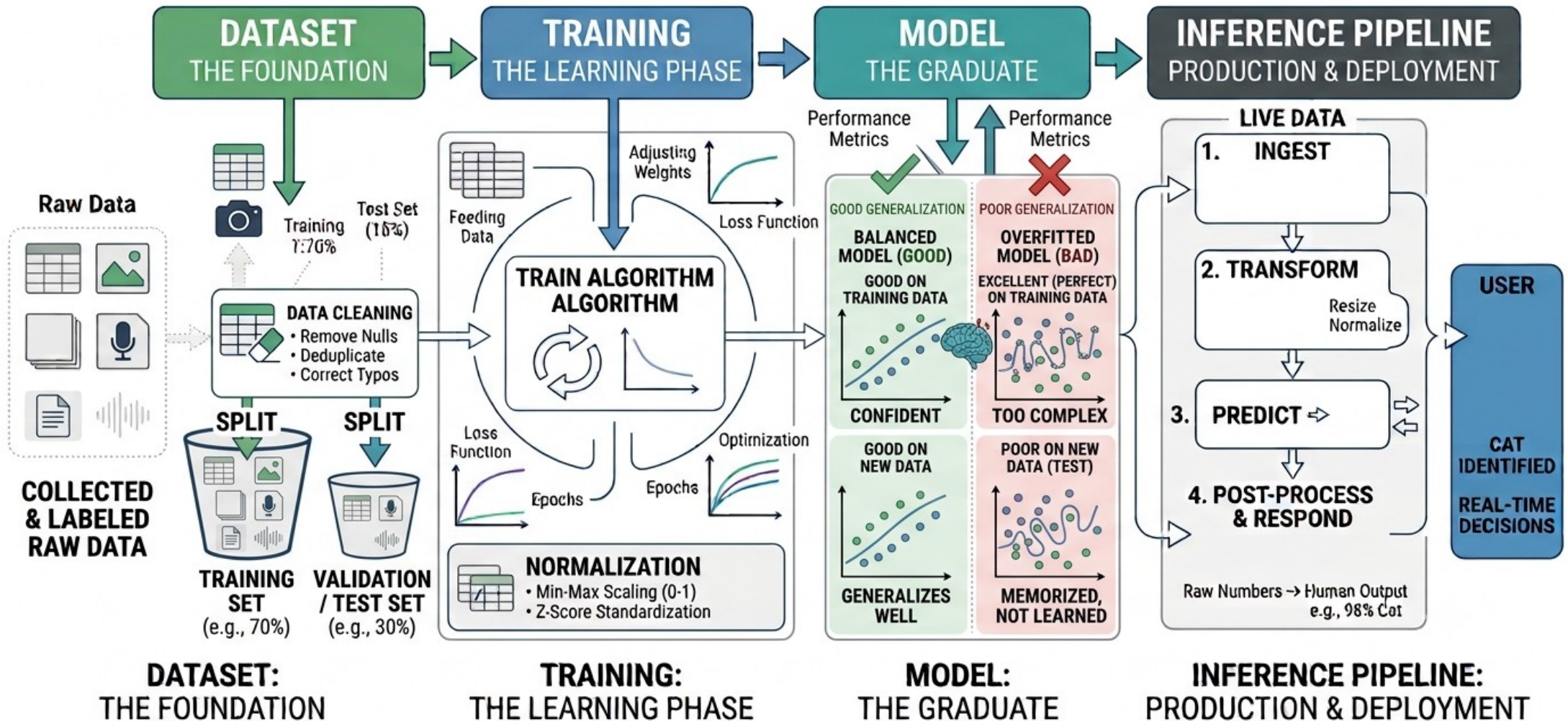
We need a lot of labeled data:

- We need to find **an automated way to collect data**
 - After building a semi automated sound recording system, I was able to collect thousands of sample data.
- We need the data to be **self labeled**.
 - Suppose that , using a monocular (single-camera) system, we try to extract the Bearing and Range of an object :
 - We record camera video,
 - We record AIS data of the objects around us, that correspond to the field of view of camera
 - We record FlightRadar etc. data also.
 - Thank to the AIS and FlightRadar data, we can say that our data is labeled (semi-labeled).

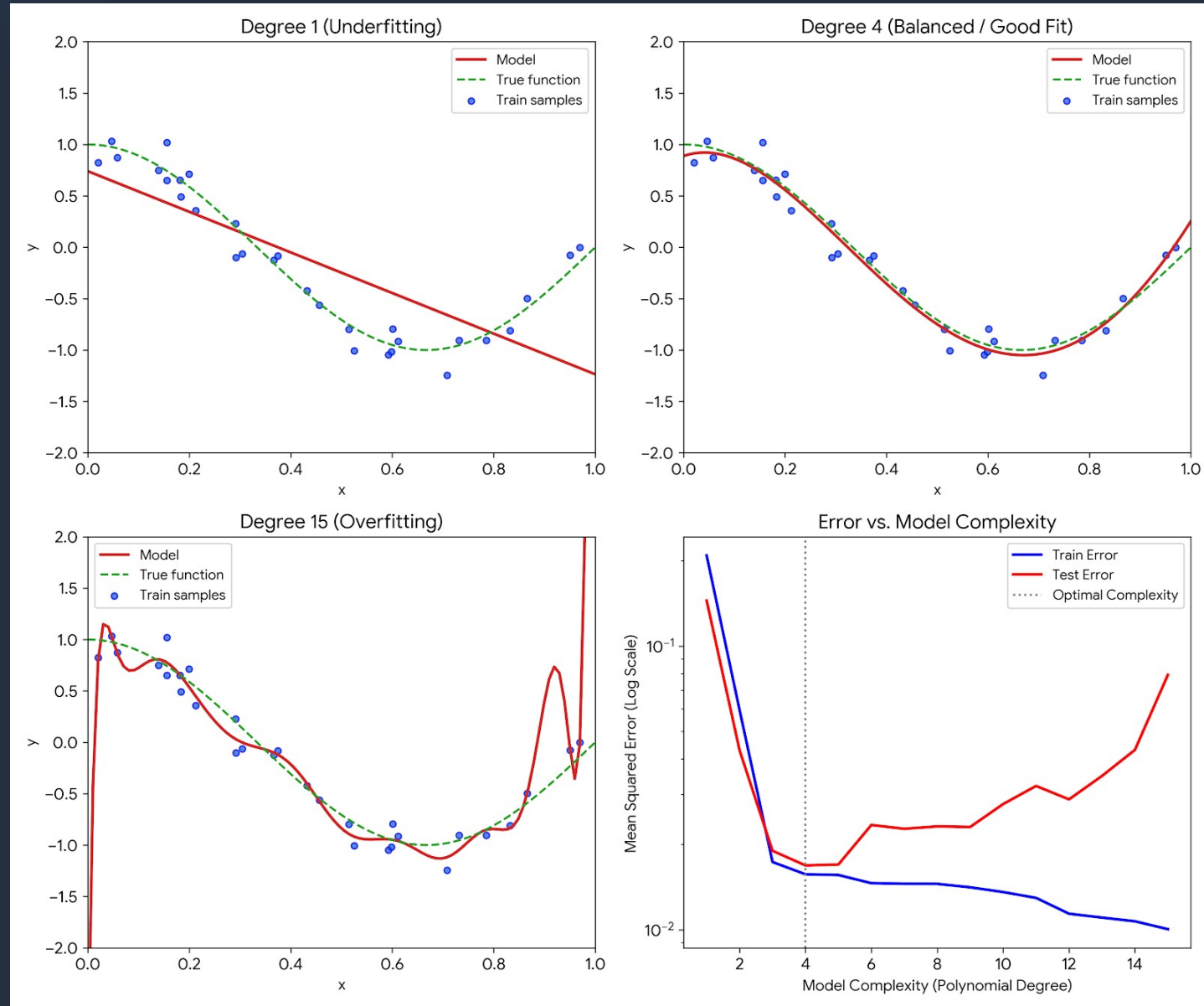


2.2 - Train/test split, overfitting (intuition, not math-heavy)

UNDERSTANDING ML PERFORMANCE & OVERFITTING



2.2 - Train/test split, overfitting (intuition, not math-heavy)



2.2 - Preventing overfitting

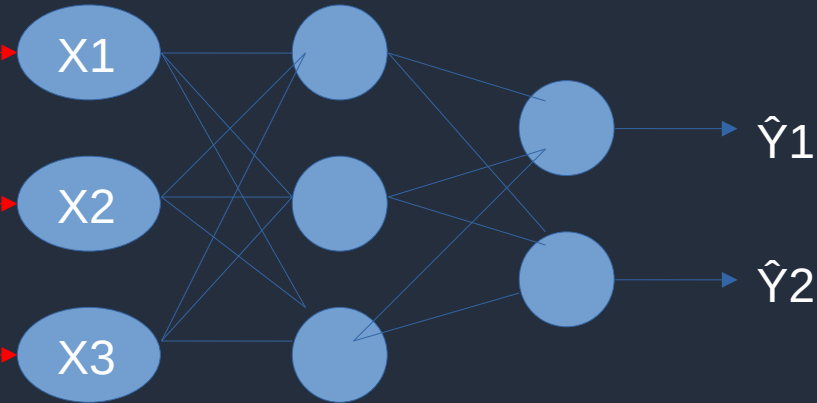
1. Regularization : Constraining the Model, L1, L2, Elastic, Dropout
2. Modifying the Data : Collect More Data, Augmentation, Feature Selection/Reduction
3. Adjusting Training Process : Early Stopping, Cross-Validation
4. Architectural Control and Ensembling : Reduce Model Capacity, **Ensemble Methods (Multiple Small models instead of 1 big model)**



2.3 – Loss Function Intuition

Loss Function = Distance Function = Metric
Loss = Error = Distance

X			Y	
X1	X2	X3	Y1	Y2
1	2	3	4	5
11	12	13	14	15
21	22	23	24	25
31	32	33	34	35
41	42	43	44	45



$$L_1 = |\hat{Y}_1 - 4| + |\hat{Y}_2 - 5|$$

$$L_2 = \text{sqrt}(|\hat{Y}_1 - 4|^2 + |\hat{Y}_2 - 5|^2)$$

$$L = -(4 \log(\hat{Y}_1) + 5 \log(\hat{Y}_2))$$

Some of Loss Functions n = output vector length (in this case 2):

Minkowski Distance :
$$L_p = \left(\sum_{i=0}^n |Y_i - \hat{Y}_i|^p \right)^{\frac{1}{p}}$$
 , L_1 = Manhattan Distance , L_2 = Euclidean Distance

Cross Entropy :
$$L = - \sum_{i=0}^n Y_i \cdot \log(\hat{Y}_i)$$
 , Softmax , Log Loss , n=number of classes , output is one-hot encoded



2.3 – Loss Function Intuition

MAE : Mean Absolute Error

RMSE : Root Mean Squared Error (A type of Euclidean Distance)

Cosine Distance : Used in LLMs, comparing two vectors $X \cdot Y / |X| \cdot |Y| = \cos$ between X and Y vectors.

KL Divergence : VAE, GAN (Besides normal loss, the distance from distribution is added to loss)

Mahalanobis Distance : Distance to a group of data points (1-N)

Hinge Loss : Used in SVMs

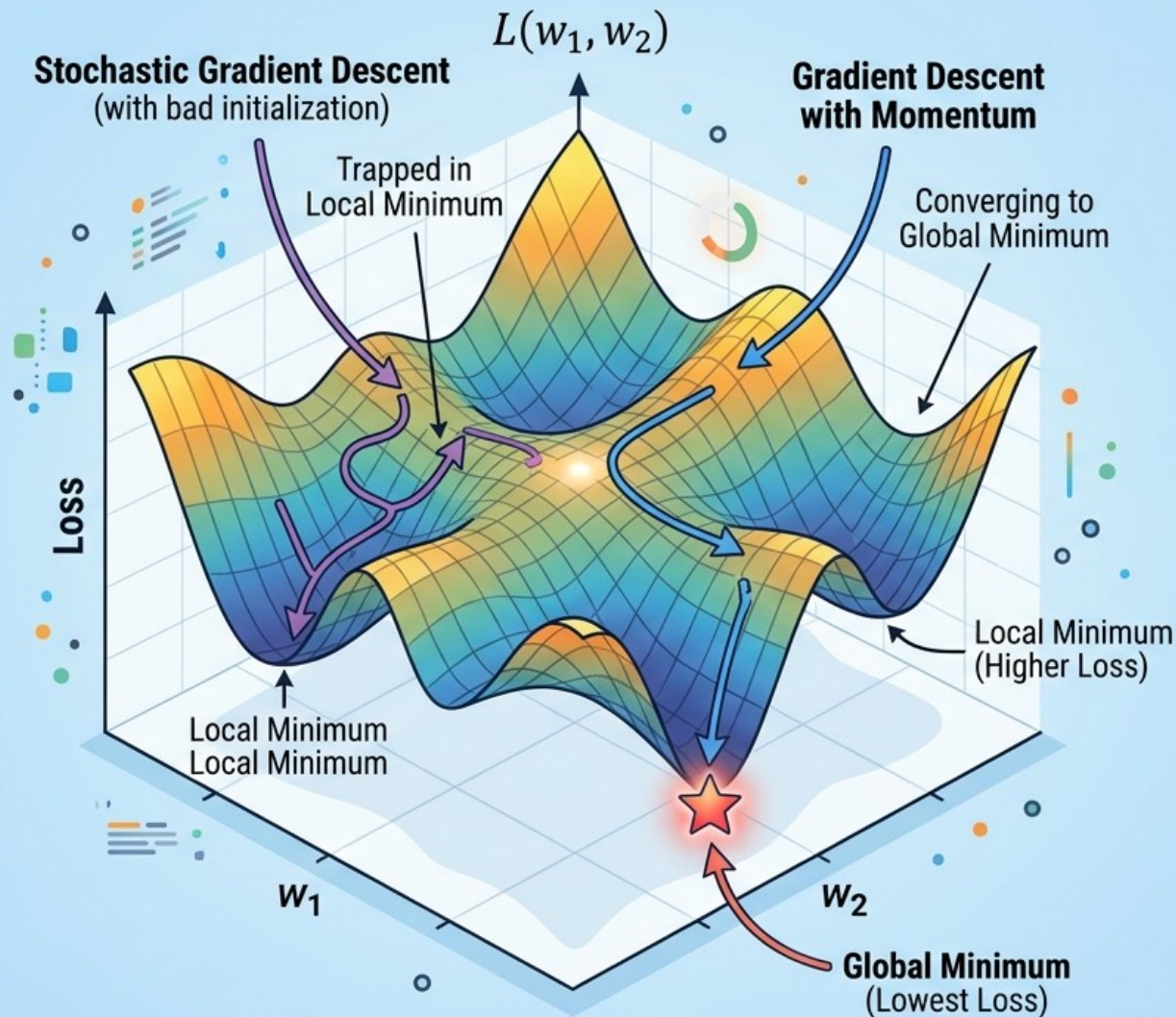
SSIM (Structural Similarity) : Used to compare images, sound, ...

PINN : Loss functions in physics informed neural network is physical formula like wave equation

Dice, Perceptual, Total Variations, (Any Metric corresponding to problem nature)



2.3 – Loss Function Intuition



2D LOSS FUNCTION GLOBAL MINIMUM SEARCH

Objective:

Find the set of parameters (w_1, w_2) that minimizes the objective function $L(w_1, w_2)$ over its entire domain.



Challenges:

- **Local Minima** (multiple dips)
- **Saddle Points** (flat regions with different slopes)
- **Plateaus** (very flat areas)
- **Complex Non-Convex Landscapes**

Common Optimization Algorithms for Global Search:



Gradient Descent with Momentum:

Accelerates past local minima



Adam (Adaptive Moment Estimation):

Adapts learning rates, good for sparse gradients



Simulated Annealing / Genetic Algorithms:

Probabilistic and evolutionary approaches, suitable for complex landscapes



2.4 - Feature vs representation learning

Feature Learning (Extraction) : Learn useful predictors.

Representation Learning : Learn a useful encoding of data.

Aspect	Feature Learning	Representation Learning
Goal	Learn useful features from raw data	Learn a meaningful representation of the data
Scope	Narrower	Broader
Output	Features for a specific task	Embeddings or latent representations that capture data structure
Typical Use	Classification, regression	Transfer learning, generation, clustering, downstream tasks
Relationship	A subset of representation learning	Includes feature learning and more



2.4 - Feature vs representation learning

Traditional approach: Human designs features such as edges, corners, texture descriptors (e.g., SIFT or HOG).

Feature learning: A neural network learns filters that detect edges, textures, object parts, etc., during training.

Representation learning: Aims to learn a transformation of the data into a representation that captures important underlying structure.

A good representation should ideally:

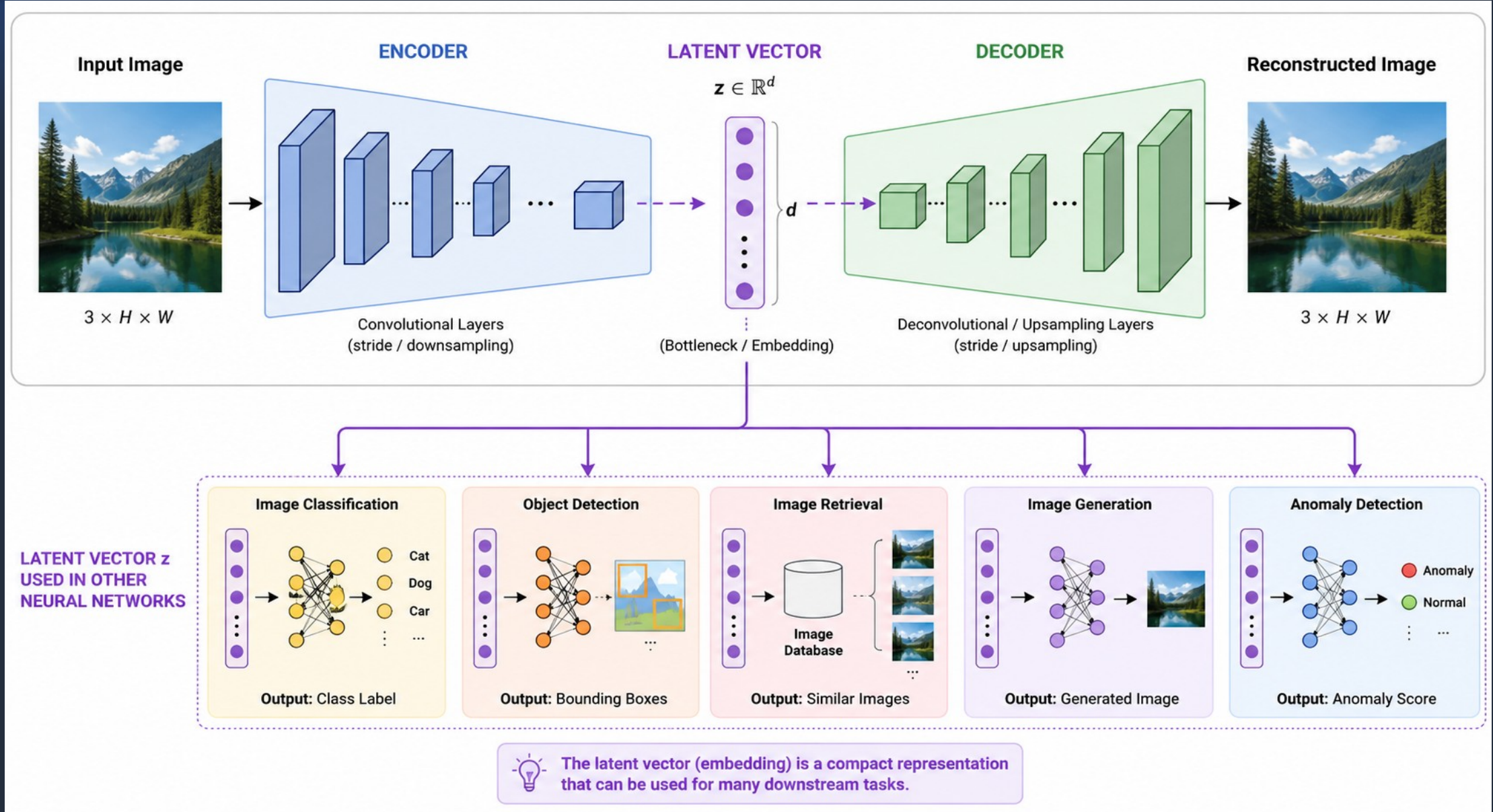
- Preserve relevant information.
- Discard irrelevant variation (noise).
- Be useful for multiple downstream tasks.
- Capture semantic relationships.

Examples include:

- Latent vectors from autoencoders
- Word embeddings (e.g., Word2Vec)
- Contextual embeddings from BERT
- Vision embeddings from contrastive learning models



2.4 - Feature vs representation learning

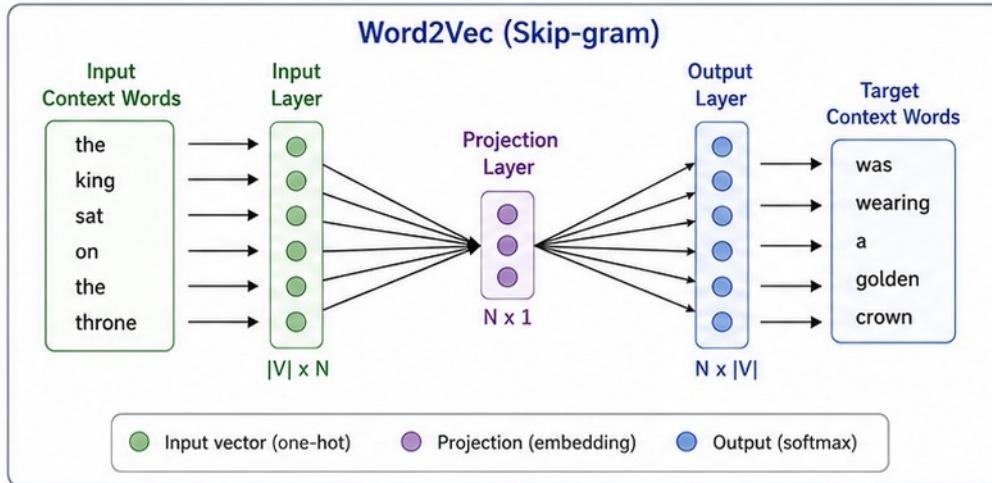


2.4 - Feature vs representation learning

Word2Vec

Words are mapped to dense vectors in a continuous vector space where semantically similar words are close to each other.

Word2Vec (Skip-gram)



Each word is represented as a dense vector (embedding)

Word	v_1	v_2	v_3	...	v_n (N dimensions)
queen	0.21	-0.13	0.34	...	0.12
woman	0.19	-0.24	0.31	...	0.07
man	-0.11	0.22	-0.28	...	-0.09
king	0.22	-0.11	0.35	...	0.10

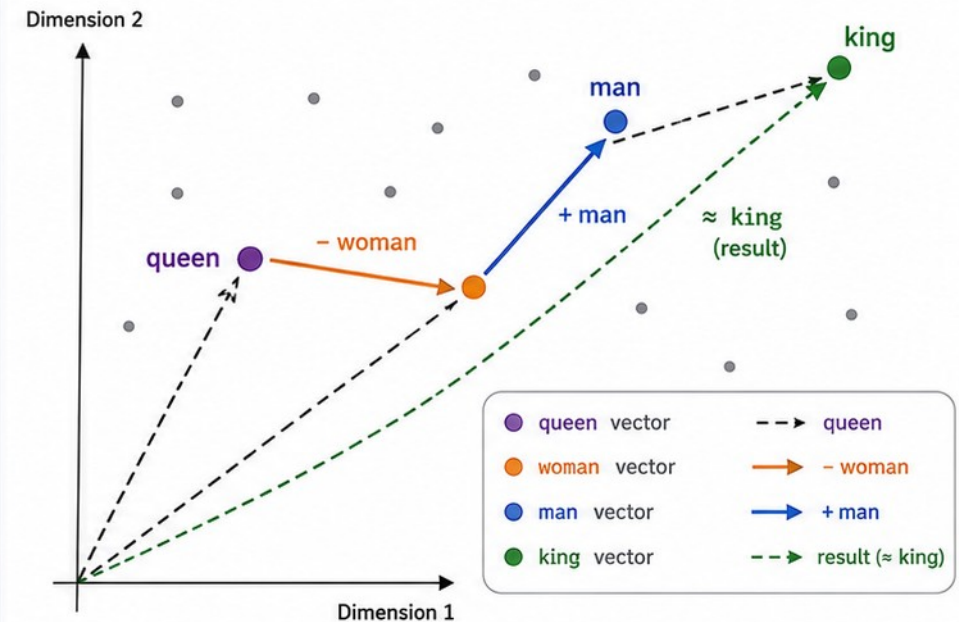
(N is typically 100–300)

Vector Arithmetic Example

$$\text{queen} - \text{woman} + \text{man} = \text{king}$$

In the vector space, the following relationship holds:

$$\vec{\text{king}} \approx \vec{\text{queen}} - \vec{\text{woman}} + \vec{\text{man}}$$



Key Idea

Word2Vec learns word embeddings so that words appearing in similar contexts have similar vectors.

This enables meaningful vector arithmetic: $\text{queen} - \text{woman} + \text{man} = \text{king}$.



2.5 - ML Model Evaluation : False Positive, Precision, Recall, Accuracy, Confusion Matrix

Suppose we have a “machine learning model” that predicts if a track is friend or foe (IFF):

FOE=POSITIVE, FRIEND=NEGATIVE

- If the track is ACTUALLY FRIEND and the model PREDICTED as FRIEND : count as TRUE NEGATIVE
- If the track is ACTUALLY FRIEND and the model PREDICTED as FOE : count as FALSE POSITIVE
- If the track is ACTUALLY FOE and the model PREDICTED as FRIEND : count as FALSE NEGATIVE
- If the track is ACTUALLY FOE and the model PREDICTED as FOE : count as TRUE POSITIVE

Confusion Matrix For Binary Classification

	Predicted Positive	Predicted Negative
Actual Positive	Number of True Positives (TP)	Number of False Negatives (FN)
Actual Negative	Number of False Positives (FP)	Number of True Negatives (TN)



2.5 - ML Model Evaluation : False Positive, Precision, Recall, Accuracy, Confusion Matrix

	Predicted Positive	Predicted Negative
Actual Positive	True Positives (TP)	False Negatives (FN)
Actual Negative	False Positives (FP)	True Negatives (TN)

$$PRECISION = \frac{TP}{TP + FP}$$

$$RECALL = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \cdot PRECISION \cdot RECALL}{PRECISION + RECALL}$$

$$ACCURACY = \frac{TP + FP}{ALL}$$

$$ALL = TP + FP + TN + FN$$

HIGH PRECISION : Model should not miss any FOE, but some of the FOES may be FRIEND actually.

HIGH RECALL : Model may miss some FOEs, but should be sure that no actual FRIEND may be reported as FOE.

HIGH ACCURACY : Model should not miss any FOE, and also should be sure that no actual FRIEND may be reported as FOE.

F1 :

- Completely ignores True Negatives (TN).
- This metric is used in lieu of ACCURACY, when the data is heavily imbalanced.
- Balance between catching positives and being right about them.



2.5 - ML Model Evaluation : False Positive, Precision, Recall, Accuracy, Confusion Matrix

Confusion Matrix : A confusion matrix is a table used to evaluate the performance of a **classification model**.

Each X-Y-Cell Filled By :
How many of detected Class X was actually Class Y ?

Suppose our model will classify the vessels on the sea :

We suppose that it is sufficient to classify the vessels into 3 classes :

- 1. Naval Vessel
- 2. Cargo Vessel
- 3. Passenger Vessel

	Naval Vessel Predicted	Cargo Vessel Predicted	Passenger Vessel Predicted
Naval Vessel Actual	80	30	5
Cargo Vessel Actual	16	650	5
Passenger Vessel Actual	4	20	190

Suppose we made 1000 experiments, 100 were said to be Naval Vessel but actually only 80 of them were Naval, 16 of them were cargo, 4 of them were Passenger ship. For the cargo and the passenger vessels, the numbers are shown above.



2.5 - ML Model Evaluation : False Positive, Precision, Recall, Accuracy, Confusion Matrix

CONFUSION MATRIX $\neq$ CORRELATION MATRIX

A confusion matrix evaluates the performance of a classification model by comparing predicted classes to actual, ground-truth classes.

	Naval Vessel	Cargo Vessel	Passenger Vessel
Naval Vessel	80	30	5
Cargo Vessel	16	650	5
Passenger Vessel	4	20	190

A correlation matrix measures and visualizes the statistical relationships (how they move together) between numerical variables in an ACTUAL dataset.

	Vessel has gun	Vessel has radar	Vessel has rooms	Vessel has portholes
Vessel has gun	1.0	0.8	0.1	0.1
Vessel has radar	0.8	1.0	0.1	0.1
Vessel has rooms	0.1	0.1	1.0	0.9
Vessel has portholes	0.1	0.1	0.9	1.0



2.5 - ML Model Evaluation : False Positive, Precision, Recall, Accuracy, Confusion Matrix

CONFUSION MATRIX IS USED IN THE MODEL EVALUATION PHASE

**CORRELATION MATRIX IS USED IN THE DATA CLEANING/FEATURE
EXTRACTION PHASES**



3 - Core ML Types (practical framing)

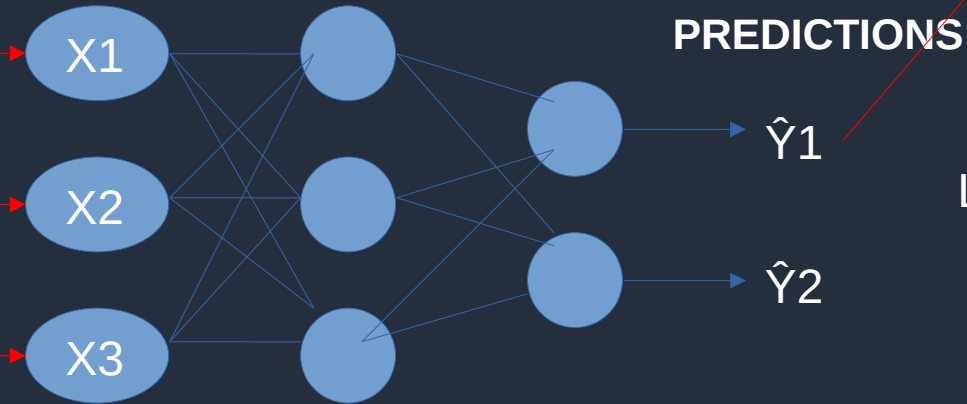
1. *Supervised learning (classification/regression)*
2. *Unsupervised learning (clustering, embeddings idea)*
3. *Reinforcement learning (high-level only)*
4. *Classical ML Techniques : Regression, KNN, SVM, Decision Tree, Random Forest, PCA, XGBoost, Ensemble Learning, K-Means clustering, DBScan clustering.*



3.1 – Supervised Learning (Classification/Regression)

! LABELED REAL DATA !

X (input)			Y (labels)	
X1	X2	X3	Y1	Y2
1	2	3	4	5
11	12	13	14	15
21	22	23	24	25
31	32	33	34	35
41	42	43	44	45



These two values are compared.

$$\text{LOSS} = |\hat{Y}_1 - 4| + |\hat{Y}_2 - 5|$$

LEARNING =
WEIGHT ADJUSTMENT

OPTIMIZER
(E.G. SGD)

SUPERVISED LEARNING = DATA IS LABELED AND THE MODEL IS LEARNING THOSE LABELS



3.1 – Supervised Learning (Classification/Regression)

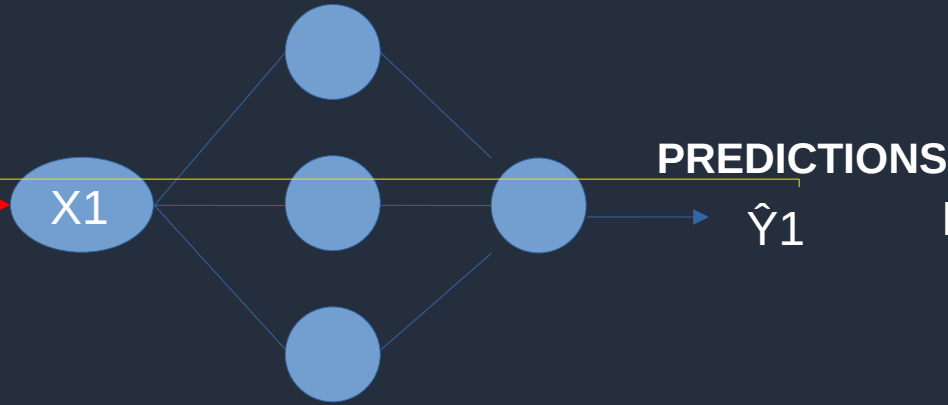
1. *Supervised Learning may be split into two types :*
2. *Regression : Curve fitting. When the target variable is a continuous, numeric value, the task is regression. The goal is to predict a specific quantity.*
3. *Classification : When the target variable is a category or a discrete value, the task is classification. The goal is to predict which class or bucket an input belongs to. (Binary/Multi)*



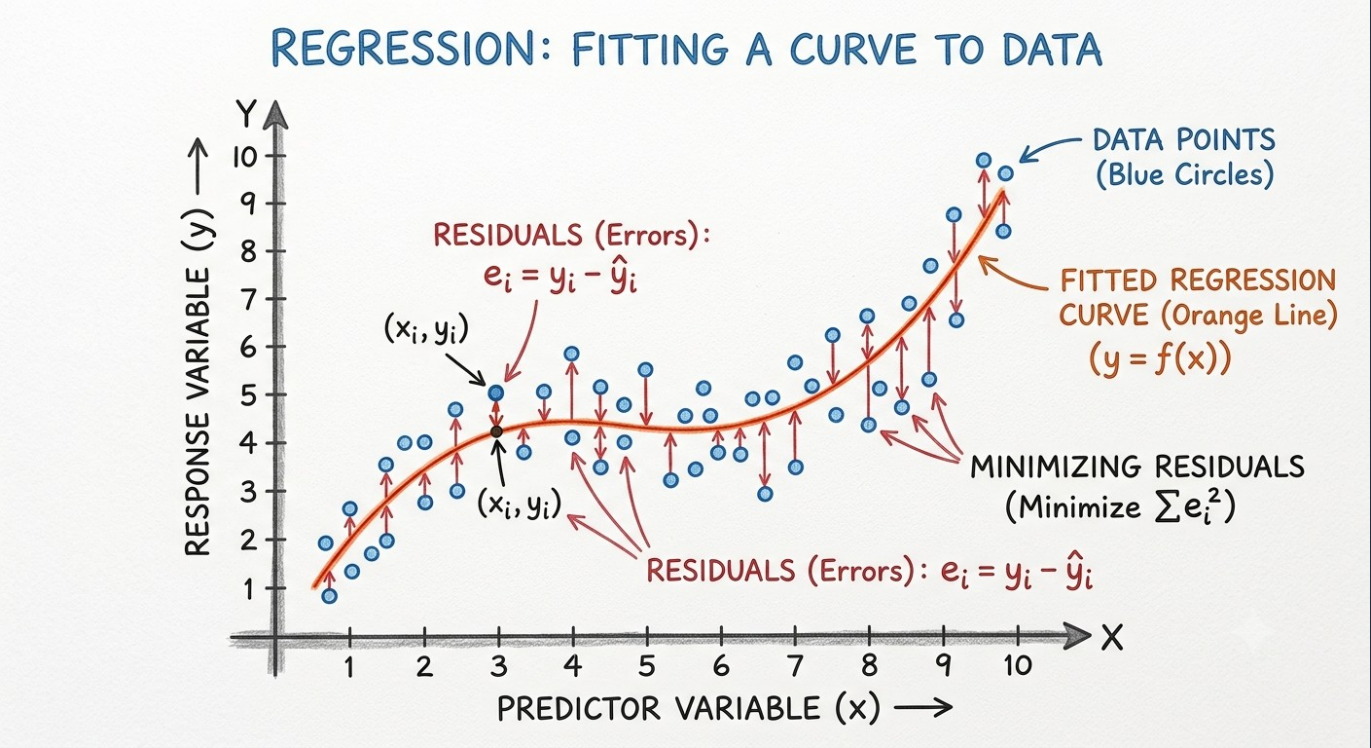
3.1 – Supervised Learning - Regression

REAL DATA

X1	Y1
0.8	0.9
1	1.4
1.2	1.7
1.5	1.5
2	2.5



LOSS = $|\hat{Y}_1 - 4|$



3.1 – Supervised Learning - Classification

One-hot-Encoding

1	0	0
---	---	---

3 classes, vector length is 3, vector is composed of 1 and 0's.

Class-1 is 100

Class-2 is 010

Class-3 is 001

REAL DATA

X1	Y1
0.8	1
1	1
1.2	2
1.5	2
2	3

X1

PREDICTIONS

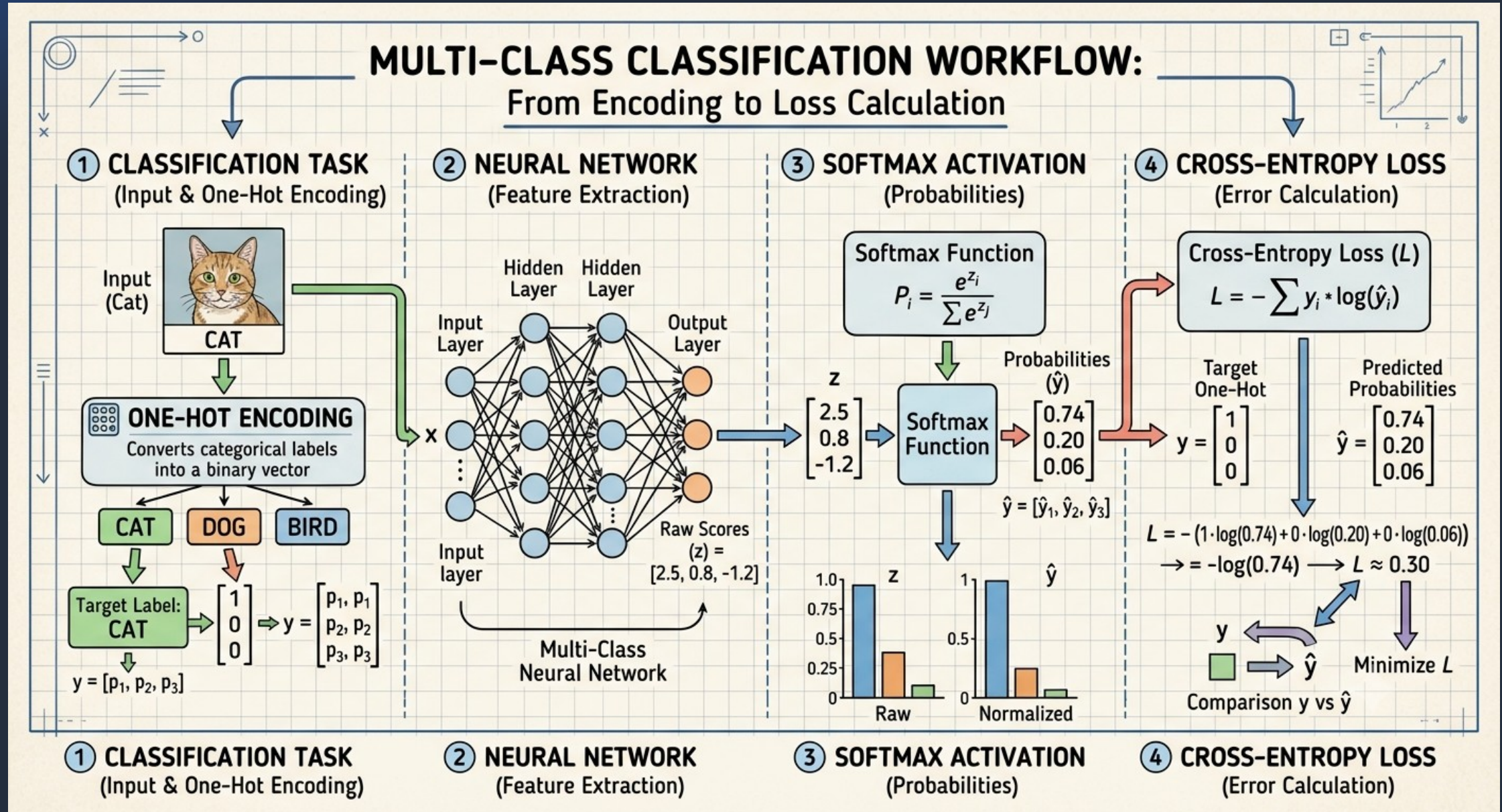
$\hat{Y}_1$

One-hot-Encoded
Softmax

$$\text{LOSS} = \text{Cross Entropy} \\ = \sum Y_1 \log(\hat{Y}_1)$$

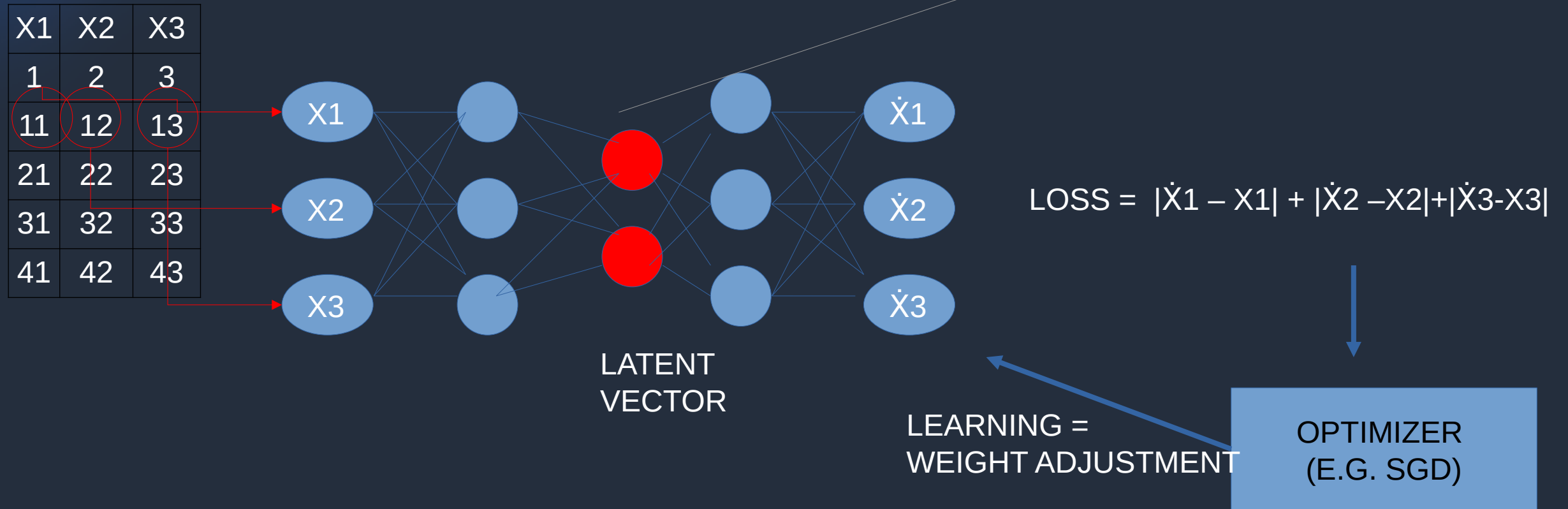


3.1 – Supervised Learning - Classification



3.2 – Unsupervised Learning (Embedding)

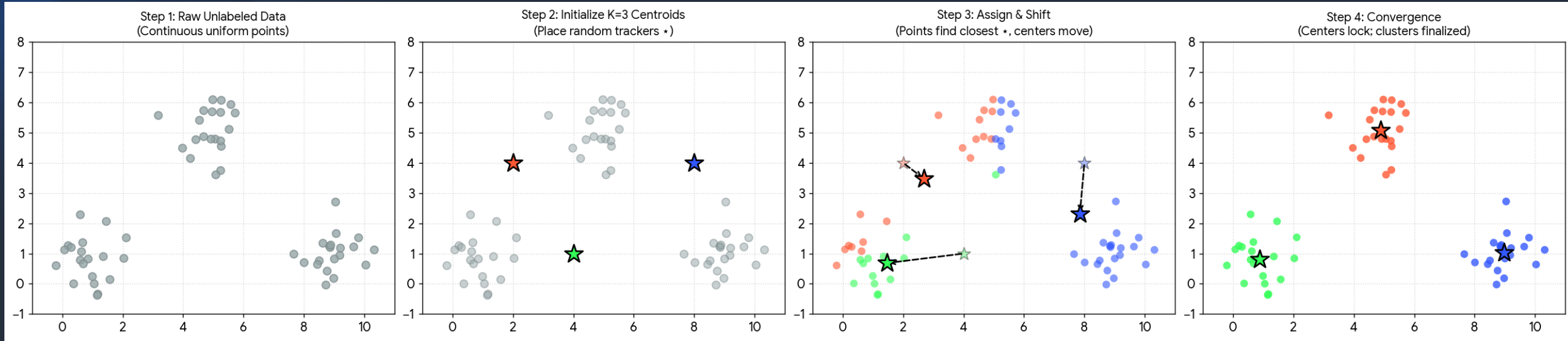
! NOT-LABELED REAL DATA !



UNSUPERVISED LEARNING = DATA IS NOT-LABELED AND THE MODEL IS LEARNING DATA FEATURES



3.2 – Unsupervised Learning (Clustering)



Step 1: Raw Unlabeled Data: The dataset starts entirely gray. The computer only tracks the coordinates (X_1 , X_2) without knowing what category anything belongs to.

Step 2: Initialize K=3 Centroids: You select the value of K. The algorithm seeds initial tracker points, known as centroids (shown as colored stars), across the data area.

Step 3: Assign & Shift:

Assignment: Each gray data point identifies which colored star is closest to it and temporarily adopts that color group.

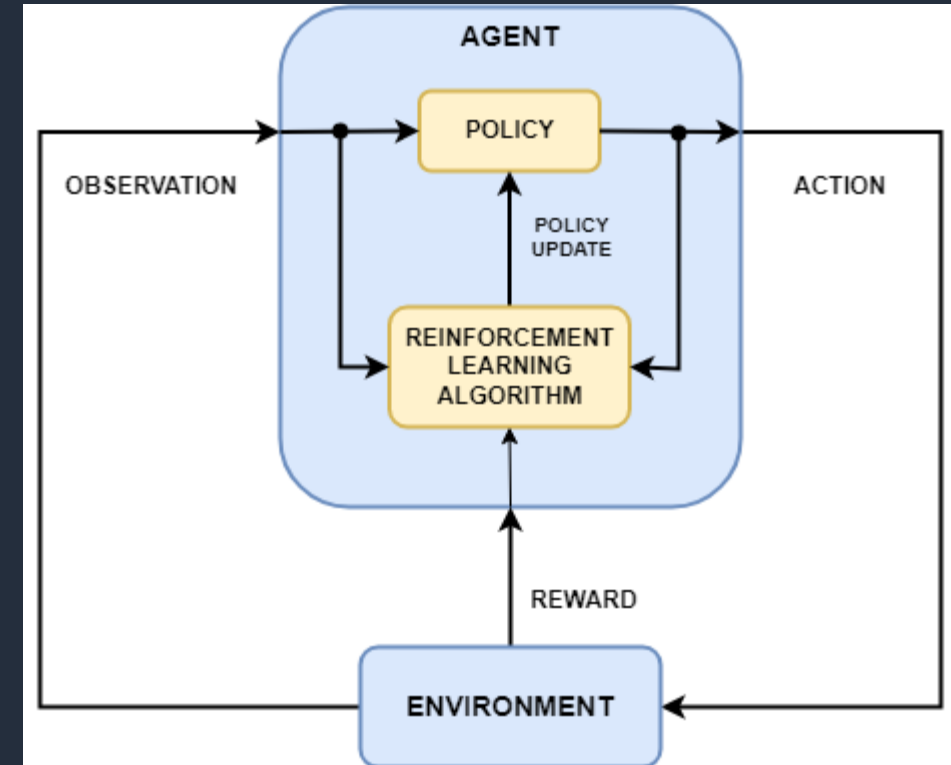
Shift: Each star calculates the geometric center (the mean coordinate) of its newly claimed colored dots and shifts directly along a vector trajectory (indicated by the dashed arrows).

Step 4: Convergence: The assign-and-shift cycle loops automatically until the stars stop moving. The borders lock, yielding isolated, color-coded clusters.



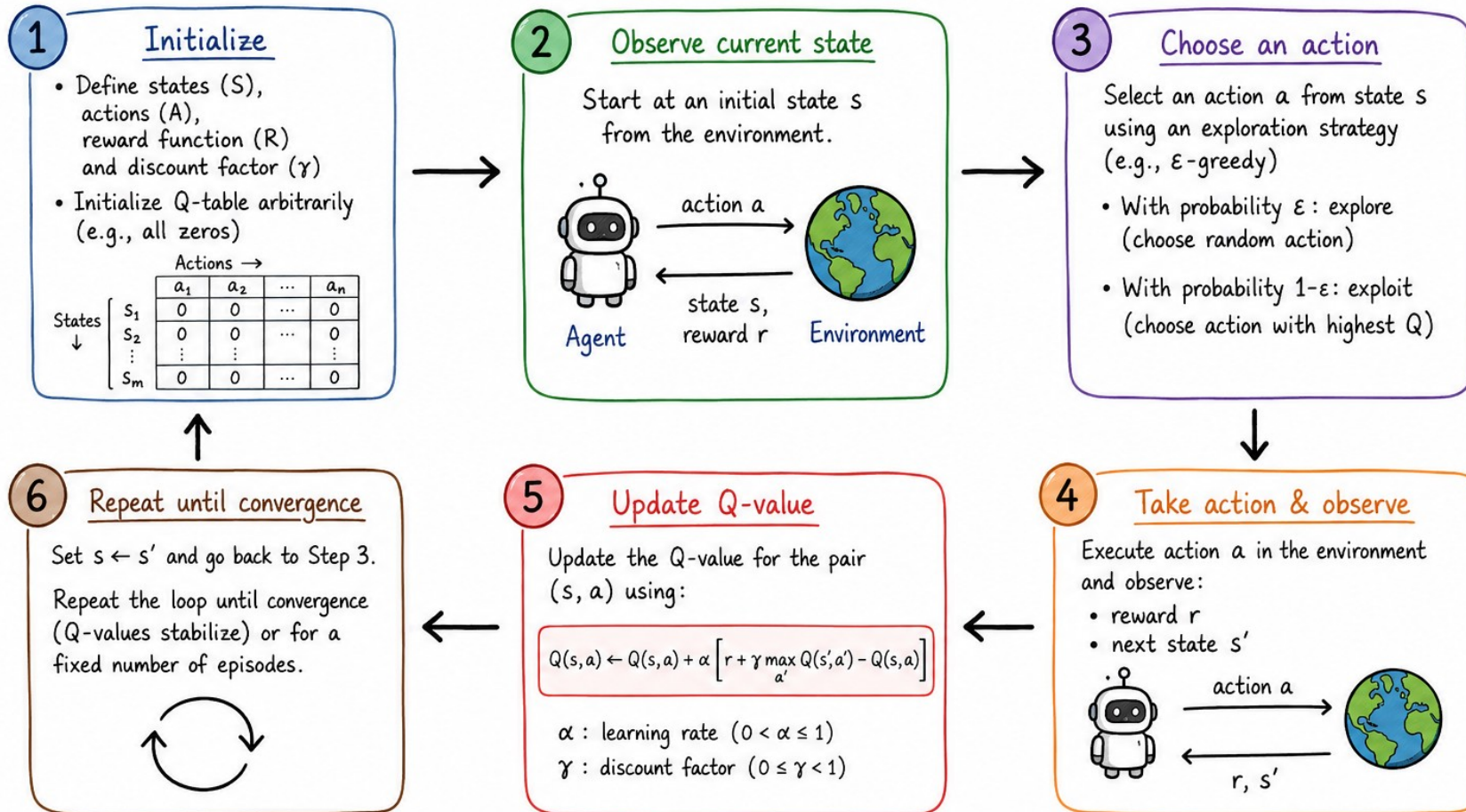
3.3 – Reinforcement Learning

- Learns through interaction with an environment by trial and error.
- Agent selects actions based on the current state to maximize cumulative reward.
- Environment provides feedback in the form of rewards and next states.
- Balances exploration and exploitation to discover optimal strategies.
- Uses a policy or value function (e.g., Q-values) to improve decision-making over time.
- Applications include robotics, game playing, autonomous driving, recommendation systems, and resource optimization.
- <https://www.youtube.com/shorts/ty7RamY3gZs> (INVERTED PENDULUM)
- <https://www.youtube.com/shorts/gNHdsEINyRg>



3.3 – Reinforcement Learning - Q-learning

Q-LEARNING: STEPS



Goal: Learn the optimal Q-values, so the agent can choose actions that maximize cumulative reward.



3.3 – Reinforcement Learning - Q-learning

Q-Learning: Learning from Experience



Q-Table
(Agent's Memory)

State	↑	↓	←	→
(0,0)	2.4	-0.5	-1.2	5.8
(1,1)	3.1	7.2	-2.0	4.5
(2,3)	6.3	8.1	9.5	-3.1
⋮	⋮	⋮	⋮	⋮

● High value (good action)
● Low value (bad action)

Bellman Update

$$Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

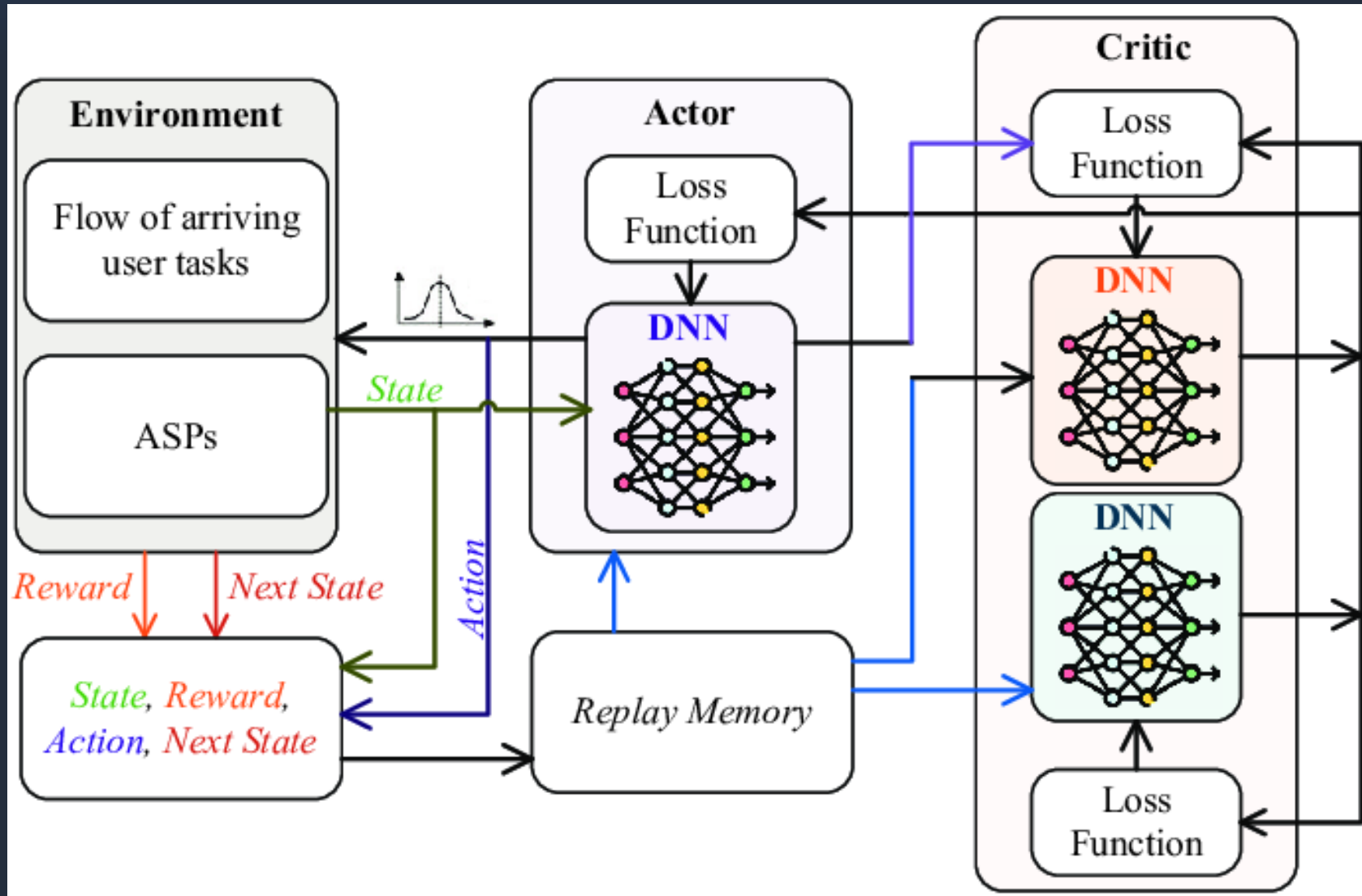
● α = Learning rate (0.1)
● r = Immediate reward
● γ = Discount factor (0.95)
● $\max Q(s',a')$ = Best future value
Updates Q-Table

Epsilon-Greedy: ϵ = 10% explore (random)

90% exploit (best Q-value)

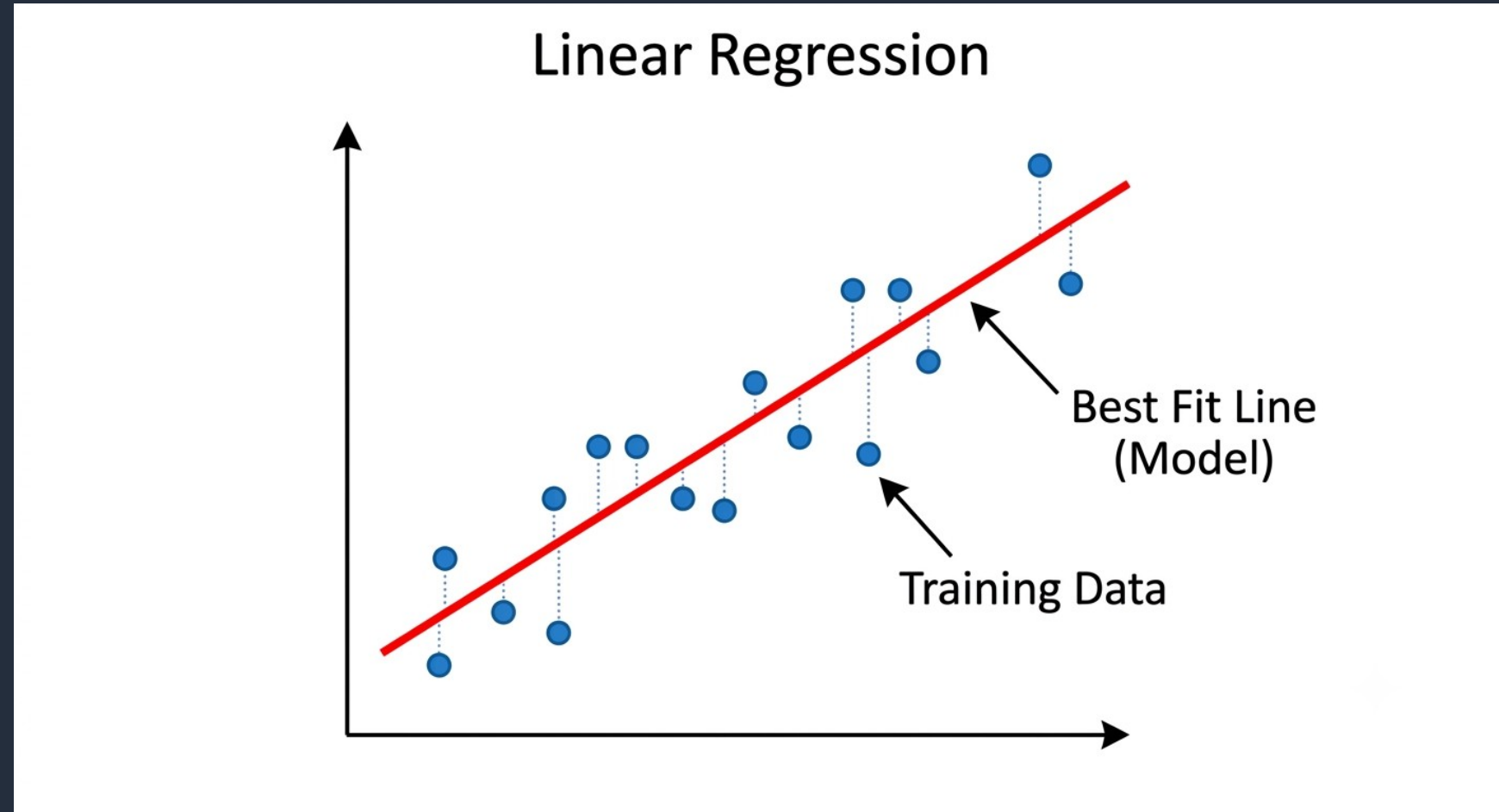


3.3 – Reinforcement Learning – Soft-Actor-Critic (SAC)



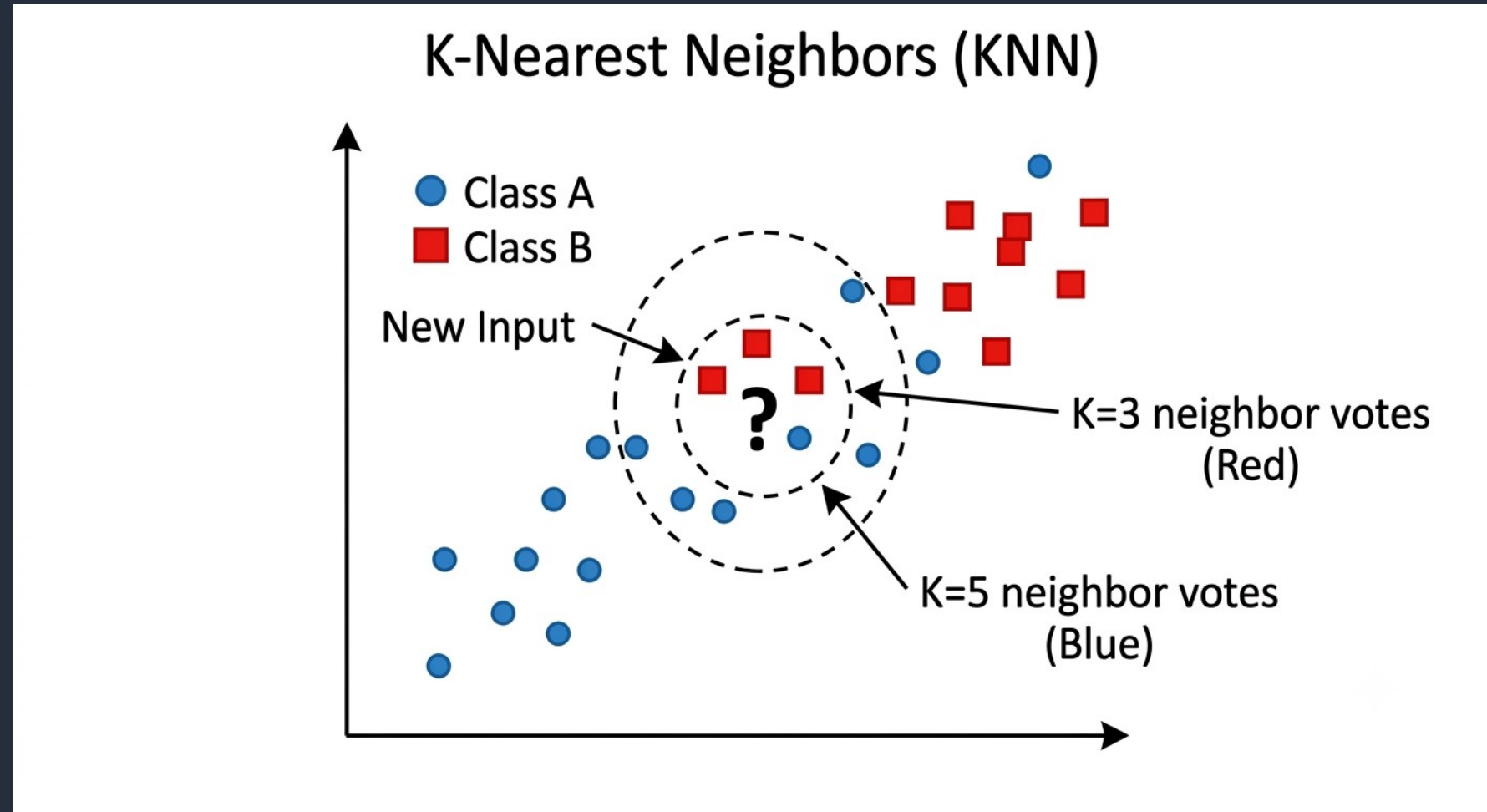
3.4 – Classical ML Techniques - Regression

- Curve Fitting
- Minimize the Errors (Residuals, dotted lines)



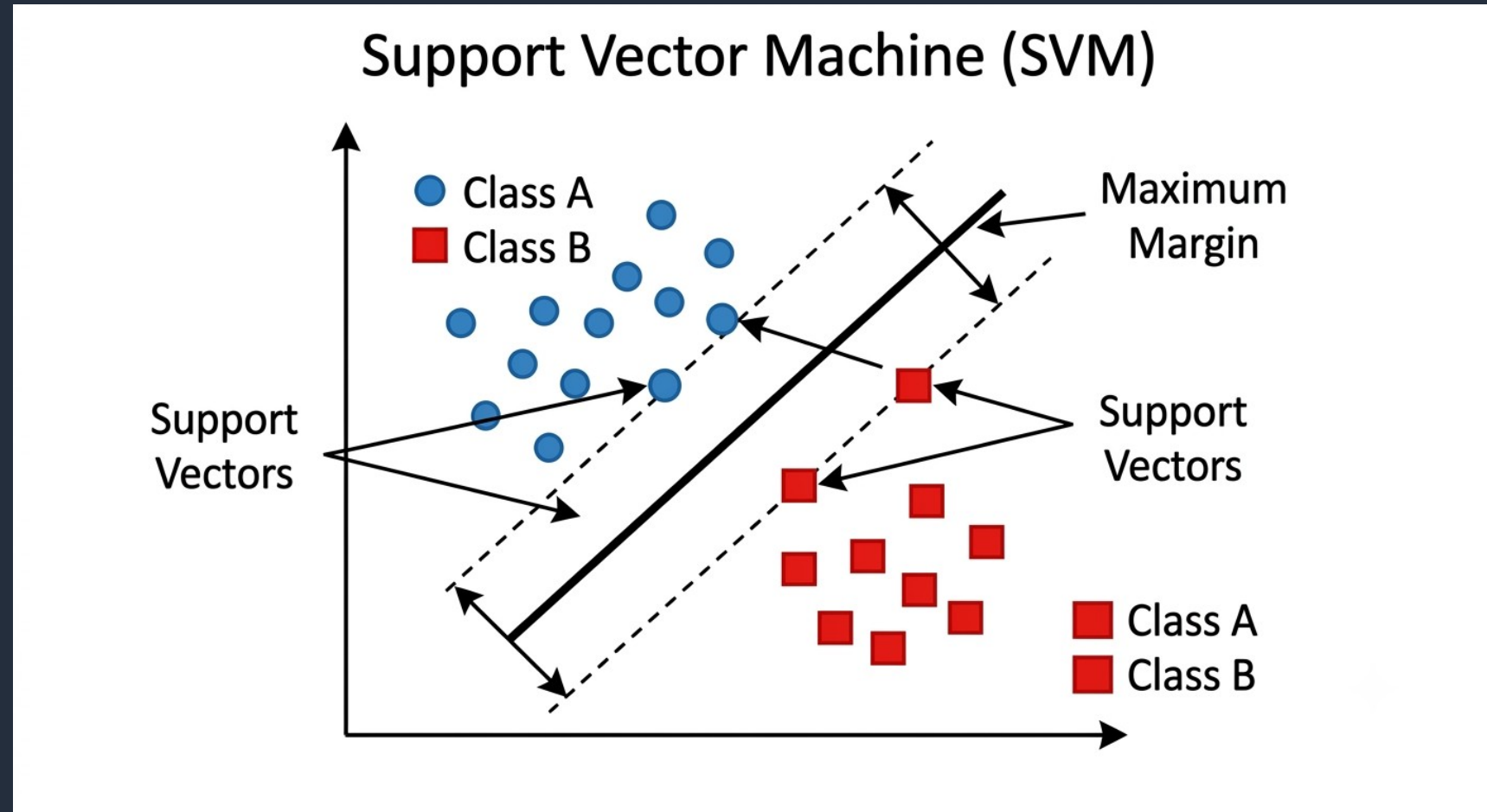
3.4 – Classical ML Techniques - KNN

- The inner dashed circle shows classification for $K=3$ (the three nearest neighbors): It finds 2 red and 1 blue, classifying the point as Red
- The outer dashed circle shows $K=5$: It finds 2 red and 3 blue, classifying the point as Blue. This highlights how sensitive KNN is to the choice of K



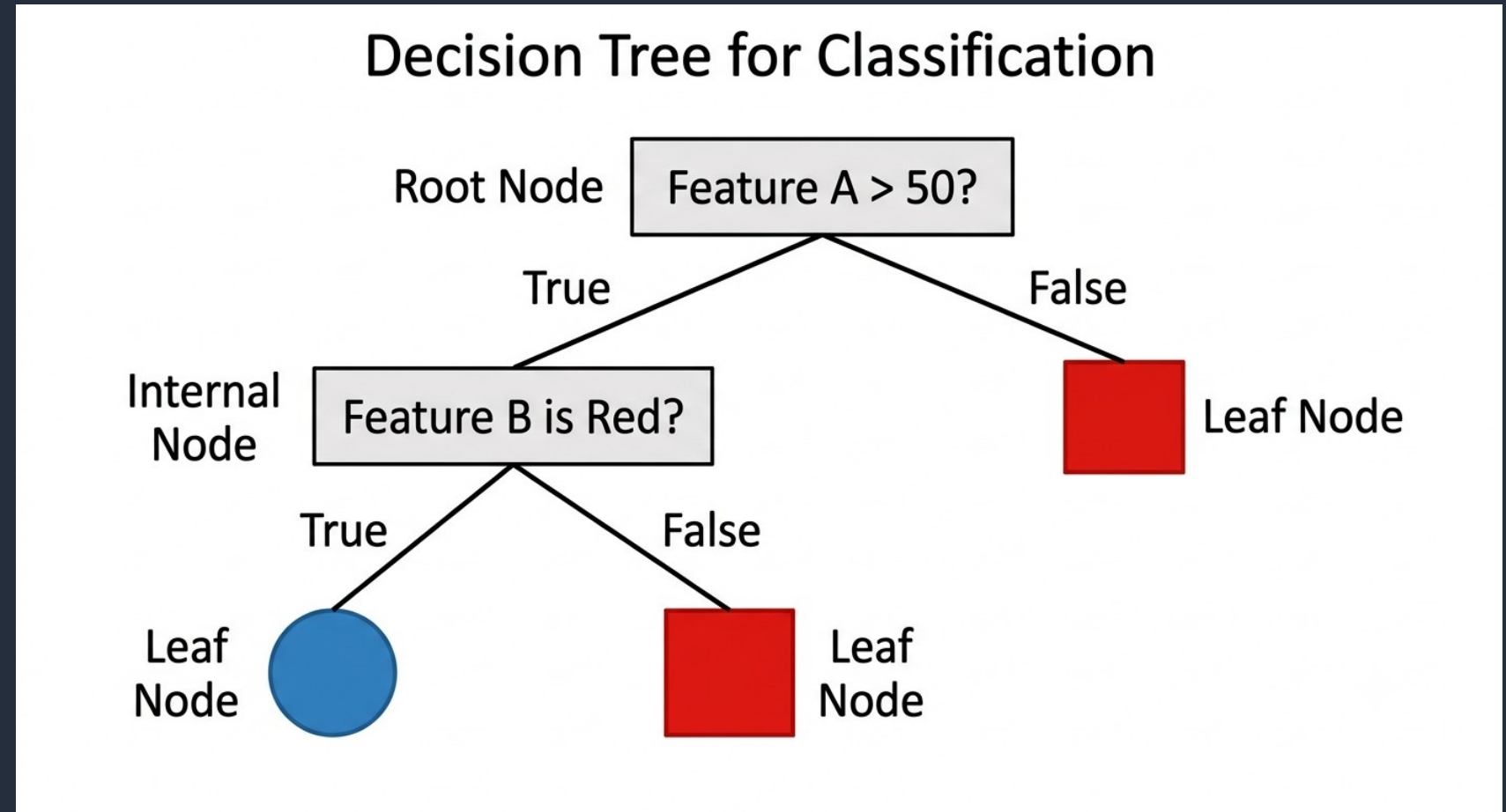
3.4 – Classical ML Techniques - SVM

- Two groups of points (blue vs. red) are separated by a solid black line (the decision boundary, or hyperplane).
- Two dashed parallel lines create a boundary 'margin.' The points that touch these dashed lines (labeled 'Support Vectors') are the critical data points that define the margin.
- SVM works by maximizing the width of this margin, providing the most robust separation.



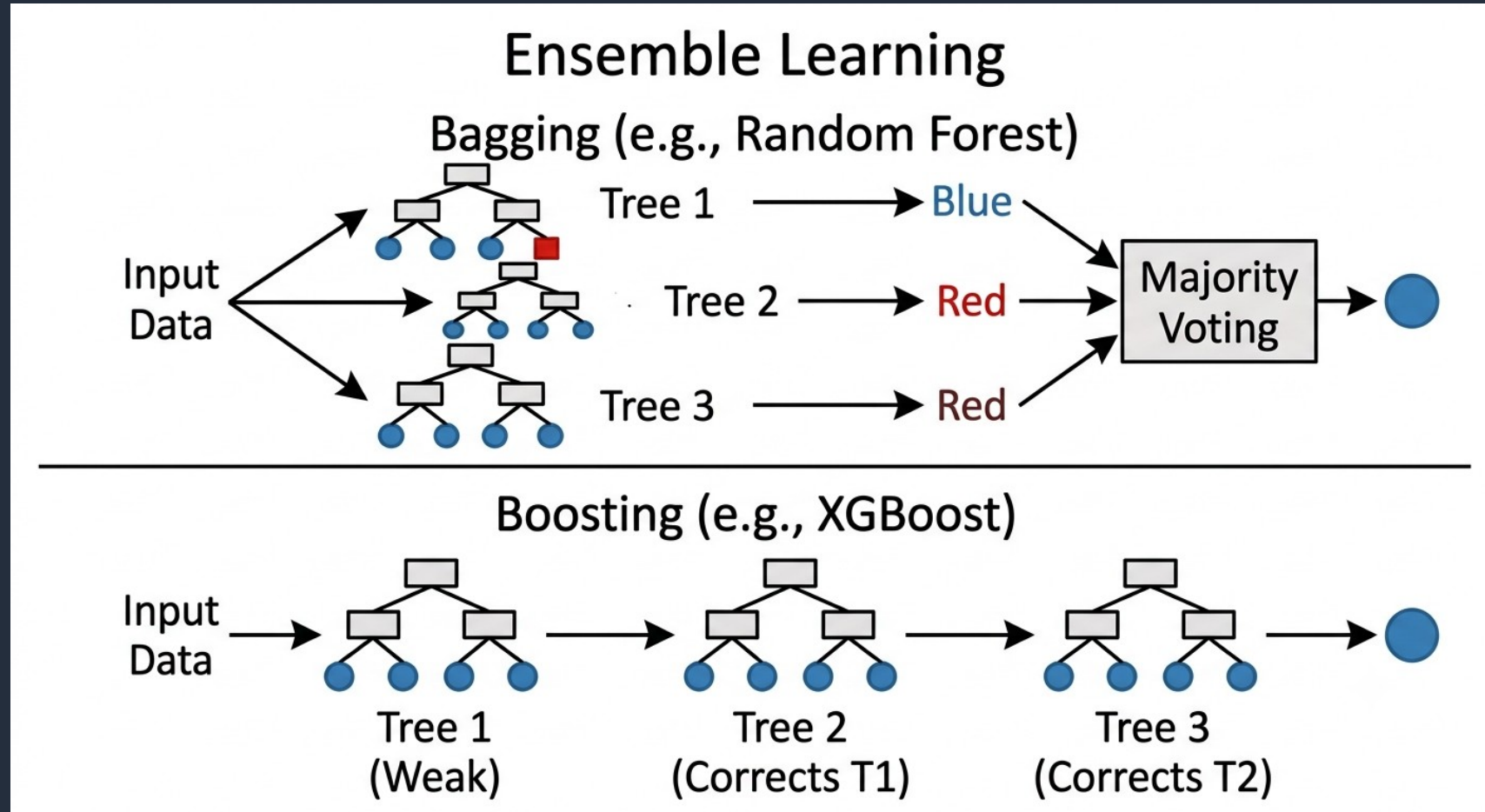
3.4 – Classical ML Techniques - Decision Trees

- The tree starts at a 'Root Node' making the first logical split (e.g., 'Feature A > 50?').
- If True, it follows the left branch; if False, the right.
- The data flows through internal 'nodes' (decisions) until it reaches a 'Leaf Node,' which provides the final classification (Red/Blue)



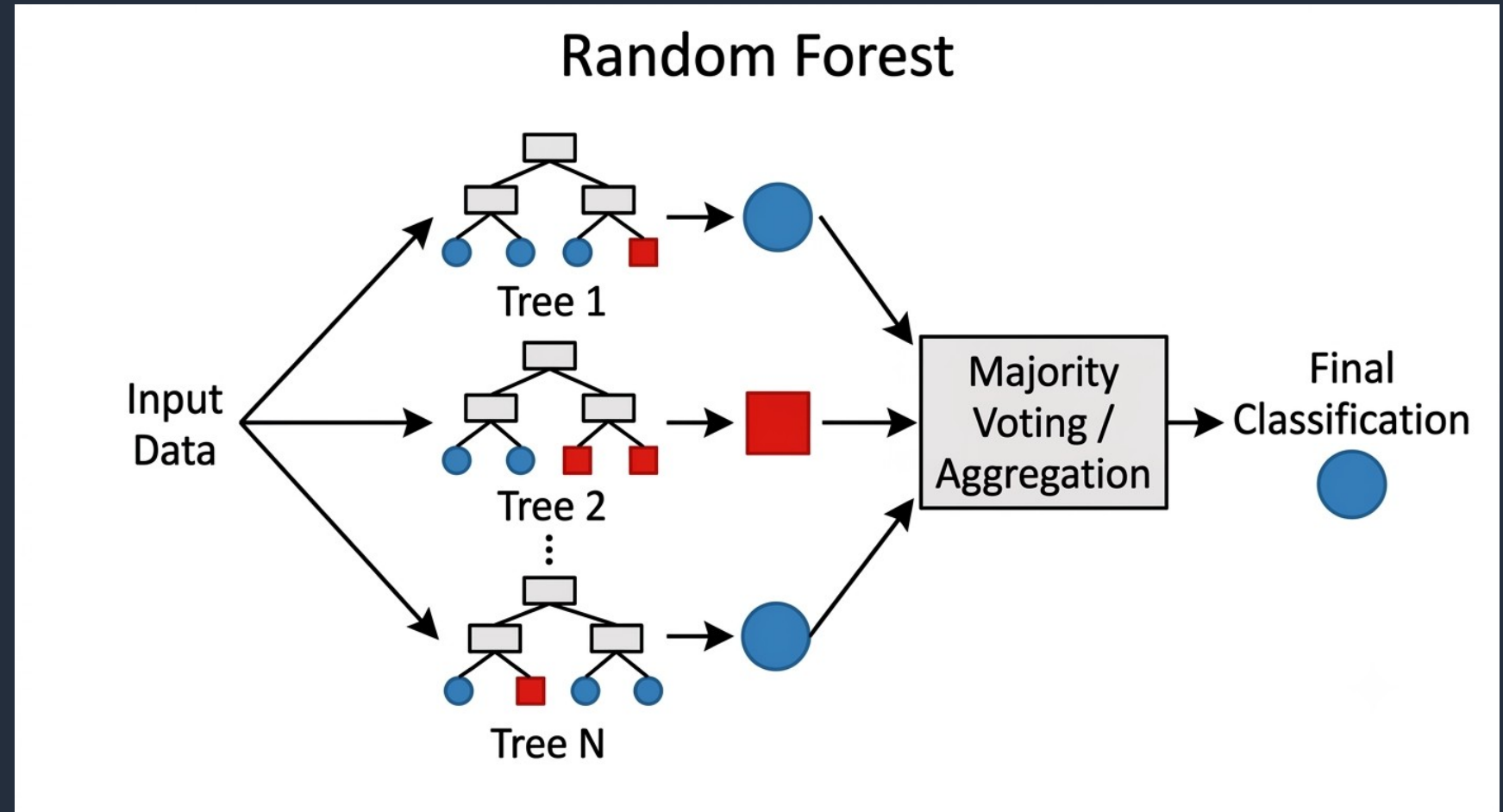
3.4 – Classical ML Techniques – Ensemble Learning

- Bagging (Top): Multiple models (e.g., small trees) run in parallel (Tree 1, 2, 3). They are independent, and their final predictions are combined via 'Majority Voting'.
- Boosting (Bottom): Models run sequentially (Tree 1 -> 2 -> 3). Tree 2 is trained to correct the errors of Tree 1, and Tree 3 corrects the combined errors of 1 and 2.



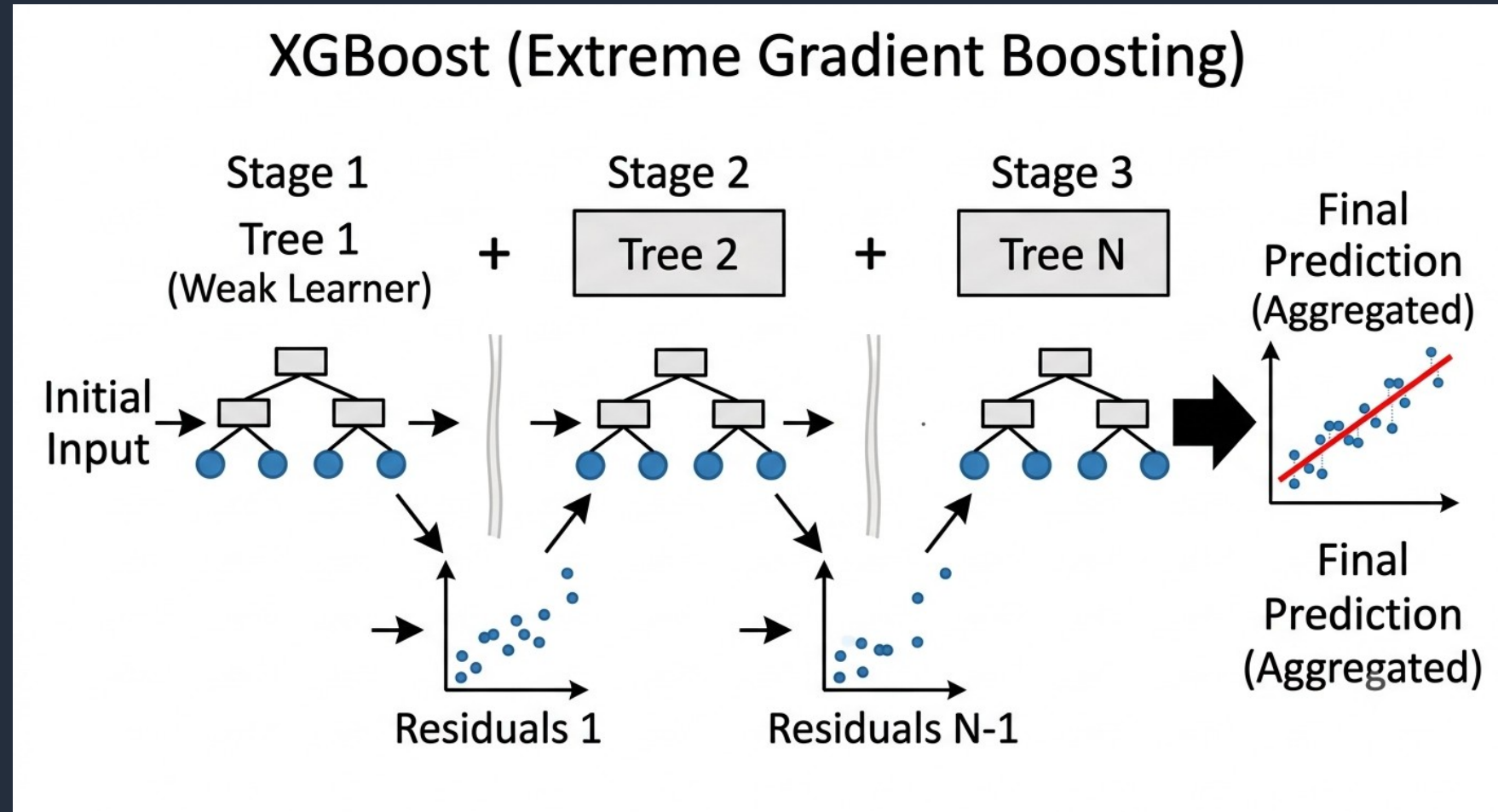
3.4 – Classical ML Techniques – Ensemble Learning - Random Forest

- Multiple small decision trees ('Tree 1' through 'Tree N') are trained independently on random subsets of the data.
- When new input data is received, each tree makes a prediction (e.g., Tree 1 predicts Blue, Tree 2 predicts Red, Tree N predicts Blue).
- These predictions feed into a central 'Majority Voting' box. The final model prediction is the result of the vote (in this case, Blue).



3.4 – Classical ML Techniques – Ensemble Learning - XGBoost

- Tree 1 is trained on the data and makes predictions. The 'Residuals 1' scatter plot shows the initial errors.
- Tree 2 is then trained specifically to predict those residuals. The additive nature is shown ('+') as the combination of Tree 1 and Tree 2 reduces the error (fewer residuals in 'Residuals 2').
- This process repeats for N trees, sequentially minimizing the error until a highly accurate 'Final Prediction' is achieved.

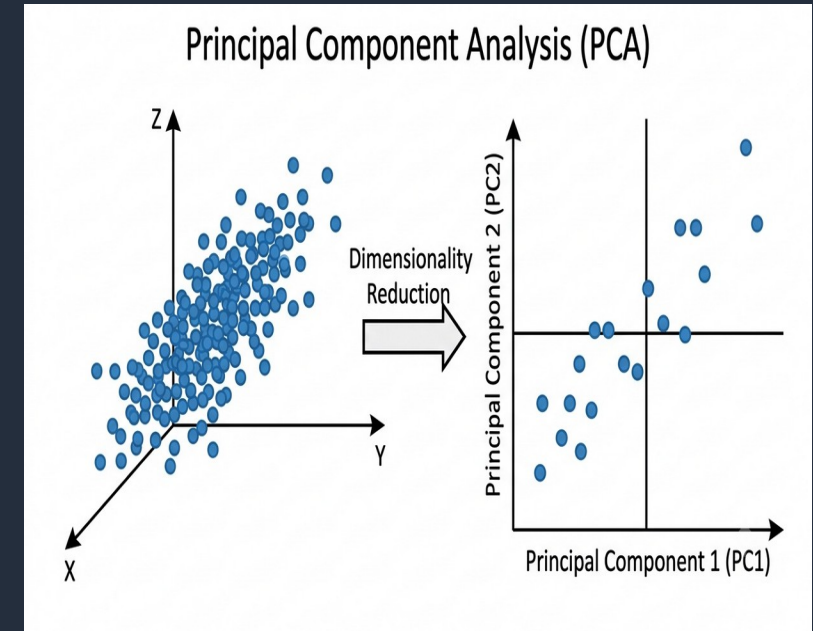


3.4 – Classical ML Techniques – Dimensionality Reduction - PCA

- Suppose that we have multi-dimensional X (input).
- We want to eliminate some of them (columns)
- We use Singular Value Decomposition linear algebra technique to select which columns (features, $X[i]$) represents best the data.

X (input)			Y (labels)	
X1	X2	X3	Y1	Y2
1	2	3	4	5
11	12	13	14	15
21	22	23	24	25
31	32	33	34	35
41	42	43	44	45

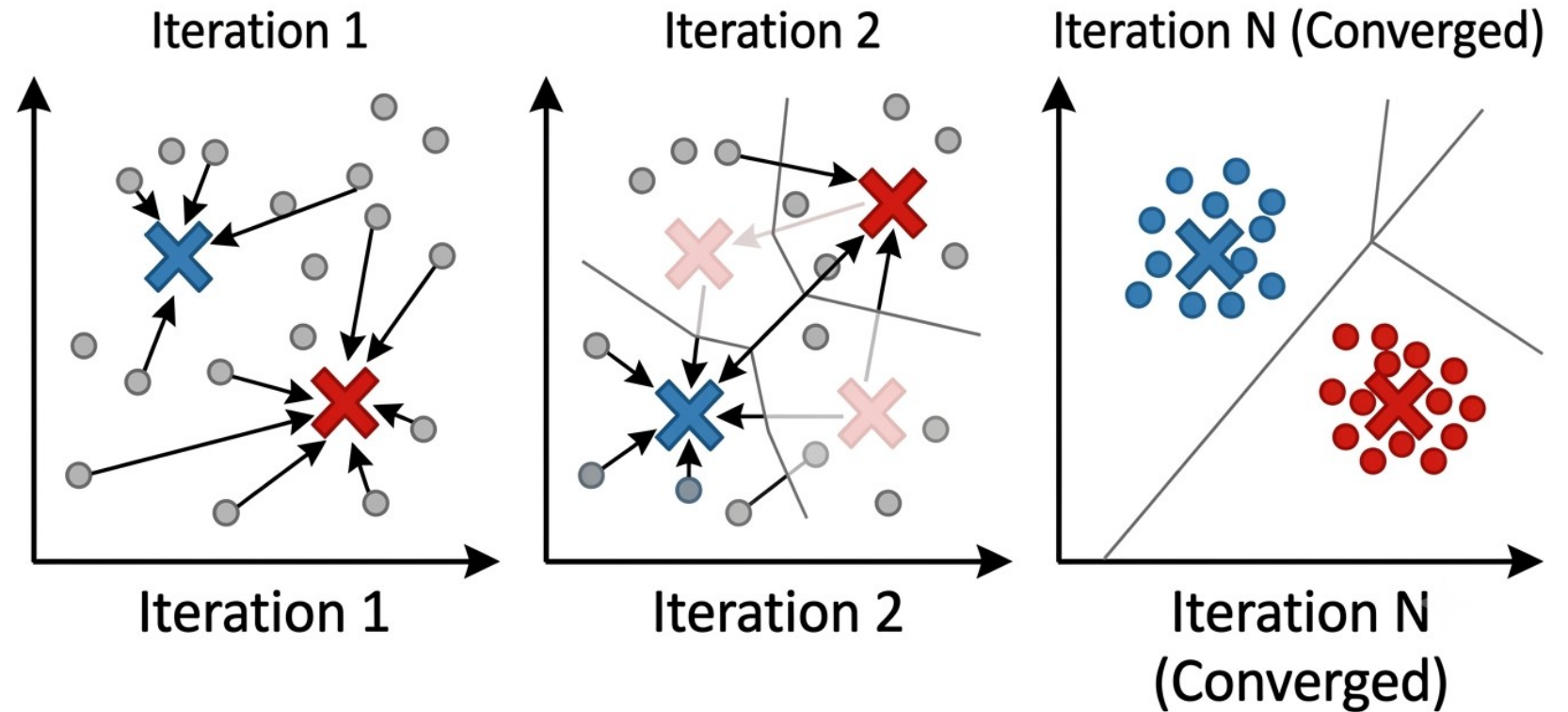
Select one or more columns that represent best the data.



3.4 – Classical ML Techniques – Clustering - KMeans

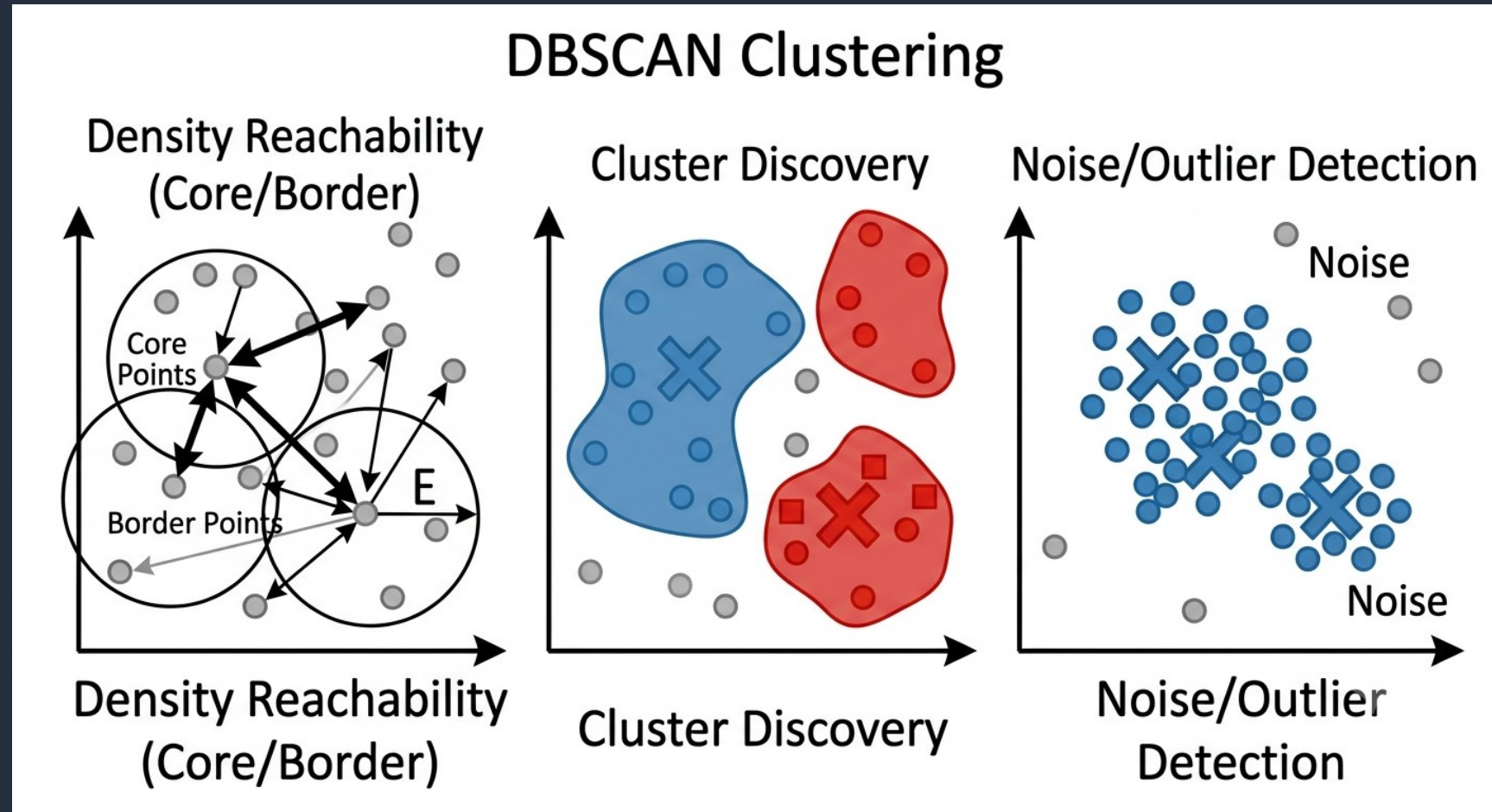
- Iteration 1: $K=2$, Two centroids (X) are randomly placed among unlabeled (grey) data. Points are assigned to the closest X, forming initial, messy clusters (faint red/blue boundaries).
- Iteration 2: The centroids move to the center of their assigned points. The points are re-assigned, and the clusters start to define themselves.
- Iteration N (Converged): The centroids have stopped moving. The points are clearly separated into stable Red and Blue clusters (re-using the established class colors from Image 1), with optimized Voronoi-like boundaries.

K-Means Clustering



3.4 – Classical ML Techniques – Clustering - KMeans

- DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is an unsupervised clustering algorithm that finds clusters based on the density of data points. It is excellent at finding arbitrary shapes and identifying noise (outliers)..



4 - Neural Networks Intuition

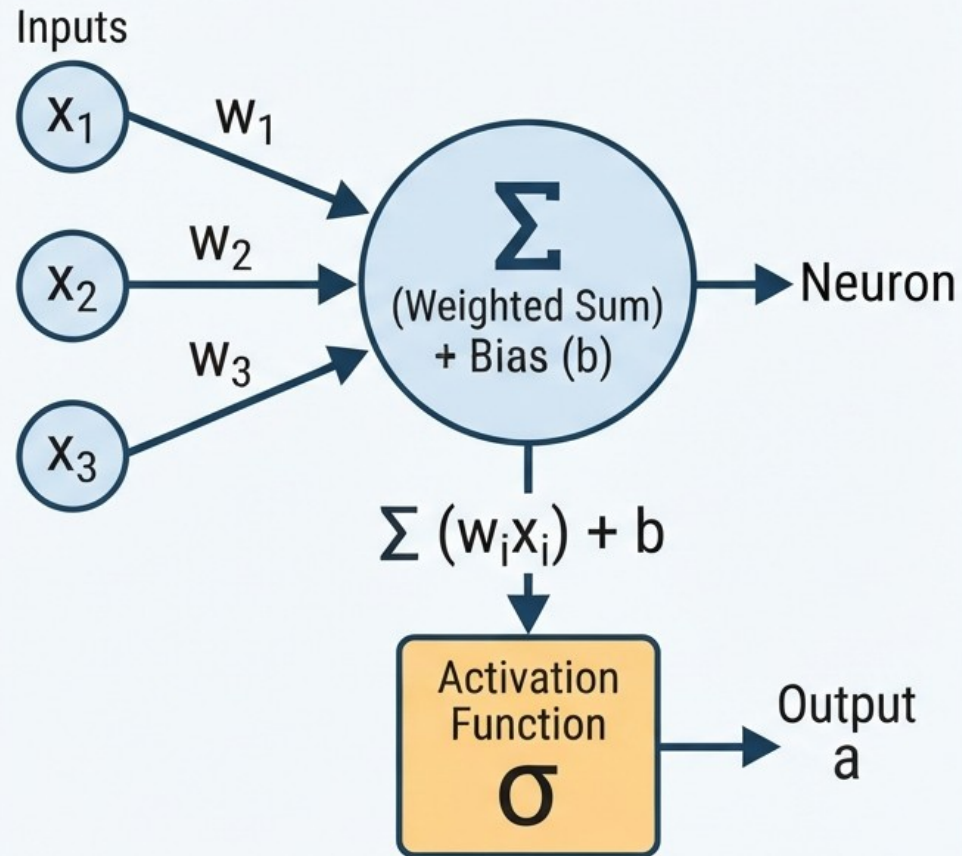
1. *Neurons, layers, activations (ReLU, softmax conceptually)*
2. *Why deep learning works better for perception/language*
3. *Classical DL Techniques : CNN, RNN, LSTM, GNN, RBM, VAE, GAN, KAN, Transformers, Diffusion*
4. *Bias/Variance problem, Regularization and smooth learning*
5. *Preventing Overfitting (Memorization) : Early Stop, Dropout, Weight Regularization, Data Augmentation (**Very important), Simplify Model, Batch Normalization*
6. *Data Augmentation*
7. *Preventing Vanishing Gradient : Residuals (Skip Connections), Use RELU/LeakyRELU Activation Fn., Batch Normalization*
8. *Cross Entropy, Learning Rate Momentum*
9. *Transfer Learning*
10. *Problem Solving Using Neural Networks*



4.1 – Neurons, Layers, Activations

NEURAL NETWORK NEURON: STRUCTURE & MATHEMATICAL FOUNDATION

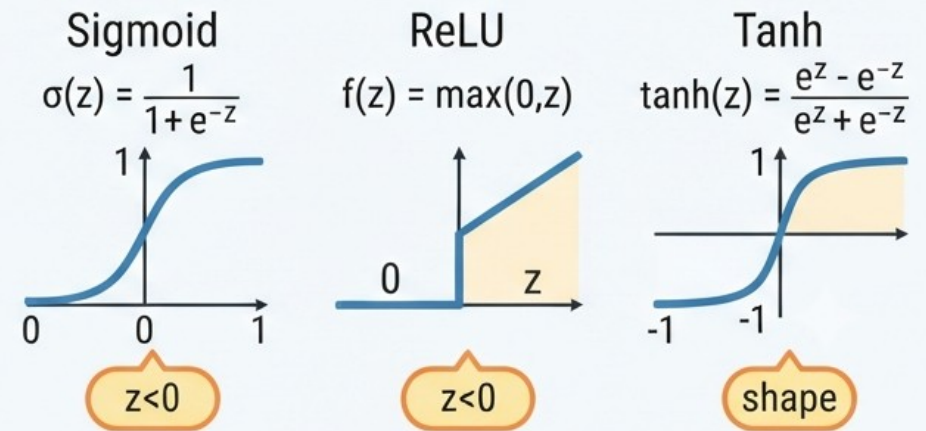
1. Neuron Structure & Function



2. Matrix Multiplication Representation

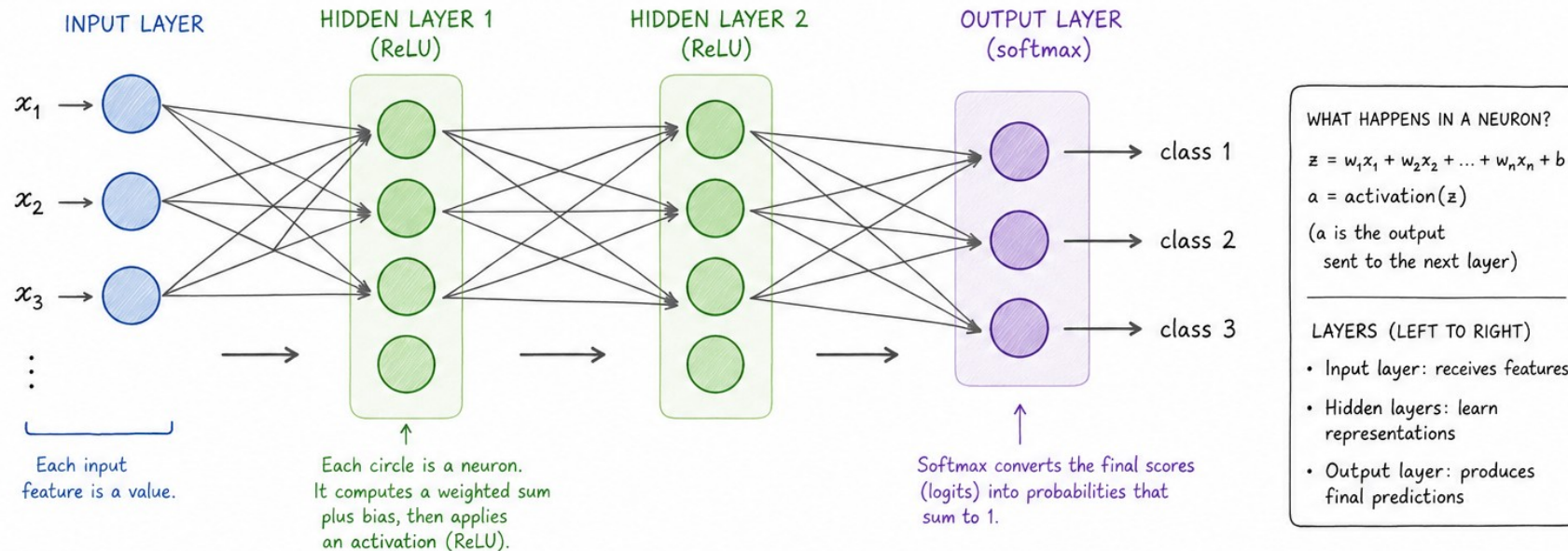
$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \times \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}^T + (b) = z$$
$$z = \mathbf{w}^T \mathbf{x} + b$$

3. Activation Functions



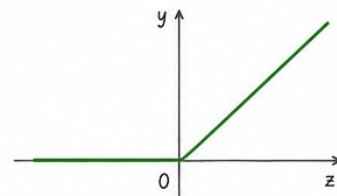
4.1 – Neurons, Layers, Activations

Neurons, layers, activations (ReLU, softmax conceptually)



ACTIVATIONS (CONCEPTUALLY)

ReLU (Rectified Linear Unit)



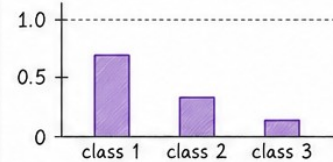
$$\text{ReLU}(z) = \max(0, z)$$

- Output is 0 if $z < 0$
- Output is z if $z > 0$
- Introduces non-linearity
- Helps the model learn complex patterns

SOFTMAX (for multi-class outputs)

Logits (raw scores)

$$z = [1.2 \quad 0.1 \quad -0.7]$$



Probabilities (sum to 1)

$$p = [0.70 \quad 0.25 \quad 0.05]$$

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

- Converts any real-valued scores into probabilities
- All outputs are between 0 and 1
- All probabilities sum to 1

Input features → Learned representations (via ReLU layers) → Class probabilities (via softmax)



4.1 – Neurons, Layers, Activations

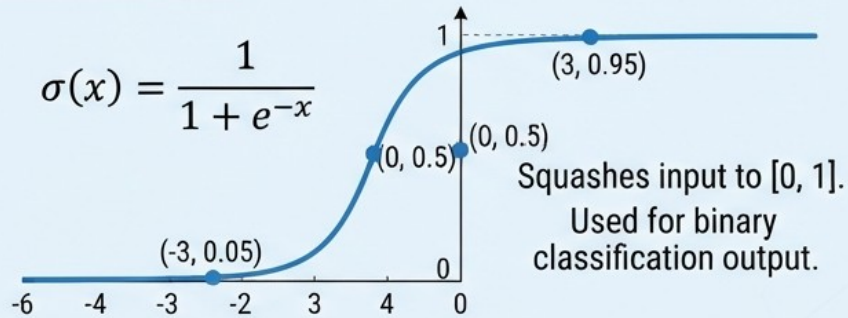
TANH USED IN
SIGNAL
GENERATOR
NETWORKS

ACTIVATION FUNCTIONS IN DEEP LEARNING

SIGMOID

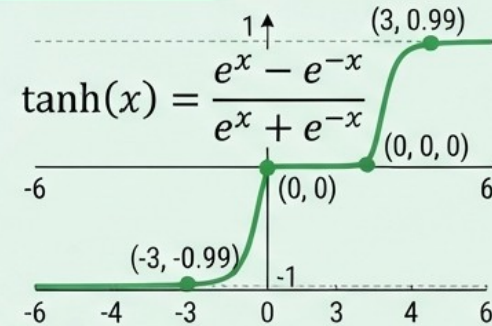
Sigmoid (Logistic) Function

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$



TANH

Hyperbolic Tangent (tanh) Function



RELU

Rectified Linear Unit (ReLU)

$$f(x) = \max(0, x)$$

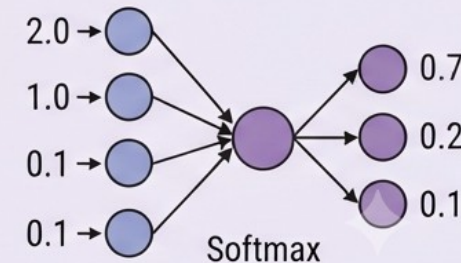


SOFTMAX

Softmax Function

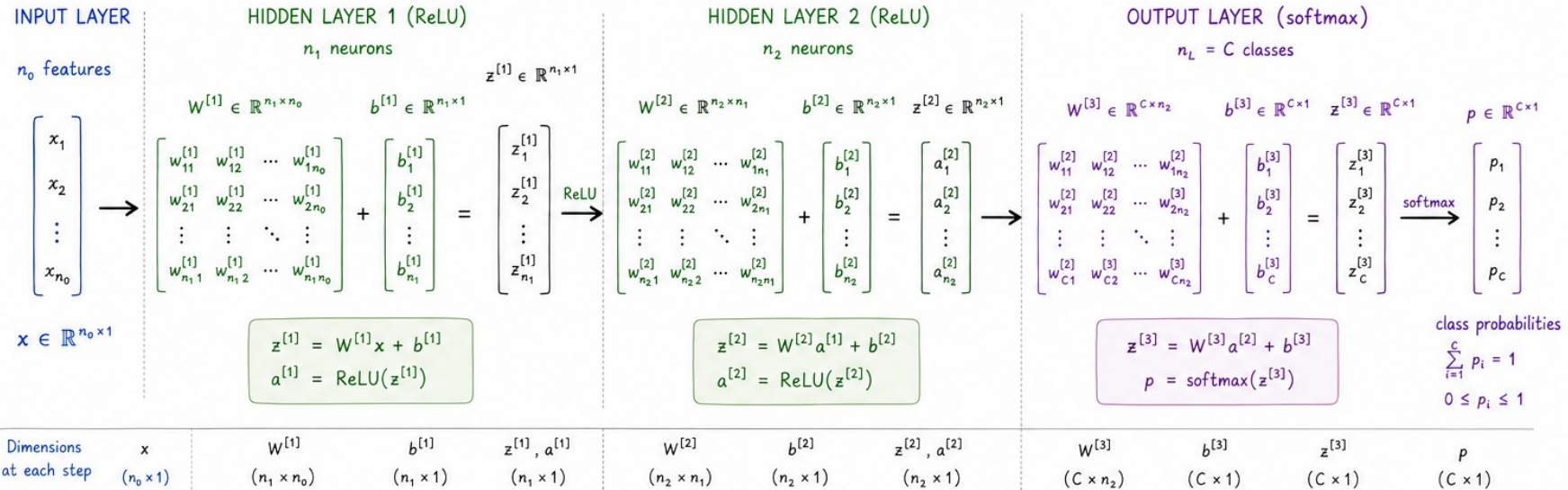
$$\sigma(z_i) = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

Converts raw scores to probabilities for multi-class classification.

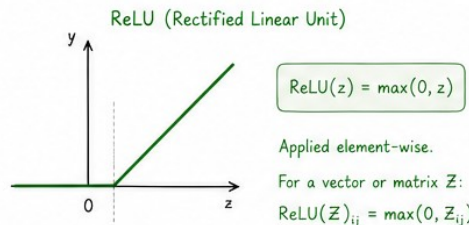


4.1 – Neurons, Layers, Activations

Neurons, layers, activations (ReLU, softmax conceptually) with matrix operations



ACTIVATIONS



Example

$$Z = \begin{bmatrix} -1.2 & 0 & 2.5 \\ 3.1 & -0.7 & 0 \\ -2.0 & 4.2 & -0.1 \end{bmatrix}$$

$$\text{ReLU}(Z) = \begin{bmatrix} 0 & 0 & 2.5 \\ 3.1 & 0 & 0 \\ 0 & 4.2 & 0 \end{bmatrix}$$

Softmax (for multi-class outputs)

Given logits $z \in \mathbb{R}^C$ (a vector)

$$z = \begin{bmatrix} 1.2 \\ 0.1 \\ -0.7 \end{bmatrix}$$

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}}$$

i	z_i	e^{z_i}
1	1.2	3.3201
2	0.1	1.1052
3	-0.7	0.4966

$\sum e^{z_j} = 4.9219$

$p = \text{softmax}(z)$
(probabilities)

$$p = \begin{bmatrix} 0.675 \\ 0.224 \\ 0.101 \end{bmatrix}$$

$\sum_i p_i = 1$

- Converts raw scores (logits) to probabilities
- All outputs in (0, 1)
- Sum of all probabilities equals 1

Forward pass (compact view)

$$x^{[0]} = x \rightarrow \begin{cases} z^{[l]} = W^{[l]}x^{[l-1]} + b^{[l]} \\ a^{[l]} = \text{ReLU}(z^{[l]}) \end{cases} \quad (l = 1, \dots, L-1) \rightarrow \begin{cases} z^{[L]} = W^{[L]}x^{[L-1]} + b^{[L]} \\ p = \text{softmax}(z^{[L]}) \end{cases}$$

Where:

- $W^{[l]}$: weights
- $b^{[l]}$: biases
- $x^{[l]}$: activations from layer l
- L : index of the output layer

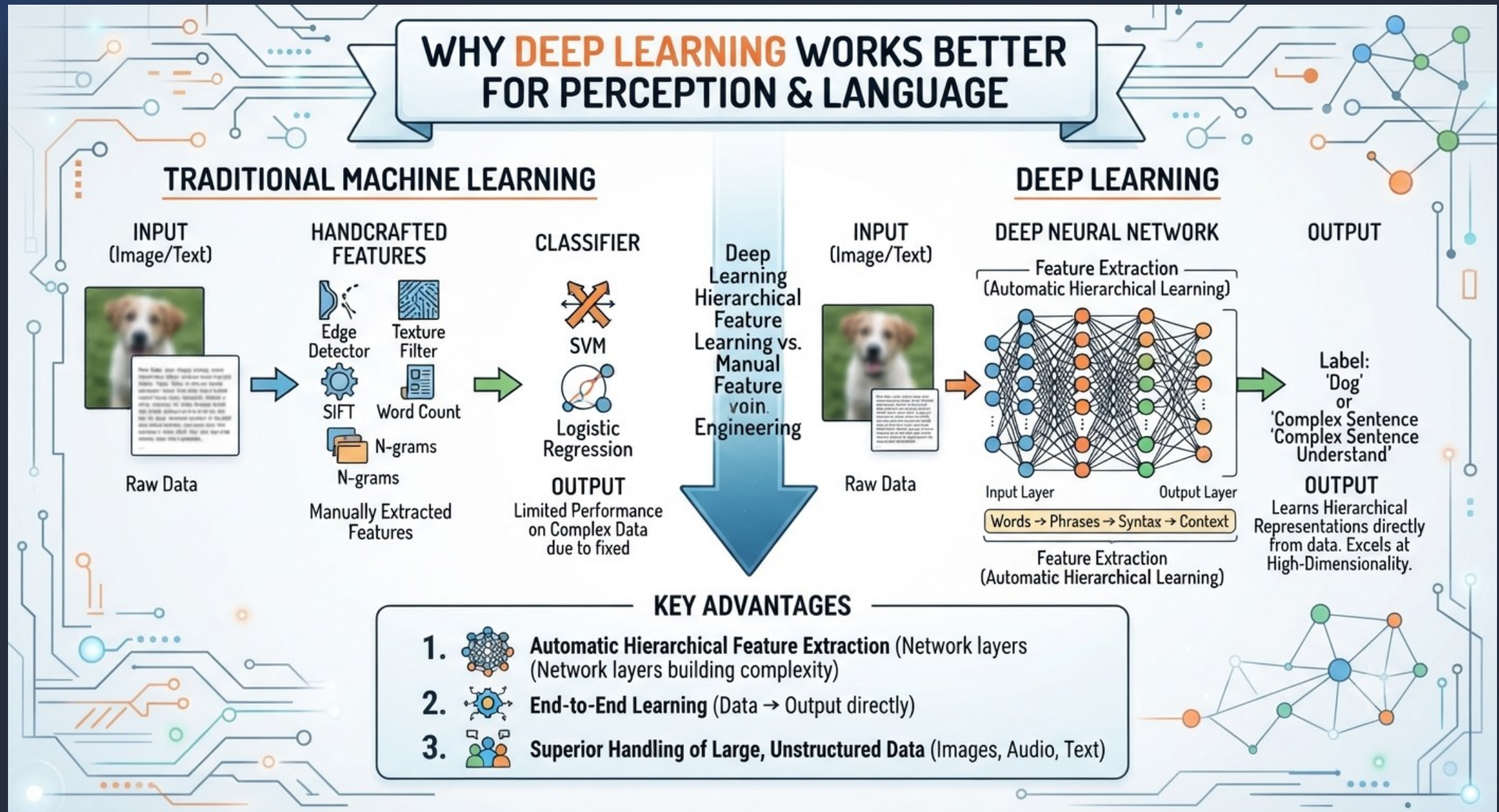
Example sizes

$n_0 = 4$ (input features)
 $n_1 = 5$ (hidden layer 1)
 $n_2 = 3$ (hidden layer 2)
 $C = 3$ (classes)

$W^{[1]} \in \mathbb{R}^{5 \times 4}$
 $W^{[2]} \in \mathbb{R}^{3 \times 5}$
 $W^{[3]} \in \mathbb{R}^{3 \times 3}$



4.2 – Why Deep Learning Works Better For Perception/Language



4.3 – Classical DL Techniques

1. *Fully Connected*
2. *CNN*
3. *RNN*
4. *LSTM*
5. *GCN*
6. *RBM*
7. *VAE*
8. *GAN*
9. *KAN*
10. *Transformers*
11. *Diffusion*



4.3.1 – Classical DL Techniques – Fully Connected

UNDERSTANDING A FULLY CONNECTED NEURAL NETWORK: ARCHITECTURE, PROPERTIES, & MATRIX OPERATIONS

PROPERTIES

1 ALL-TO-ALL CONNECTIONS

Every neuron in one layer connects to every neuron in the next.

2 LAYER TYPES

- Input
- Hidden (features)
- Output (predictions)

3. PARAMETERS

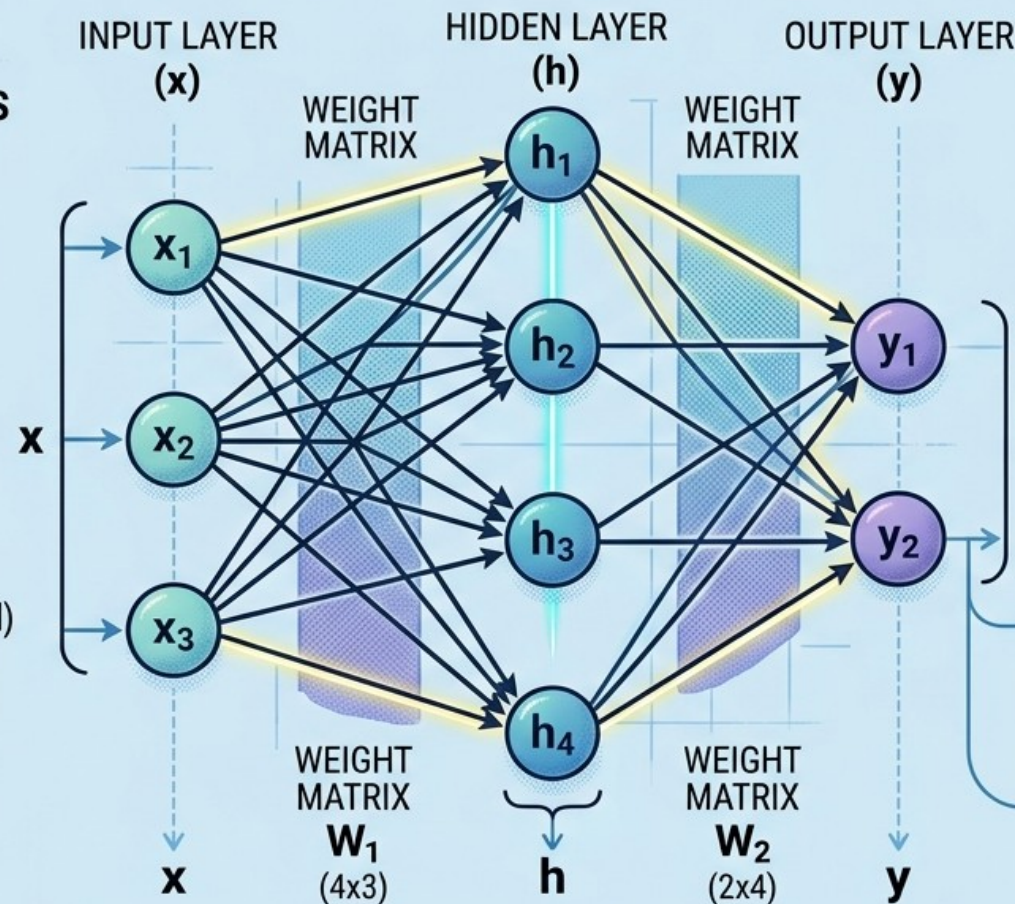
- Weights (strength of connection)
- Biases (activation threshold)

4. NO SPATIAL STRUCTURE

Does not preserve input spatial relationships.

5. DENSE CONNECTIONS

Computationally intensive for large inputs.



MATRIX OPERATIONS

(FORWARD PASS)

$$h = f_1(x \cdot W_1^T + b_1)$$

$$y = f_2(h \cdot W_2^T + b_2)$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \times \begin{bmatrix} x_1 & x_2 & x_3 \\ \text{ } & \text{ } & \text{ } \\ \text{ } & \text{ } & \text{ } \end{bmatrix} + \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix} = \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \\ b_{14} \end{bmatrix}$$

(1x3) (3x4) (1x4) (1x4)

$$y = f_2(h \cdot W_2^T + b_2)$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \times \begin{bmatrix} x_1 & x_2 & x_3 \\ \text{ } & \text{ } & \text{ } \\ \text{ } & \text{ } & \text{ } \end{bmatrix} + \begin{bmatrix} b_{11} \\ b_{12} \\ b_{14} \end{bmatrix} = \begin{bmatrix} \text{ } & \text{ } \\ \text{ } & \text{ } \\ \text{ } & \text{ } \end{bmatrix}$$

(1x3) (3x4) (1x4) (1x4)

ACTIVATION FUNCTION

$$g(z) = f(z) \quad \begin{cases} \text{ReLU or} \\ \text{Sigmoid} \end{cases}$$



4.3.2 – Classical DL Techniques – CNN

UNDERSTANDING A CONVOLUTIONAL NEURAL NETWORK (CNN): ARCHITECTURE, PROPERTIES, MATRIX OPERATIONS, & APPLICATIONS.

CNN PROPERTIES & CONCEPTS

1 RECEPTIVE FIELD & LOCAL CONNECTIVITY

 Visual filter over on input (autot to obtain innit).

2 WEIGHT SHARING

- Filters with shared
- Filters with d#shared
- Filters shared parameters

3. TRANSLATION INVARIANCE

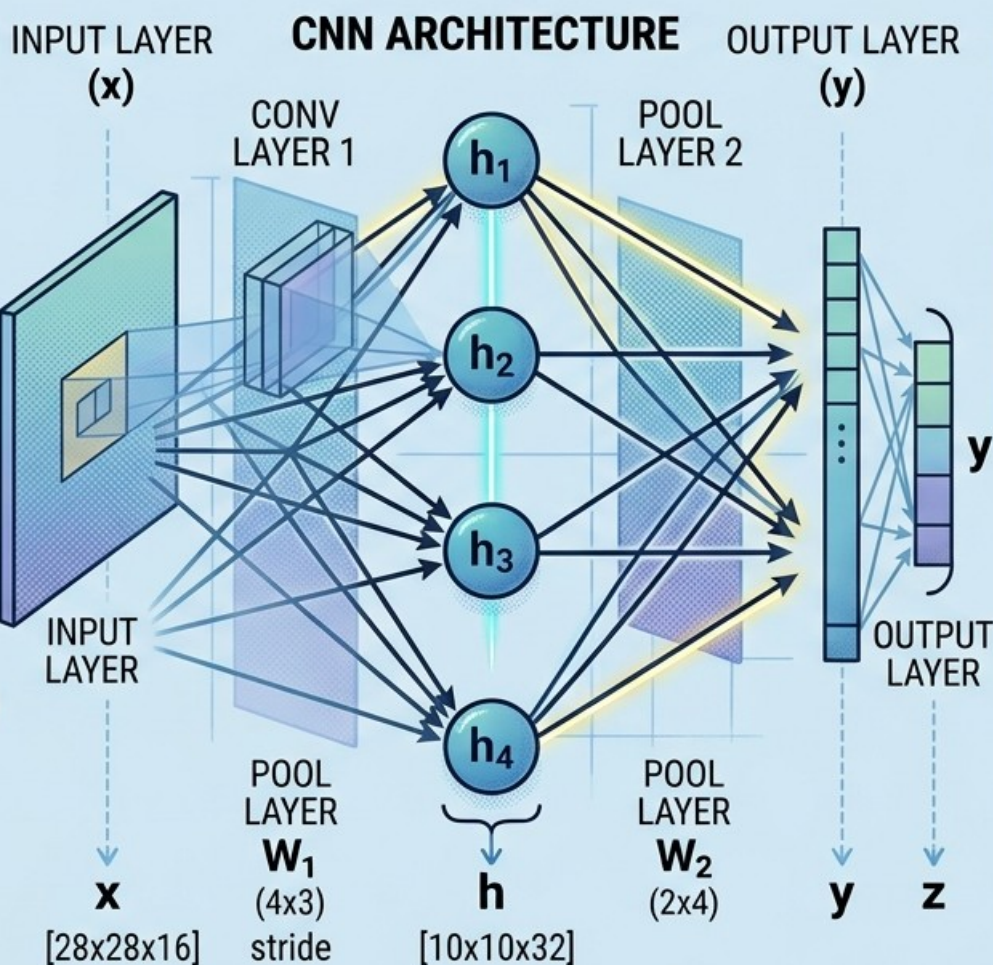
- Weights (strength of connection)
- Biases (activation threshold)

4. DIMENSIONALITY REDUCTION

 Does not preserve input (Pooling/Stride)

5. NON-LINEARITY

 Computationalasive (ReLU, etc.)



MATRIX OPERATIONS (FOR CONVOLUTION)

CONVOLUTION $\rightarrow$ FEATURE MAP

$$y_{ij} = f\left(\sum_{sub} x_{sub} \cdot w\right) + b$$

A 3x3 input matrix $\begin{bmatrix} x_1 & x_2 & x_3 \\ x_2 & x_3 & x_3 \\ x_2 & x_3 & x_3 \end{bmatrix}$ is multiplied by a 3x3 weight matrix $\begin{bmatrix} w_1 & w_2 & w_3 \\ w_1 & w_2 & w_3 \\ w_1 & w_2 & w_3 \end{bmatrix}$ to produce a 1x4 output vector $\begin{bmatrix} \Sigma & \Sigma & \Sigma & \Sigma \end{bmatrix}$, resulting in the value 1.

POOLING OPERATIONS

MAX-POOLING $\rightarrow$ $\max$ $\rightarrow$ max value

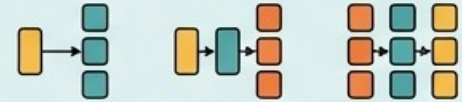
APPLICATION AREAS



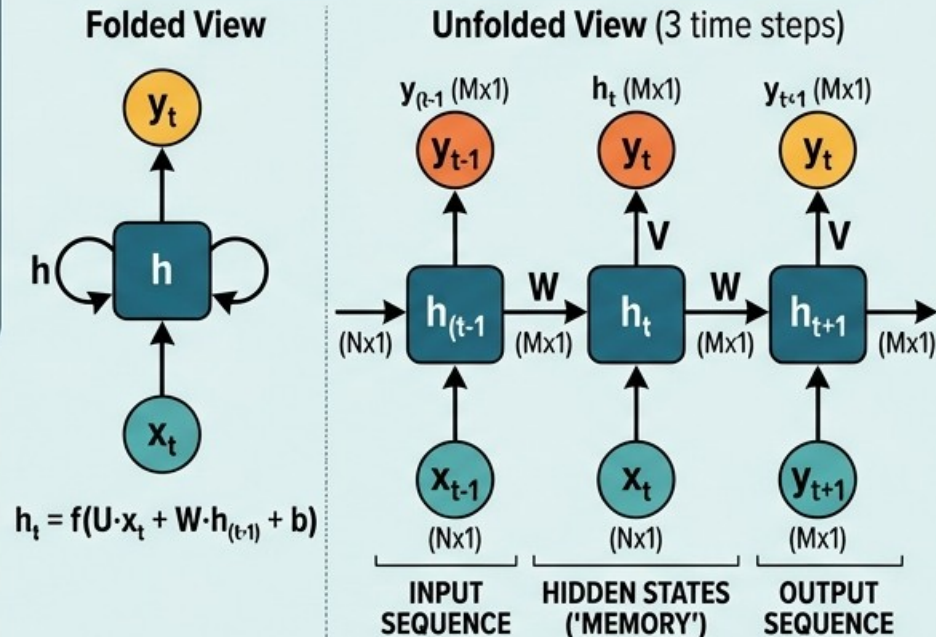
4.3.3 – Classical DL Techniques – RNN

UNDERSTANDING A RECURRENT NEURAL NETWORK (RNN): ARCHITECTURE, PROPERTIES, MATRIX OPERATIONS, & APPLICATIONS

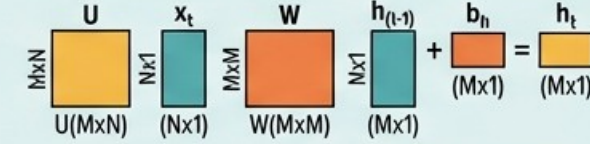
RNN PROPERTIES

- 1 SEQUENTIAL DATA PROCESSING**
Operates on data over time.
- 2 INTERNAL MEMORY**
 h_t stores information from previous steps
- 3 VARIABLE-LENGTH SEQUENCES**
Can handle inputs of varying lengths.
- 4 PARAMETER SHARING**
The same U, V, W, b are used for all steps.
- 5 TYPES (examples)**

One-to-Many Many-to-One Many-to-Many

RNN ARCHITECTURE (FOLDED & UNFOLDED)



MATRIX OPERATIONS (FORWARD PASS)

- 1 Hidden State Update**
 $[h_t] = f([U] \cdot [x_t] + [W] \cdot [h_{(t-1)}] + [b_h])$

- 2 Output Calculation**
 $[y_t] = g([V] \cdot [h_t] + [b_y])$

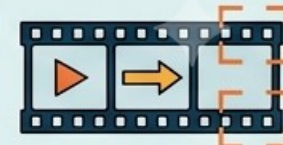

RNN APPLICATION AREAS

- 1 MACHINE TRANSLATION**

- 2 SPEECH RECOGNITION**

- 3 TIME-SERIES FORECASTING**

- 4 SENTIMENT ANALYSIS**

- 5 VIDEO ANALYSIS**




4.3.4 – Classical DL Techniques – LSTM

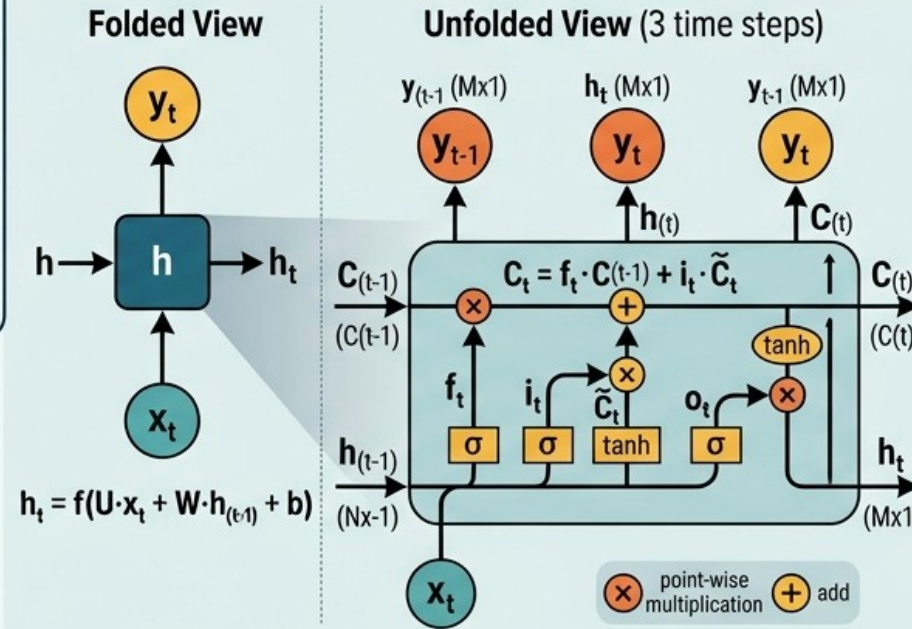
UNDERSTANDING A LONG SHORT-TERM MEMORY (LSTM) NETWORK: ARCHITECTURE, PROPERTIES, MATRIX OPERATIONS, & APPLICATIONS

LSTM PROPERTIES

- 1 SOLVES VANISHING GRADIENTS**
Remembers over longer time
- 2 COMPLEX GATE MECHANISMS**
Forget, input, output gates
- 3 LONG-TERM & SHORT-TERM MEMORY**
Distinct C and H states
- 4 EXPLICIT FORGETTING & REMEMBERING**
gating operations
- 5 SEQUENCE DATA PROCESSING**



LSTM ARCHITECTURE (UNFOLDED & CELL DETAIL)



MATRIX OPERATIONS (FORWARD PASS)

- 1 Forget Gate**
 $[f_t] = \sigma([W_f] \cdot [h_{t-1}, x_t] + [b_f])$
 - 2 Input Gate**
 $[i_t] = \sigma([W_i] \cdot [h_{t-1}, x_t] + [b_i])$
 - 3 Output Gate**
 $[o_t] = \sigma([W_o] \cdot [h_{t-1}, x_t] + [b_o])$
 - 4 Input Candidate**
 $[\tilde{C}_t] = \tanh([W_c] \cdot [h_{t-1}, x_t] + [b_c])$
- 3 POINT-WISE OPERATIONS**
- $C_t = C_{(t-1)} \otimes f_t + i_t \otimes \tilde{C}_t$
- $h_t = o_t \otimes \tanh(C_t)$

LSTM APPLICATION AREAS

- 1 ADVANCED MACHINE TRANSLATION**
- 2 VOICE-ACTIVATED ASSISTANT**
- 3 STOCK MARKET FORECASTING**
- 4 TEXT SUMMARIZATION**
- 5 VIDEO ACTION RECOGNITION**



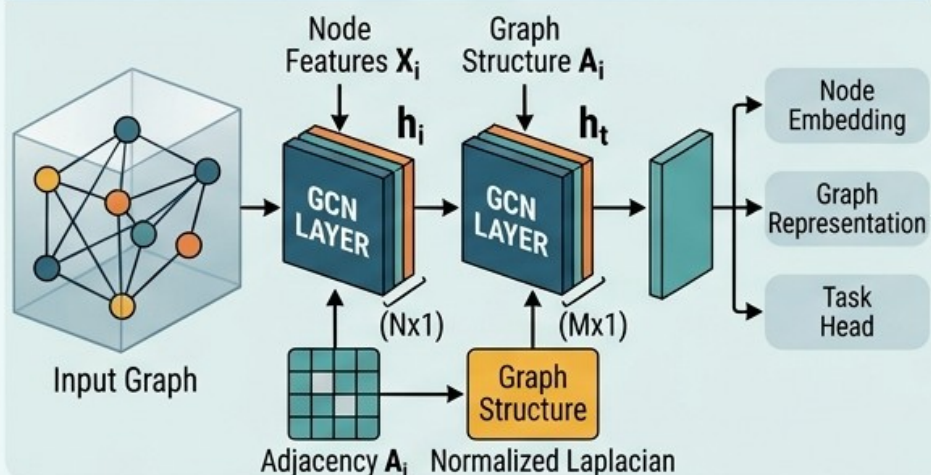
4.3.5 – Classical DL Techniques – GCN

UNDERSTANDING GRAPH CONVOLUTIONAL NETWORKS (GCN): ARCHITECTURE, PROPERTIES, MATRIX OPERATIONS, & APPLICATIONS

GCN PROPERTIES

- 1 SPATIAL REASONING**
Overcoming Euclidean constraints of CNNs.
- 2 MESSAGE PASSING**
Node features propagate over the graph.
- 3 WEIGHT SHARING**
One convolution kernel per graph, not per node.
- 4 LAYER-WISE LEARNING**
Hierarchical representation over multiple convolutions.

GCN ARCHITECTURE (FEATURE & STRUCTURE)



MATRIX OPERATIONS (GCN LAYER BREAKDOWN)

1. Feature Transformation

$$H^{(l)} \times W^{(l)} \rightarrow W_H$$

Input Features $H^{(l)}$ (NxK) × Weights $W^{(l)}$ (MxK) → W_H (NxK)

2. Graph Smoothing & Aggregation

$$\hat{A} \times W_H \rightarrow Z_{\text{raw}}$$

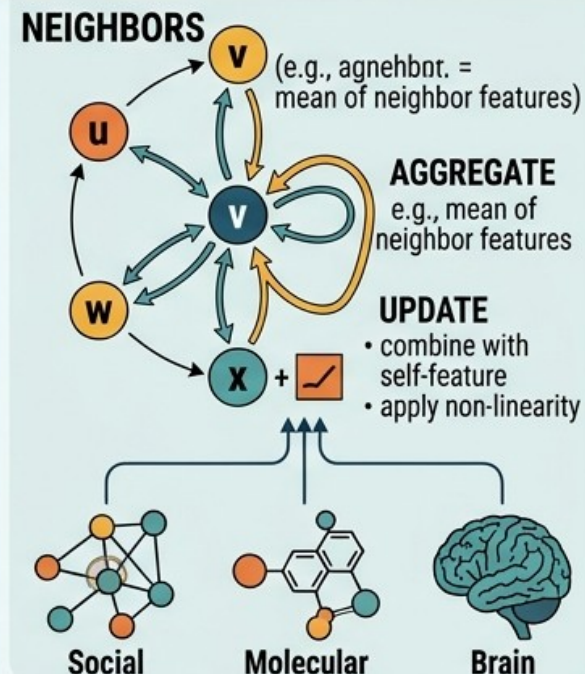
Normalized Adjacency $\hat{A}$ (NxN) × W_H (MxK) → Z_{raw} (NxK)

3. Non-Linear Activation

$$H^{(l+1)} = \sigma([Z_{\text{raw}}] + [b])$$

NORMALIZED ADJACENCY ($\tilde{A}$) $\tilde{A} = \left(\frac{1}{\sum_i \sum_j A_{ij}} (A + I) \right) \times \text{Adjacency Matrix}$

GCN LAYER DETAIL



GCN APPLICATION AREAS

- 1 DRUG DISCOVERY**
Small molecule binding sites. Result:
- 2 SOCIAL NETWORK ANALYSIS**
Person node grid, Connection types, Community detection.
- 3 RECOMENDER SYSTEMS**
Interaction graph.
- 4 TRAFFIC FORECASTING**
Predictions for specific points.
- 5 FINANCIAL FRAUD DETECTION**
Predictions for specific points.



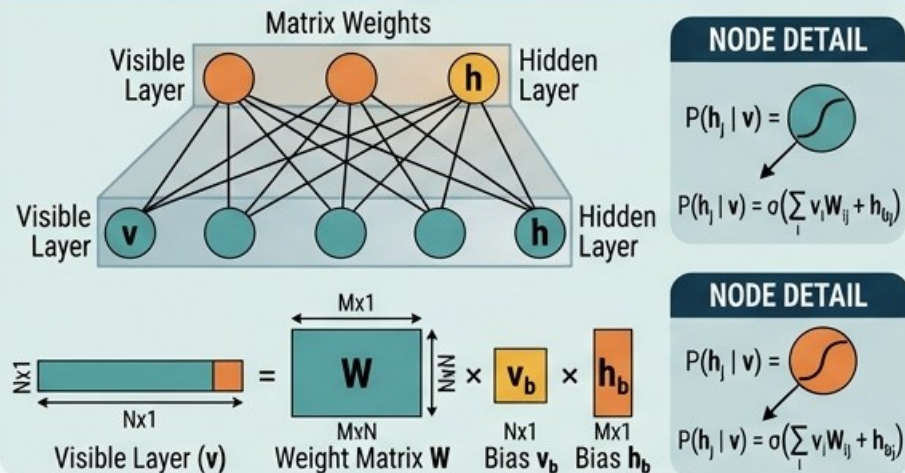
4.3.6 – Classical DL Techniques – RBM

UNDERSTANDING RESTRICTED BOLTZMANN MACHINES (RBM): ARCHITECTURE, PROPERTIES, MATRIX OPERATIONS, & APPLICATIONS

RBM PROPERTIES

- 1 UNSUPERVISED LEARNING**
Training data is unlabeled (e.g., unlabeled image grid)
- 2 STOCHASTIC NETWORK**
Visible and hidden units are random variables
- 3 NO INTRA-LAYER CONNECTIONS**
Connections only between visible and hidden layers 
- 4 ENERGY-BASED MODEL**
Lower energy states are more probable
- 5 CONTRASTIVE DIVERGENCE**
The standard learning algorithm

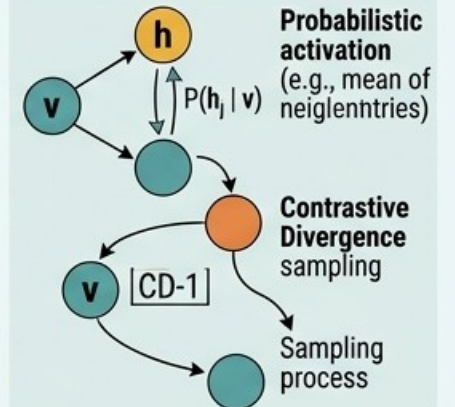
RBM ARCHITECTURE



MATRIX OPERATIONS (FORWARD & BACKWARD PASS)

- 1. Visible to Hidden Pass**
 $[v]^T (1 \times N) \times [W]^T (N \times M) + [h_b]^T (1 \times M) \rightarrow [p(h)]^T (1 \times M)$
Sigmoid applied element-wise
- 2. Hidden to Visible Pass**
 $[h]^T (1 \times M) \times [W] (M \times N) + [v_b]^T (1 \times N) \rightarrow [p(v)]^T (1 \times N)$
Sigmoid applied element-wise
- 3. Weight Update (CD-1)**
 $\Delta W = \eta * (\langle v^0 h^0 \rangle_{\text{data}} - \langle v^1 h^1 \rangle_{\text{model}})$
- 4. Energy Function**
 $E(v, h) = -\sum_i v_i v_{bi} - \sum_j h_j h_{bj} - \sum_i \sum_j v_i h_j W_{ij}$

RBM STOCHASTICITY & LEARNING DETAIL



RBM APPLICATION AREAS

- 1 High-dim input data**
Compress to low-dim features
DIMENSIONALITY REDUCTION
- 2 Feature Learning**
Learning higher-level features from raw data
FEATURE LEARNING
- 3 Matrix factorization**
Recommender systems
RECOMMENDER SYSTEMS
- 4 Generating new data samples**
e.g., new samples
GENERATIVE MODELLING
- 5 Initialization for deep belief networks**
PRE-TRAINING FOR DEEP NETWORKS



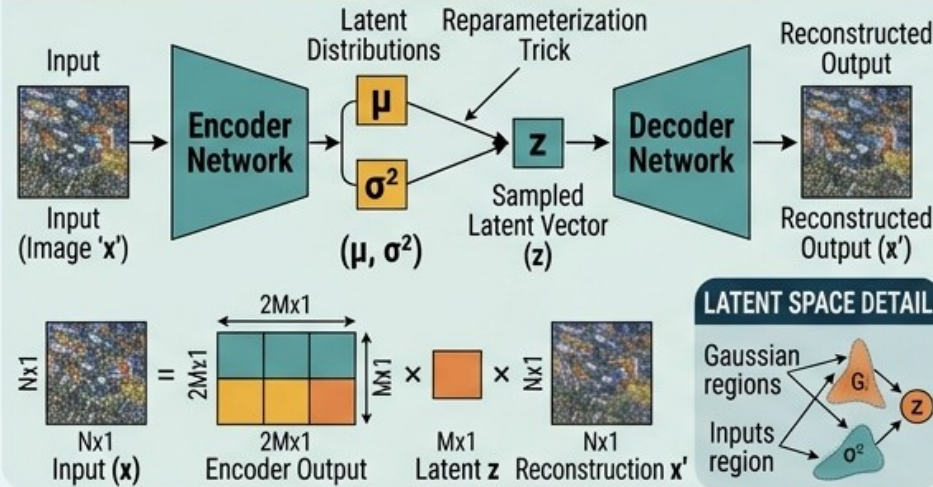
4.3.7 – Classical DL Techniques – VAE

UNDERSTANDING VARIATIONAL AUTOENCODERS (VAE): ARCHITECTURE, PROPERTIES, MATRIX OPERATIONS, & APPLICATIONS

VAE PROPERTIES

- 1 PROBABILISTIC LATENT SPACE**
Encodes inputs as probability distributions (e.g., Gaussian bell)
- 2 GENERATIVE MODELLING**
Can generate new data samples from latent space
- 3 REGULARIZED LATENT SPACE**
The latent distribution is constrained to be close to a prior (e.g., Normal $N(0,1)$)
- 4 UNSUPERVISED LEARNING**
Trained on unlabeled data to reconstruct and generate
- 5 REPARAMETERIZATION TRICK**
Enables backpropagation through a stochastic sampling process

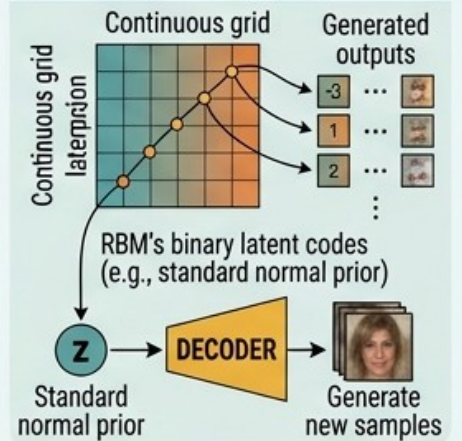
VAE ARCHITECTURE



MATRIX OPERATIONS (FORWARD PASS)

- 1. Encoder Forward Pass**
 $[x]^T (1 \times N) \times [W_{enc}]^T + [b_{enc}] \rightarrow [p(\mu, \log \sigma^2)]^T (1 \times 2M)$
- 2. Reparameterization Trick** $[z] = [\mu] + [\sigma] \odot [\epsilon]$
 $[z]^T (1 \times M) \times [W_{dec}]^T + [b_{dec}] \rightarrow [p(x'|z)]^T (1 \times N)$
Sigmoid or Linear activation
- 3. Decoder Forward Pass**
- 4. Loss Function**
Reconstruction Loss (e.g., MSE or Cross-Entropy) + KL Divergence Loss
 $KL(N(\mu, \sigma) || N(0, 1)) = -0.5 \sum (1 + \log(\sigma^2) - \mu^2 - \sigma^2)$

LATENT SPACE EXPLORATION & GENERATION



VAE APPLICATION AREAS

- 1 Noisy data**
Clean data output (e.g., denoised image)
DATA RECONSTRUCTION & DENOISING
- 2**
Generating new, unique samples (e.g., synthesized medical scans or object views)
DATA SYNTHESIS & AUGMENTATION
- 3**
Smooth transitions between images (e.g., from an open to a closed mouth face)
LATENT SPACE INTERPOLATION & MANIPULATION
- 4**
Identifying out-of-distribution data (e.g., flagged abnormal test result)
ANOMALY DETECTION
- 5**
Changing image attributes (e.g., changing daytime scene to night)
IMAGE-TO-IMAGE TRANSLATION

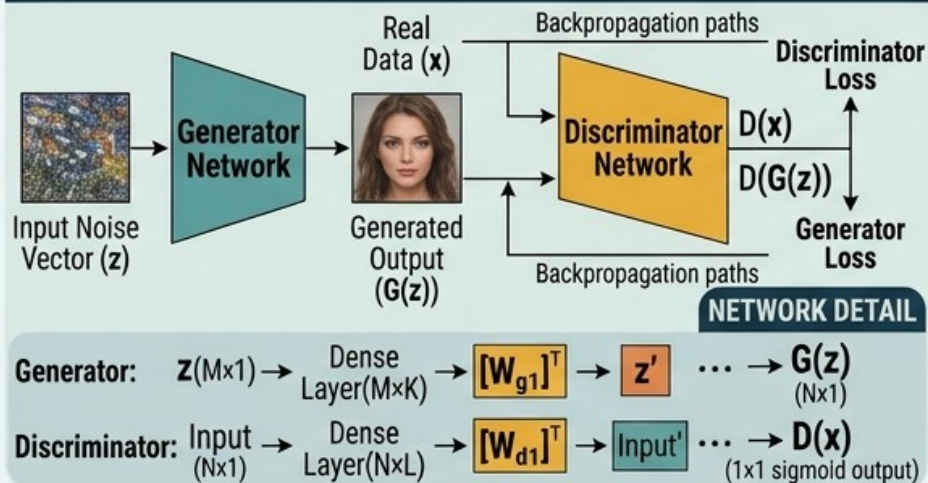
4.3.8 – Classical DL Techniques – GAN

UNDERSTANDING GENERATIVE ADVERSARIAL NETWORKS (GANs): ARCHITECTURE, PROPERTIES, MATRIX OPERATIONS, & APPLICATIONS

GAN PROPERTIES

- 1 ADVERSARIAL TRAINING**
Competition between Generator and Discriminator
- 2 GENERATIVE MODEL**
Learns to create new data samples
- 3 UNSUPERVISED LEARNING**
Trained on real, unlabeled data
- 4 LATENT SPACE EXPLORATION**
Traverses noise space to manipulate output features
- 5 MINIMAX GAME**
Generator tries to minimize $D(G(z))$, Discriminator tries to maximize it

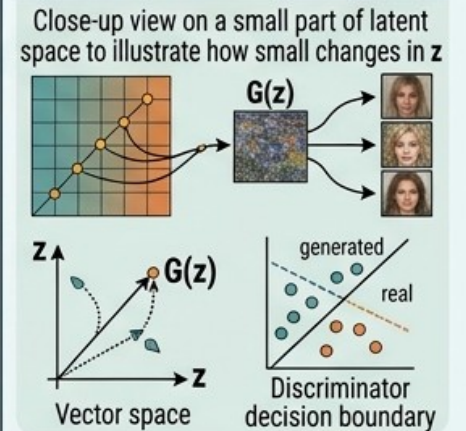
GAN ARCHITECTURE (GENERATION FLOW & LOSS)



MATRIX OPERATIONS (FORWARD PASS)

- 1. Generator Forward Pass**
 $[z]^T (1 \times M) \times [W_{g1}]^T + [b_{g1}] \rightarrow [p(G(z))]^T (1 \times N \times 1)$
Sigmoid or ReLU activation
- 2. Discriminator Forward Pass**
Real $[x]^T (1 \times N) \times [W_{d1}]^T + [b_{d1}] \rightarrow [p(D(x))]^T (1 \times 1)$
1x1 sigmoid
For Gener. $[G(z)]^T (1 \times N) \times [W_{d1}]^T + [b_{d1}] \rightarrow [p(D(G(z)))]^T (1 \times 1)$
1x1 sigmoid
- 3. Minimax Loss Function**
$$\mathcal{L}_{GAN} = \min_G \max_D \mathbb{E}_x [\log D(x)] + \mathbb{E}_z [\log(1 - D(G(z)))]$$

GAN STOCHASTICITY & ADVERSARIAL DETAIL



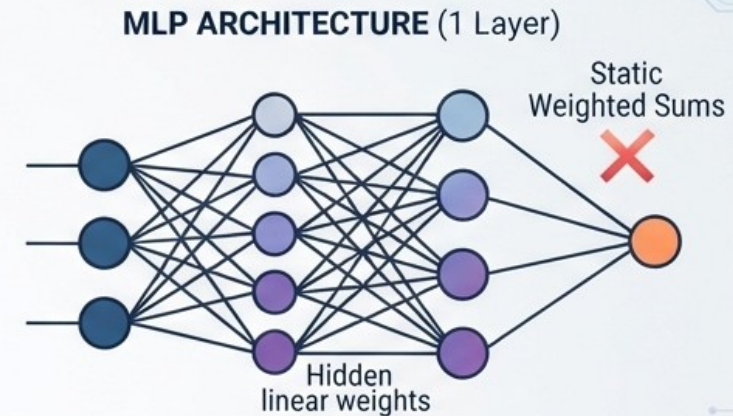
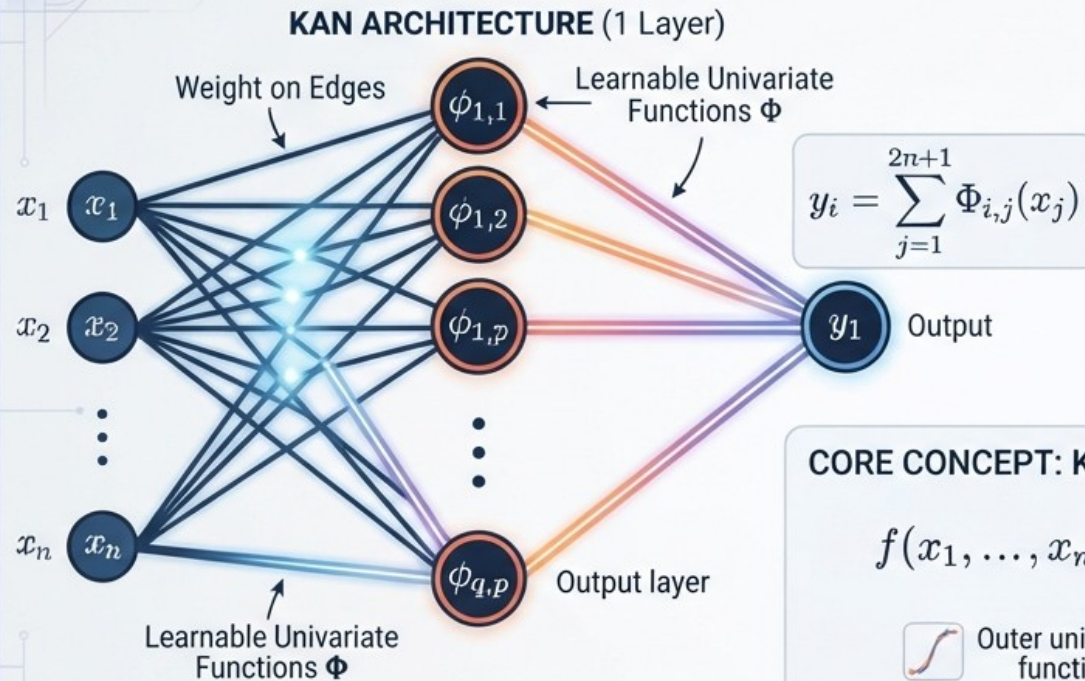
GAN APPLICATION AREAS

- 1 IMAGE GENERATION & SYNTHESIS**
Creating high-quality unique faces, landscapes, or objects (e.g., stylized generated images)
- 2 IMAGE-TO-IMAGE TRANSLATION**
Style transfer (e.g., converting a daytime photo to nighttime, painting style)
- 3 IMAGE SUPER-RESOLUTION**
Creating high-resolution details from low-resolution inputs
- 4 TEXT-TO-IMAGE GENERATION**
Creating images from descriptive text prompts
- 5 DATA AUGMENTATION**
Generating rare data samples for other models (e.g., rare disease scans)



4.3.9 – Classical DL Techniques – KAN

KOLMOGOROV-ARNOLD NETWORKS (KAN): FUNCTION DECOMPOSITION ARCHITECTURE



CORE CONCEPT: KOLMOGOROV-ARNOLD REPRESENTATION THEOREM

$$f(x_1, \dots, x_n) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^n \phi_{q,p}(x_p) \right)$$

Outer univariate function

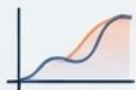
Inner univariate functions

number of input variables

Required number of basis functions in the hidden layer

KEY KAN CHARACTERISTICS

FUNCTION-BASED



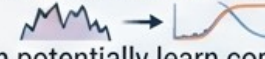
Learns arbitrary univariate functions on edges (e.g., B-splines), not fixed weights.

NO LINEAR WEIGHTS



Weights are replaced by non-linear, adaptive functions.

GREATER EXPRESSIVITY



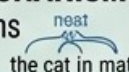
Can potentially learn complex functions with fewer parameters than MLPs in certain tasks.

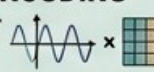


4.3 10 – Classical DL Techniques – Transformer

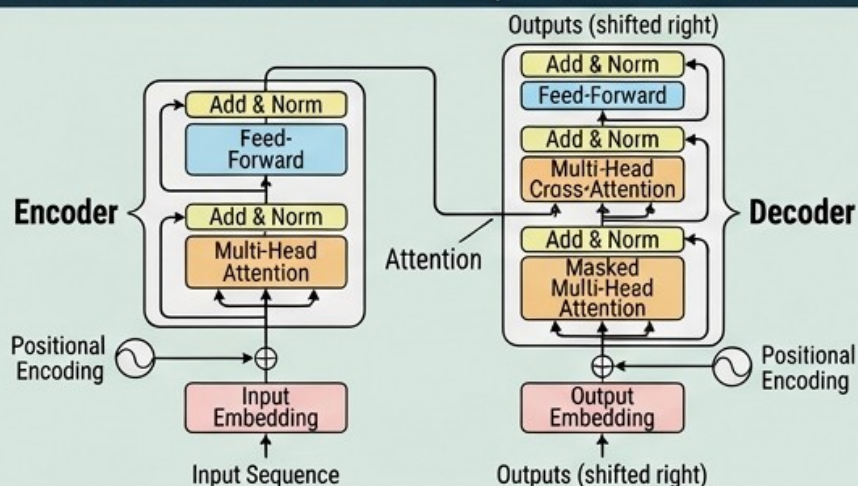
UNDERSTANDING TRANSFORMER NETWORKS: ARCHITECTURE, PROPERTIES, MATRIX OPERATIONS, & APPLICATIONS

TRANSFORMER PROPERTIES

- 1 SELF-ATTENTION MECHANISM**
Relates different positions of a single sequence

- 2 PARALLEL PROCESSING**
Decouples computation from sequential data, unlike RNNs
- 3 GLOBAL RECEPTIVE FIELD**
Every token attends to every other token

- 4 POSITIONAL ENCODING**
Introduces order information

- 5 MULTI-HEAD ATTENTION**
Learns different types of relationships simultaneously

TRANSFORMER ARCHITECTURE (ENCODER-DECODER FLOW)



MATRIX OPERATIONS (MULTI-HEAD SELF-ATTENTION)

1. Linear Transformation

$$[X]^T (T \times D) \times [W_{QK/K/V}]^T (D \times D') \rightarrow [Q/K/V]^T (T \times D')$$

$T(\text{tokens}) \times D'(\text{model dim per head})$

2. Attention Score Calculation 3. Value Aggregation

$$[A] = \text{softmax}\left(\frac{[Q] \times [K]^T}{\sqrt{d_k}}\right) \quad [\text{Attention_Head}] = [A] \times [V]$$

4. Head Concatenation & Final Linear

$$[\text{MultiHead}(Q, K, V)] = \text{Concat}(\text{head}_1, \text{head}_2, \dots) \times [W_o]$$

FFN (Feed-Forward Network)

$$[X]^T (T \times D) \times [W_1]^T (D \times D_{ff}) \rightarrow \text{ReLU} \times [W_2]^T (D_{ff} \times D) \rightarrow \text{FFN}(X')$$

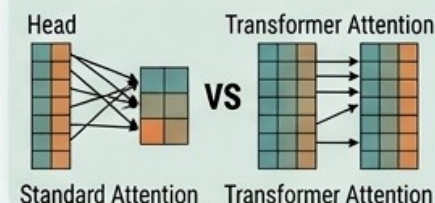
ATTENTION VISUALIZATION VISUALIZATION & DETAIL

Close-up-view on a lasst sequence to how multi-head ("heads"):

"The cat sat on the mat"

"The cat sat on the mat"

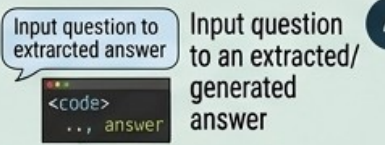
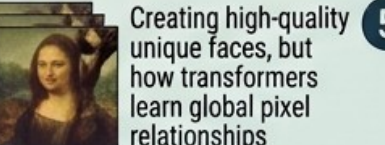
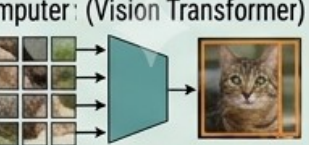
"The cat sat on the mat"



TRANSFORMER APPLICATION AREAS

- 1 Source text**

MACHINE TRANSLATION
- 2 Document**

TEXT SUMMARIZATION
- 3 Input question to extracted answer**

QUESTION ANSWERING
- 4 Input question to an extracted/generated answer**

IMAGE GENERATION & SYNTHESIS
- 5 Computer: (Vision Transformer)**

COMPUTER VISION



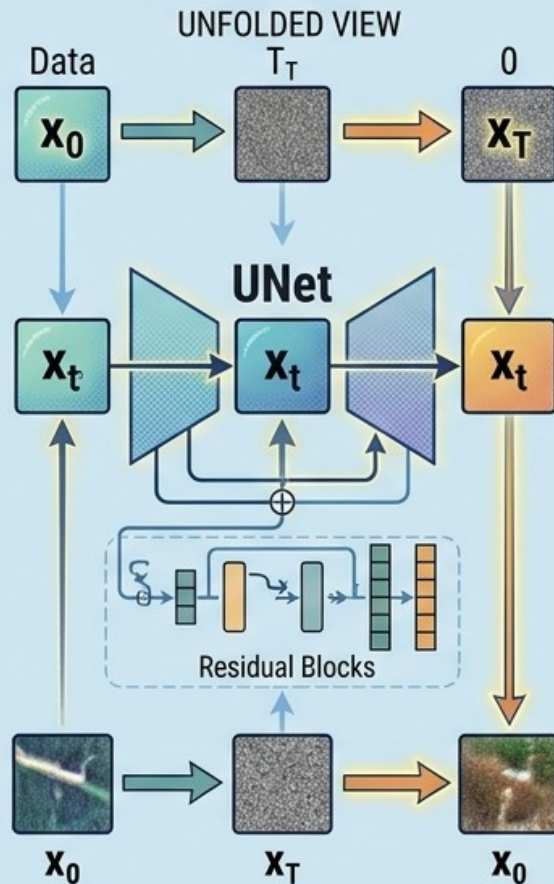
4.3.11 – Classical DL Techniques – Diffusion

UNDERSTANDING DIFFUSION MODELS: ARCHITECTURE, PROPERTIES, MATRIX OPERATIONS, & APPLICATIONS

DIFFUSION MODEL PROPERTIES

- 1 NOISE-TO-DATA GENERATION**
 - reverses noise process
- 2 STOCHASTIC DENOISING**
 - probabilistic steps
- 3 PROGRESSIVE REFINEMENT**
 - gradual detail creation
- 4 LATENT SPACE SAMPLING**
 - operates on latent vectors
- 5 DENSE CONNECTIONS**
 - e.g., inside UNet blocks

DIFFUSION PROCESS & ARCHITECTURE



MATRIX OPERATIONS (REVERSE PROCESS - DENOISING)

- 1. NOISE ESTIMATION**
 $[x_t]^T (Nx D) \times \text{UNet}(W, b) \rightarrow [\epsilon_\theta]^T (Nx D)$
(TxD) $\rightarrow$ (TxD)
- 2. LATENT STATE UPDATE**
 $[x_{t-1}] = f([x_t], [\epsilon_\theta], [\sigma_t], [z_t])$
$$= \begin{matrix} \text{Grid} & \times & \text{Matrix} & \left(\begin{matrix} \text{Vector} & + & \text{Vector} \end{matrix} \right) \end{matrix}$$

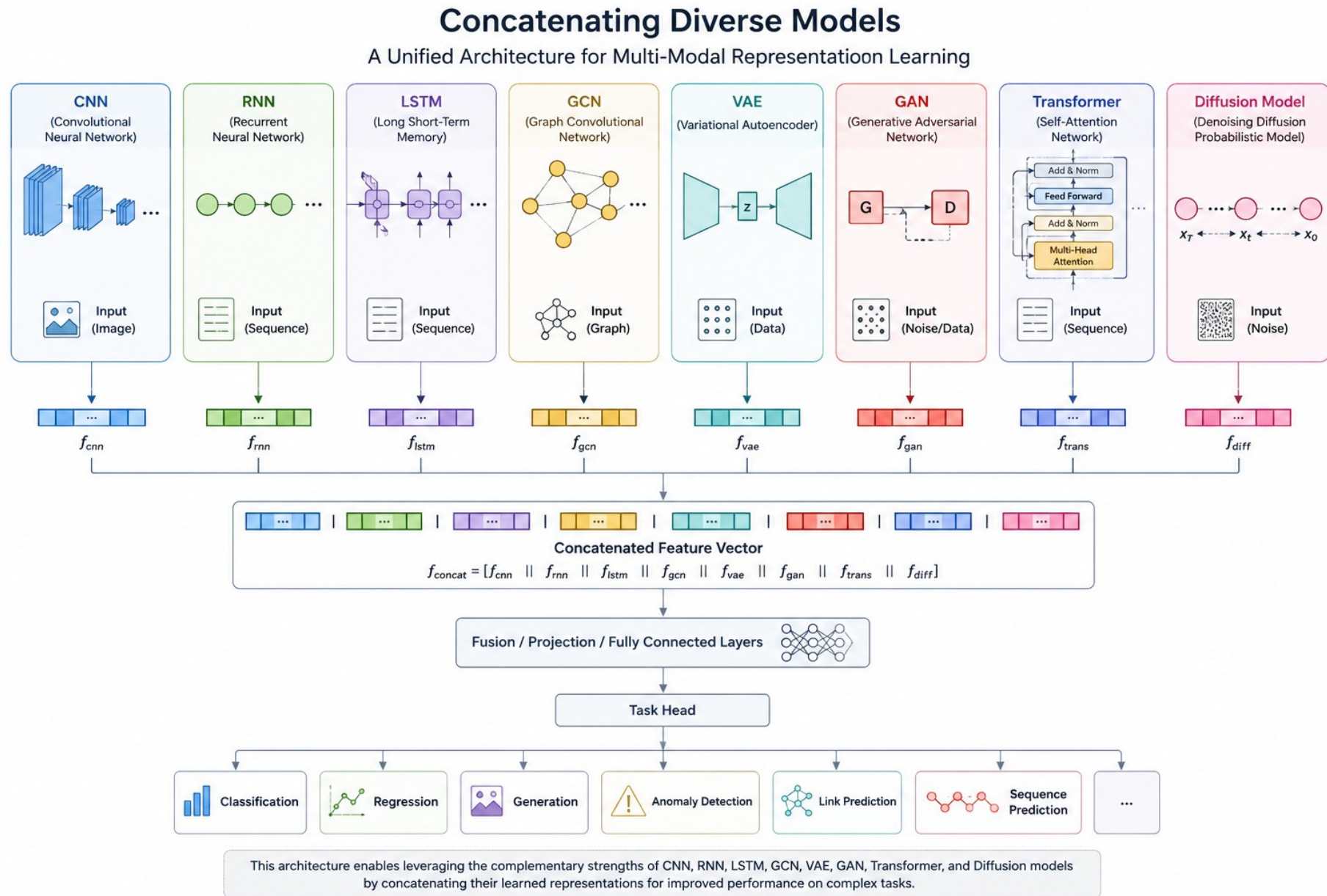
(TxD) (TxD) ($Tx4$) ($1x4$)
- 3 NOISE ACCUMULATION**
Visualization
Time
Cumulative error cumu. error
- 4 ACTIVATION FUNCTION**
ReLU or Sigmoid

DIFFUSION MODEL APPLICATIONS

- 1 IMAGE SYNTHESIS & GENERATION**
- 2 IMAGE-TO-IMAGE TRANSLATION**
- 3 IMAGE SUPER-RESOLUTION**
- 4 TEXT-TO-IMAGE GENERATION**
- 5 DATA DENOISING & REPAIR**



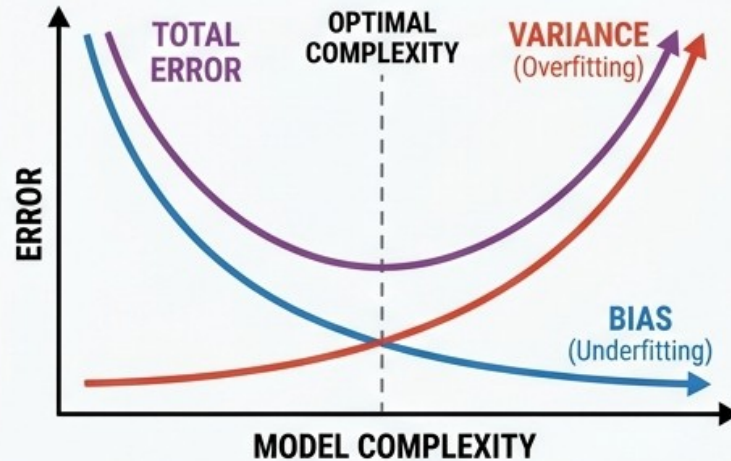
4.3.12 – Classical DL Techniques – Concatenating Models



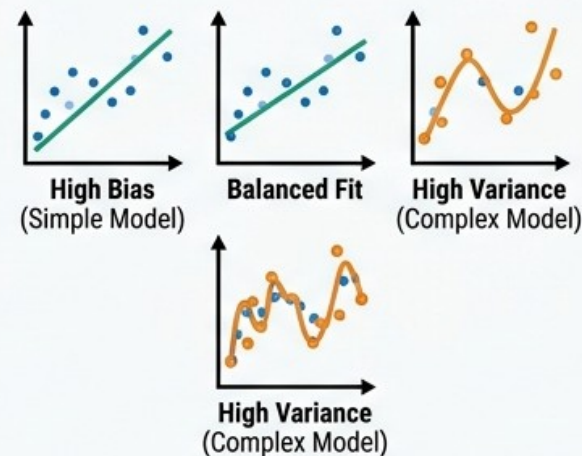
4.4 Bias/Variance Problem, Regularization and Smooth Learning

BIAS-VARIANCE PROBLEM & REGULARIZATION

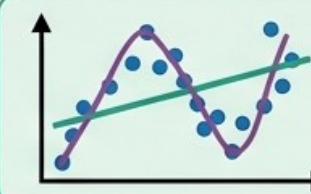
THE BIAS-VARIANCE TRADEOFF



CONCEPTUAL MODEL PERFORMANCE

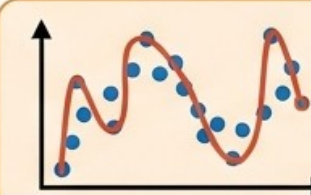


UNDERSTANDING BIAS AND VARIANCE



HIGH BIAS (Underfitting)

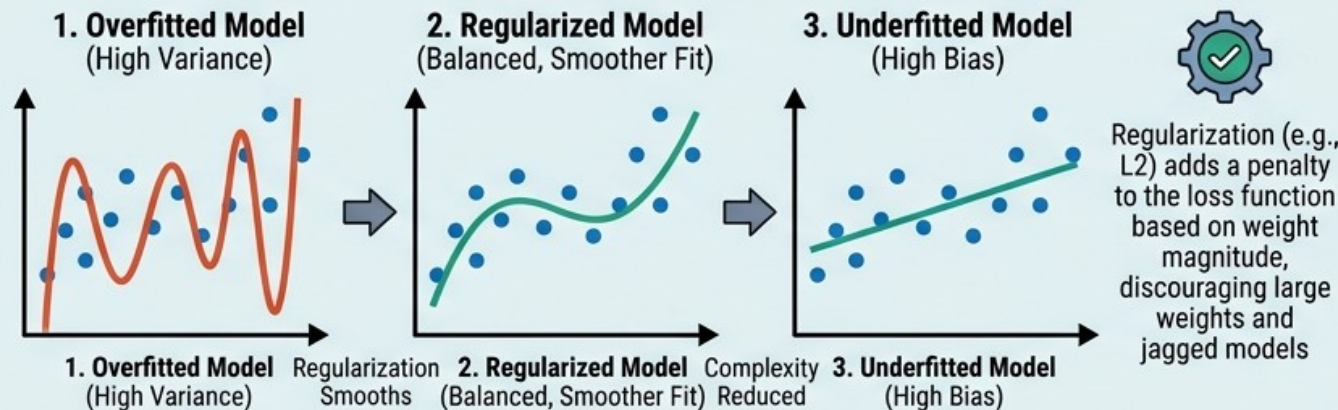
- Linear model or training points
- Perfectly passing all training points but missing new data



HIGH VARIANCE (Overfitting)

- Perfectly passing through all training model
- Perfectly fitting points but missing new data

EFFECTS OF REGULARIZATION ON MODEL SMOOTHNESS

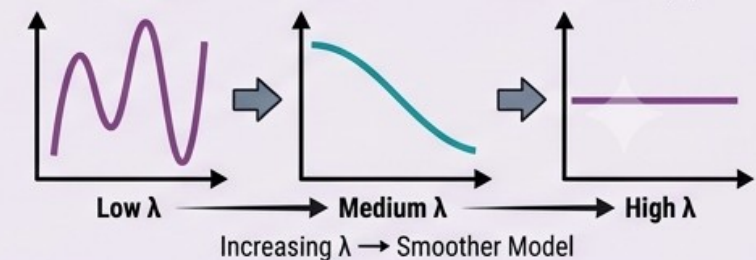


HOW REGULARIZATION REDUCES VARIANCE

LOSS FUNCTION WITH REGULARIZATION TERM

$$\text{Loss} = \text{Error} + \lambda * \text{Complexity}$$

EFFECT OF REGULARIZATION PARAMETER (λ)



4.5 – Preventing Overfitting (Memorization)

1. *Early Stopping*
2. *Dropout*
3. *Weight Regularization*
4. *Data Augmentation (**Very important)*
5. *Simplify Model*
6. *Batch Normalization*

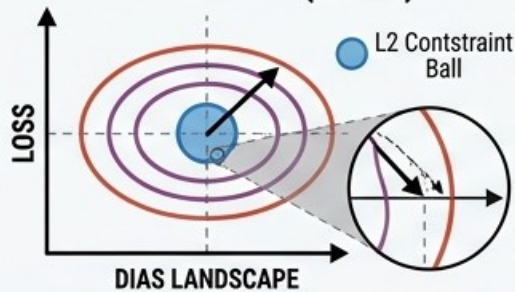


4.5 – Preventing Overfitting (Memorization)

ADVANCED REGULARIZATION TECHNIQUES & WEIGHT CONSTRAINT METHODS

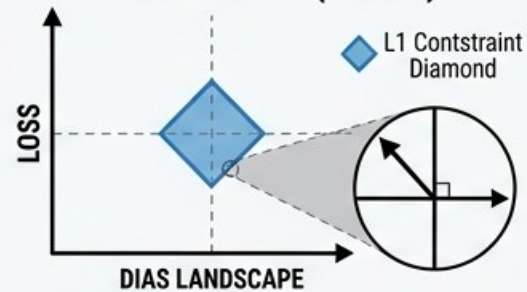
WEIGHT DECAY & PENALTY CONSTRAINTS

L2 PENALTY (RIDGE)



L2 (Ridge): Penalty $\lambda \sum w_i^2$ shrinks weights towards zero, reduces model variance.

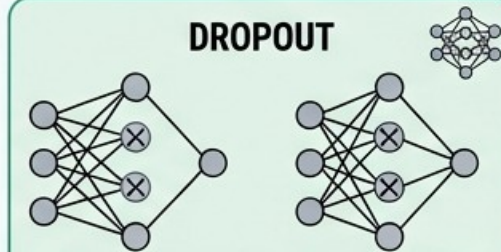
L1 PENALTY (LASSO)



L1 (Lasso): Penalty $\lambda \sum |w_i|$ induces sparsity (zero weights), effective for feature selection.

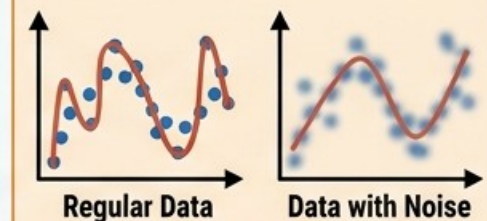
MODEL ENSEMBLING & NOISE TECHNIQUES

DROPOUT



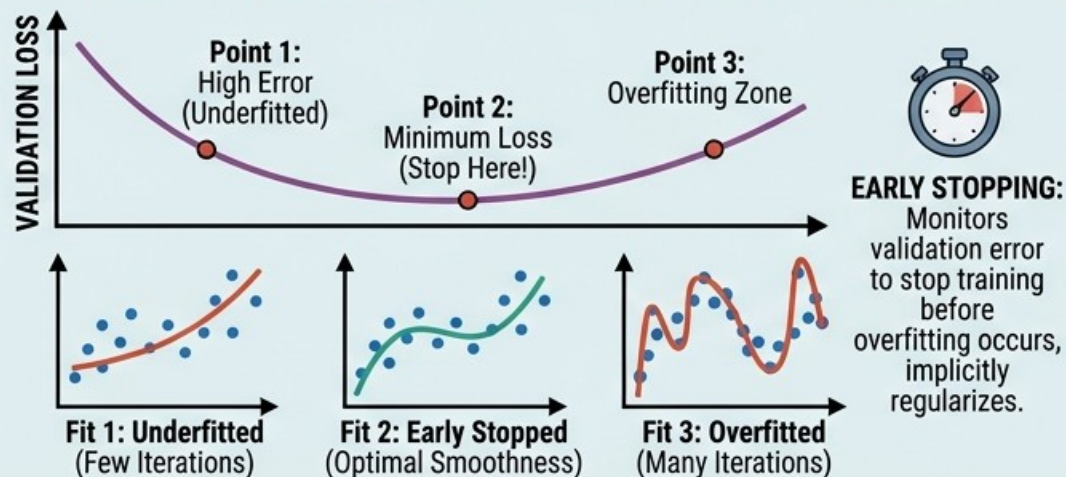
Randomly drops units during training to prevent co-adaptation, works as an ensemble method.

NOISE INJECTION



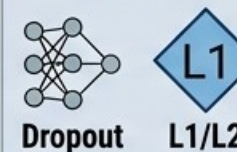
Adds noise to inputs or weights to improve robustness and generalization.

ITERATIVE REGULARIZATION & EARLY STOPPING



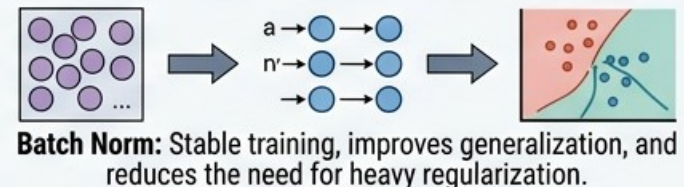
ADVANCED REGULARIZATION STRATEGIES

COMBINED TECHNIQUES



COMBINED TECHNIQUES

BATCH NORMALIZATION & GENERALIZATION



Batch Norm: Stable training, improves generalization, and reduces the need for heavy regularization.

DATA AUGMENTATION



Data Augmentation: Artificially expands dataset to increase diversity, reduces model's reliance on specific patterns.



4.6 – Data Augmentation

1. Find new ways to collect data : Automate the data collection phase if possible.
2. Generate Data Using GANs
3. Generate Data Using Monte Carlo Simulator (or Deterministic Algorithms / Unity)
4. Generate Data By Transforming (Noising, Apply Affine Transformations, Frequency Deviations, Image Manipulation Methods, Sound Manipulation Methods, ...)



4.6 – Data Augmentation

UNDERSTANDING DATA AUGMENTATION: TECHNIQUES, MECHANISMS, MATRIX OPERATIONS, & APPLICATIONS

DATA AUGMENTATION STRATEGIES

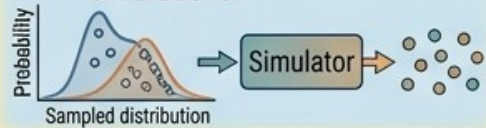
1 FINDING NEW WAYS TO COLLECT DATA

Strategies for novel data acquisition.
→ Replaces 'Early Stopping' plot.



3 GENERATING DATA USING MONTE CARLO SIMULATOR

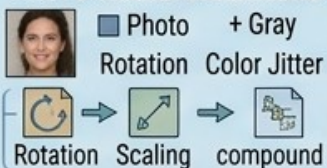
- Generating data through simulations.



4 AUGMENTING USING DATA RELATED TECHNIQUES

- Directly transforming existing data for variety. $L_{xD} = \sum_i w_i$

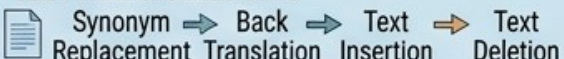
PHOTO TRANSFORMATIONS



SOUND WAVE TRANSFORMATIONS

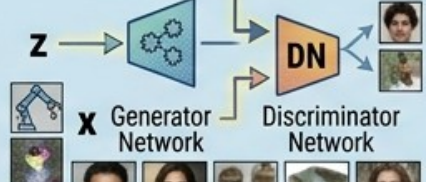


TEXT TRANSFORMATIONS

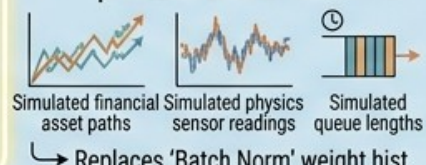


2 GENERATING DATA USING GANs

Creating synthetic data with Generative Adversarial Networks.



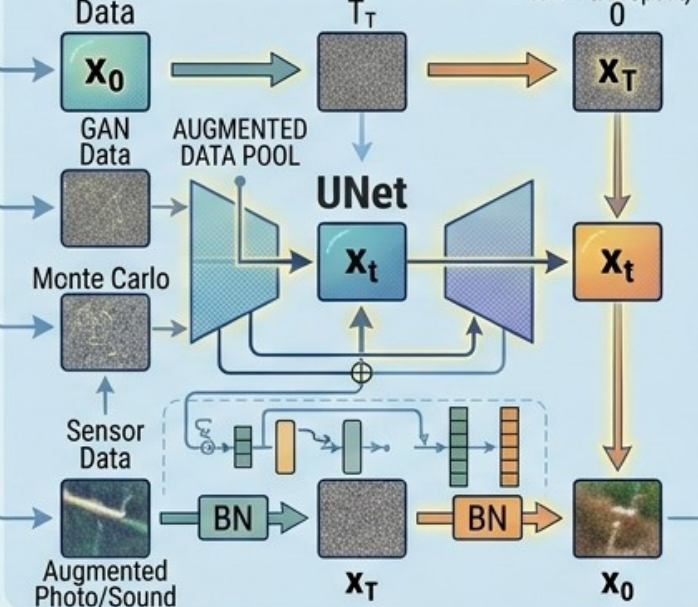
Examples "Dropout" mini-net.
• Generating data through probabilistic simulations.



→ Replaces 'Batch Norm' weight hist.

MECHANISMS & ARCHITECTURE INTEGRATION

AUGMENTED DATA → SIMPLE MODEL → (with added layers for BN & Dropout)



MATRIX OPERATIONS (AUGMENTATION EFFECTS)

1 IMAGE ROTATION EFFECT

$$[x]_{\text{rotated}} = [x] \times [R]^T \text{ where } \begin{pmatrix} 2 & 2 \\ 2 \times 2 \end{pmatrix}$$

$T \times D$ grid × 2×2 matrix = $T \times D$ rotated grid

2 SOUND PITCH SHIFT EFFECT

$$[J] = \begin{matrix} T \times D \\ T \times S \end{matrix}$$

$T \times D$ grid → $T \times S$ shifted grid

3 TEXT SYNONYM REPLACEMENT EFFECT

specific word vectors = replaced with neighbor vectors - vocabulary space

$T \times D$ grid → $(1 \times D)$ vector

AUGMENTATION METHOD APPLICATIONS

1 ADVANCED FACE CLASSIFICATION

From collected, GAN, and transformed

Face images → Classification

2 ROBUST VOICE RECOGNITION

From transformed sound files

Sound waves → Recognition

3 NATURAL LANGUAGE PROCESSING

From transformed text

Text → LSTMs

4 COMPUTER VISION TRAINING

Robustness to new data perspectives

Images → Training

5 DATA DENOISING

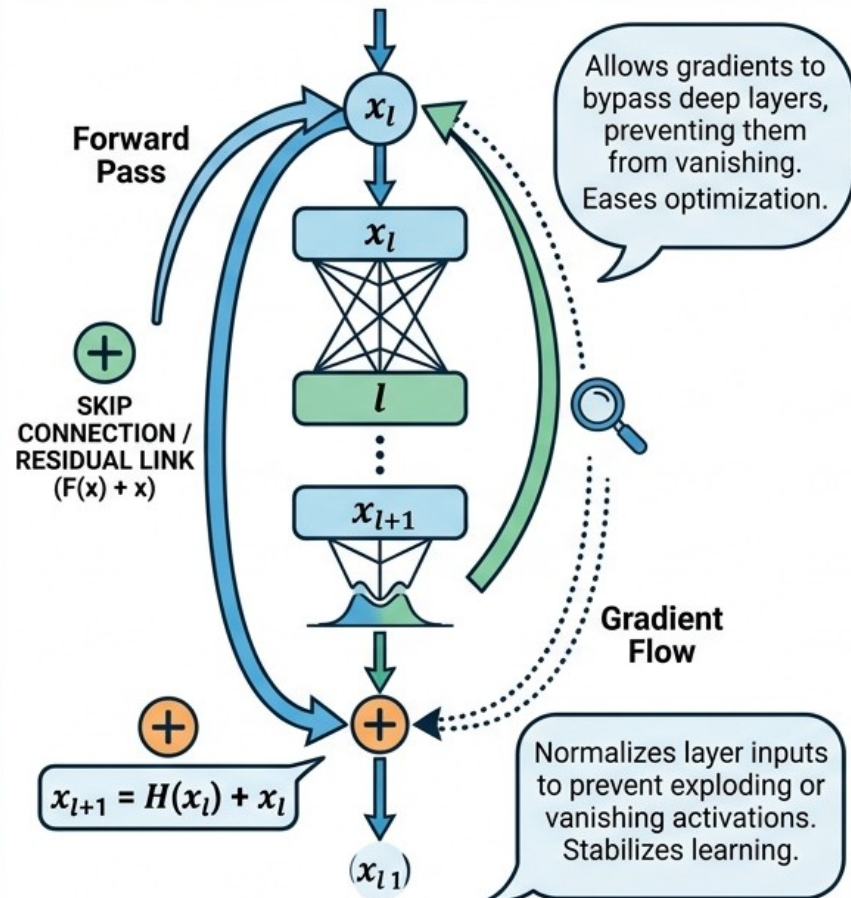
Batch Norm for consistent features

Data → Denoising

4.7 – Preventing Vanishing Gradient

1. Residuals (Skip Connections), Use RELU/LeakyRELU Activation Fn., Batch Normalization

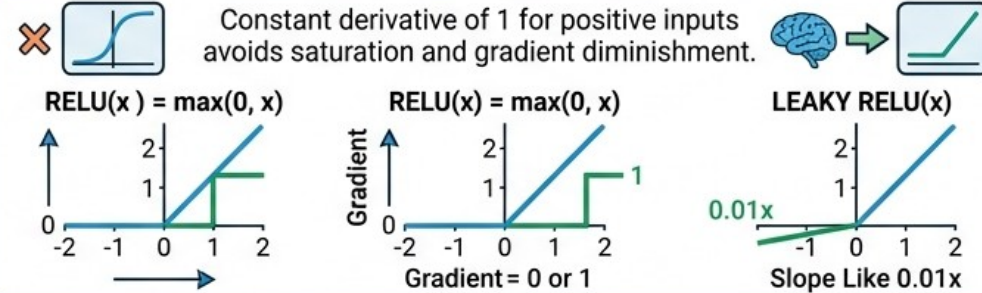
1. RESIDUALS (SKIP CONNECTIONS)



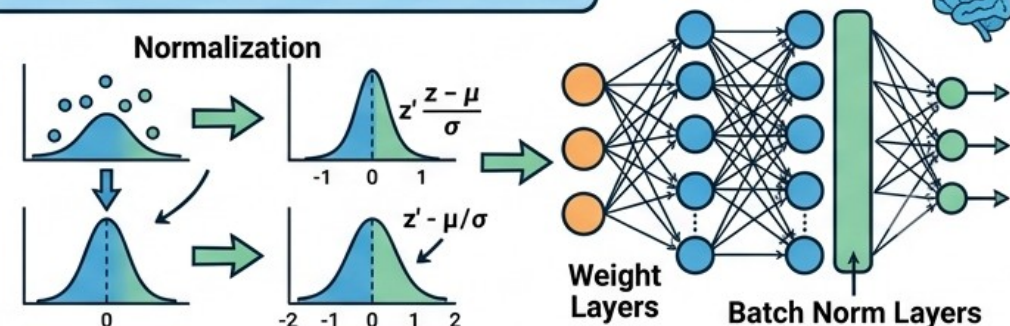
PREVENTING VANISHING GRADIENT IN DEEP NETWORKS

KEEPING GRADIENTS STRONG & HEALTHY

2. USE RELU / LEAKY RELU ACTIVATION FN.



3. BATCH NORMALIZATION



4.8 Cross Entropy, Learning Rate Momentum

≧ Cross Entropy, Learning Rate, Momentum ≦

1) Cross Entropy

Measures the difference between the predicted probabilities and the true distribution.

Example (3 classes)

Class	True (y)	Predicted (p)
Cat	1	0.70
Dog	0	0.20
Bird	0	0.10

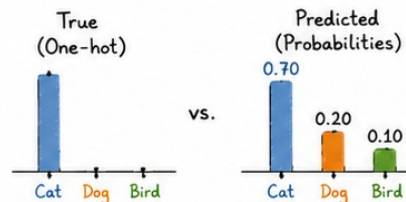
Cross Entropy Loss:

$$L = - \sum_i y_i \log(p_i)$$

For this example:

$$L = -\log(0.70) = 0.357$$

(lower is better)



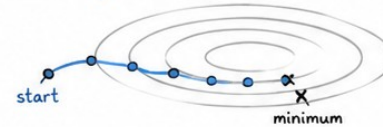
2) Learning Rate (η)

Controls the step size when updating parameters.

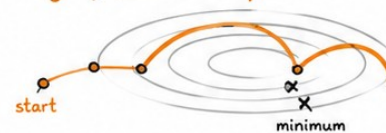
Update rule (Gradient Descent):

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t)$$

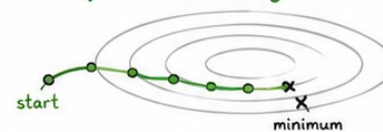
Small η : slow but stable



Large η : fast but risky



Good η : efficient convergence



3) Momentum (μ)

Helps accelerate updates in consistent directions and dampens oscillations.

Update with Momentum:

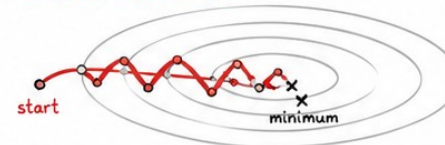
$$v_{t+1} = \mu v_t - \eta \nabla L(\theta_t)$$

$$\theta_{t+1} = \theta_t + v_{t+1}$$

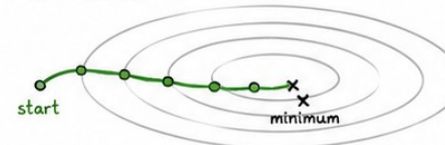
Intuition:

Momentum = accumulating velocity in the gradient direction.

Without Momentum



With Momentum



Summary

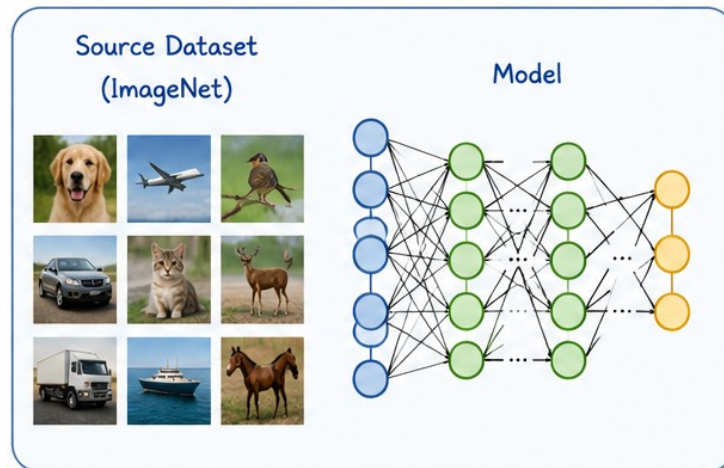
- **Cross Entropy**: measures how far predictions are from the truth.
- **Learning Rate**: determines how big each update step is.
- **Momentum**: builds velocity to go faster and smoother.



4.9 Transfer Learning

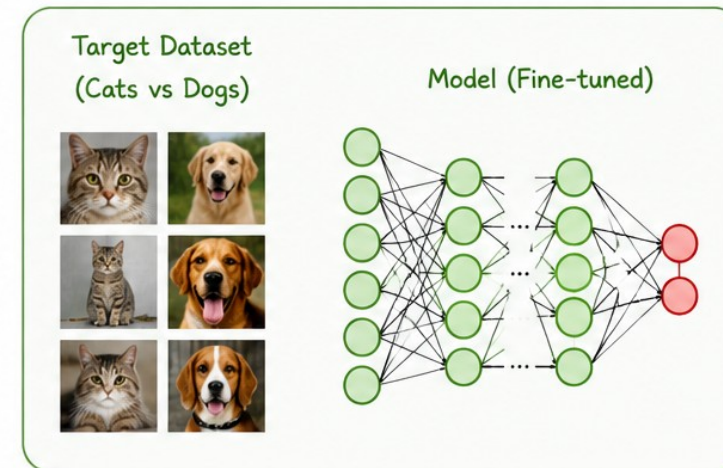
TRANSFER LEARNING

1. Pre-train on Source Task
(Large Dataset)



The model learns **general features**
(edges, textures, shapes, etc.)

2. Fine-tune on Target Task
(Smaller Dataset)



The model **adapts to the target task**
with a smaller dataset

Transfer learned
knowledge



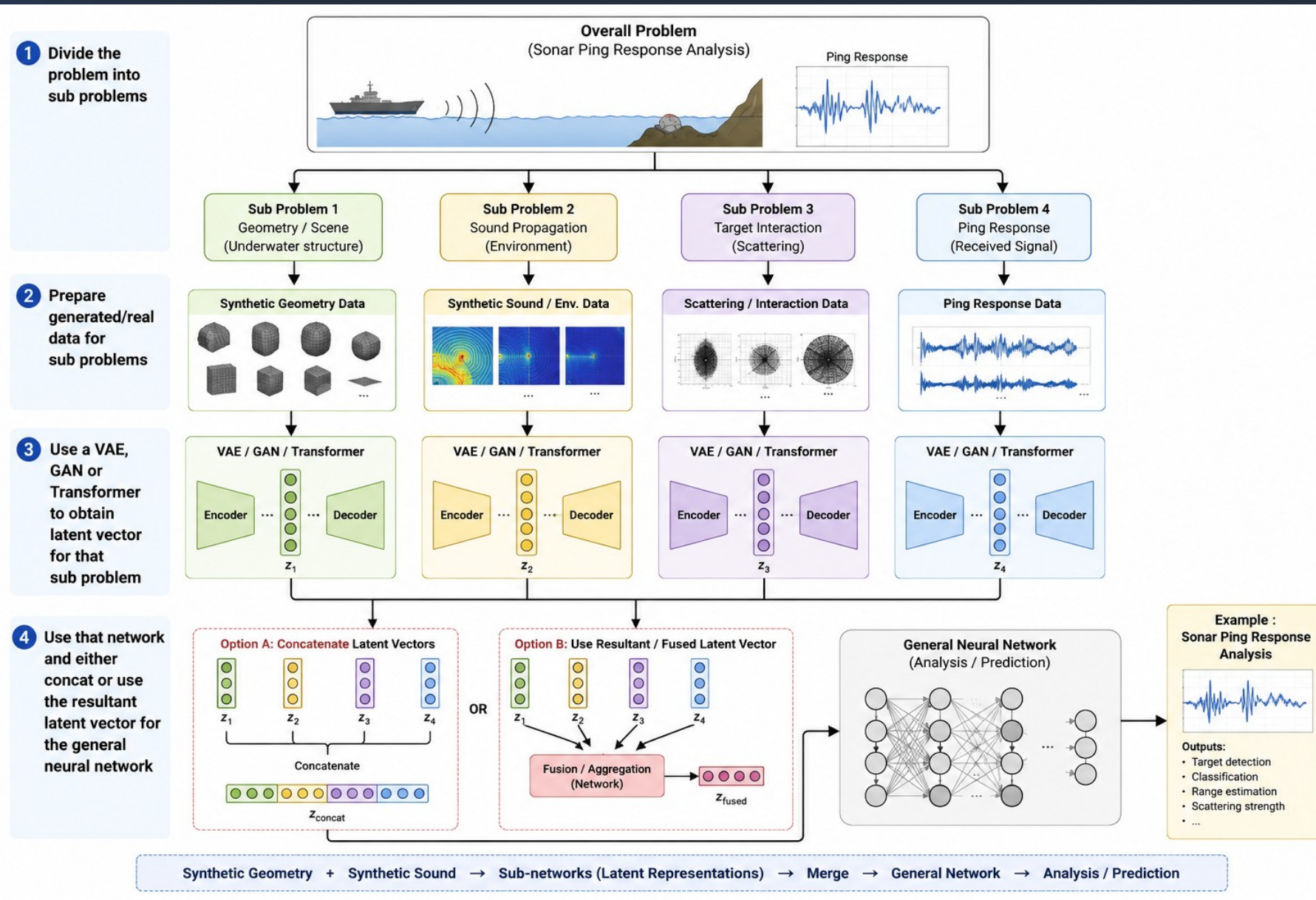
(Keep learned
weights)

Key Idea

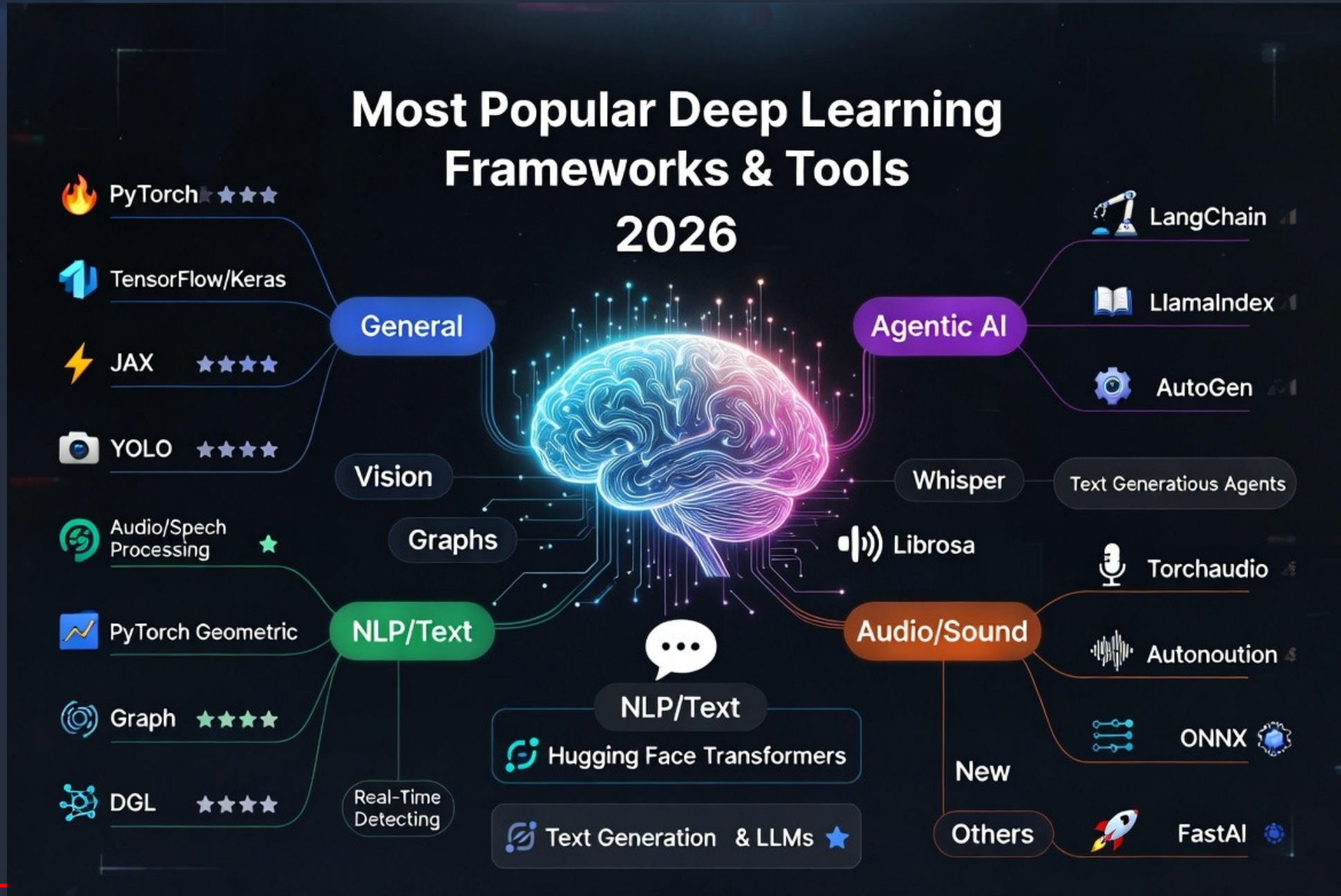
Leverage knowledge learned from a large **source task**
and apply it to a related **target task**.



4.10 Solving Problems Using Neural Networks



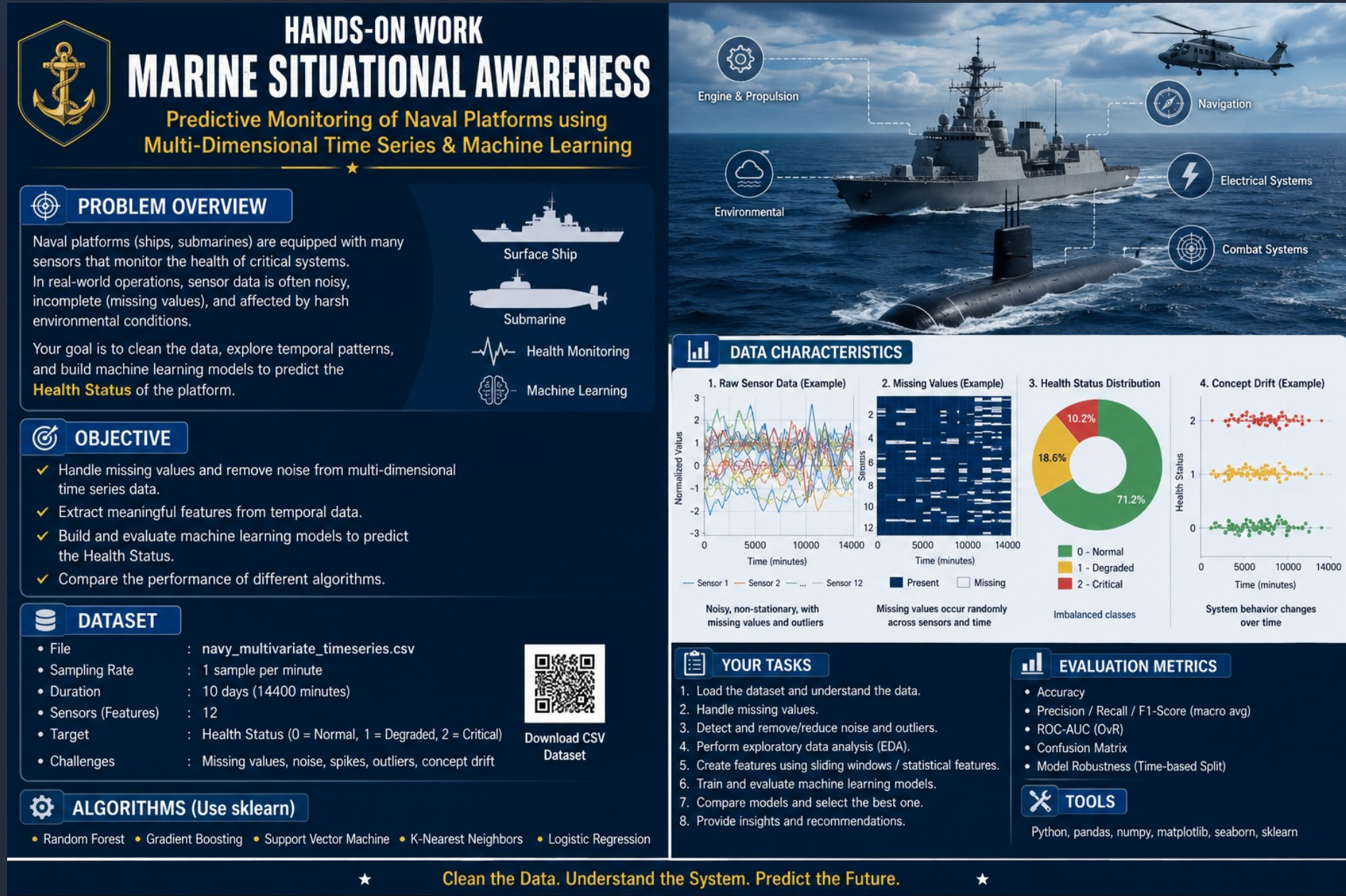
Popular Frameworks



Homework-1.0

Deliver :

1. Data Visualization using Matplotlib before and after data cleaning.
2. Enumerate meaningful features (columns)
3. Prediction results with evaluation metrics.



Homework-1.1

Problem : Predict the noise level of a room by measuring the light level

Tasks :

1. Collect Data : Write a python program to record 3-second video every hour (webcam+microphone). Leave the program to record for 3 days.
2. Clean the Data : Delete the corrupted files using a python program.
- 3.
4. Extract sound and images from video using a python program.
5. Use a CNN network that inputs image and predicts sound decibel.

Deliver :

1. Data Visualization using Matplotlib before and after data cleaning.
2. Python programs.
3. Prediction results with evaluation metrics.



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